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Diploma Thesis

Aeroelastic Tailoring of High- Aspect Ratio Wings via Tow-steering

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Περίληψη

Τις τελευταίες δεκαετίες, τα σύνθετα υλικά χρησιμοποιούνται ολοένα και περισσότερο στην αεροδιαστημική βιομηχανία, εξαιτίας των εξαιρετικών τους μηχανικών ιδιοτήτων για ελαφρές κατασκευές και την επιπλέον ευελιξία που δίνουν στον σχεδιασμό. Παραδοσιακά, στις υψηλών απαιτήσεων εφαρμογές χρησιμοποιούνται πολύστρωτες δομές με μονοδιευθυντικές στρώσεις απο ινώδες σύνθετο υλικό. Ωστόσο, με την εισαγωγή των αυτομάτων μηχανών τοποθέτησης ινών (AFP), πλέον είναι δυνατή η κατασκευή συνθέτων με ίνες που ακολουθούν καμπύλες διαδρομές εντός της κάθε στρώσης. Αυτή η διαδικασία είναι γνωστή ως "tow-steering" και τα αντίστοιχα σύνθετα ως "σύνθετα μεταβλητής δυσκαμψίας".

Τα σύνθετα μεταβλητής δυσκαμψίας προσδίδουν ακόμα μεγαλύτερη ευελιξία στον σχεδιασμό δομών, εκμεταλλευόμενα την ανισοτροπία των ινώδων συνθέτων για βελτίωση της δομικής συμπεριφοράς, χωρίς επιπλέον προσθήκη βάρους. Ωστοσο, λαμβάνοντας υπόψιν την πολυπλοκότητα των προβλημάτων αεροελαστικότητας, δεν είναι ούτε προφανές ούτε εύκολο να βρεθούν οι βέλτιστες διευθύνσεις των ινών. Για αυτό, επιβάλλεται συχνά η χρήση προηγμένων υπολογιστικών μεθόδων και αλγόριθμων βελτιστοποίησης. Στην παρούσα διπλωματική εργασία, αναπτύσσεται ενα πλάισιο δομικής βελτιστοποίησης για το αεροελαστικό μοντέλο αναφοράς πτέρυγας υψηλού εκπετάσματος uCRM-13.5. Το δομικό και το αεροδυναμικό μοντέλο της πτέρυγας υλοποιούνται στο υπολογιστικό πακέτο Nastran, ενώ για την βελτιστοποίηση χρησιμοποιείται ο αλγόριθμος MIDACO. Πιο συγκεκριμένα, η δομική συμπεριφορά της πτέρυγας μοντελοποιείται από έναν συνδυασμό Πεπερασμένων στοιχείων κελύφους και δοκού, ενώ ένα πλέγμα επίπεδων στοιχείων διπόλου (DLM) χρησιμοποιείται για τα αεροδυναμικά φορτία. Μια σειρά βελτιστοποιήσεων πραγματοποιείται για να αξιολογηθούν τα οφέλη της χρήσης των συνθέτων μεταβλητής δυσκαμψίας σε σύγκριση με τα συμβατικά σταθερής δυσκαμψίας σύνθετα. Κατά την διάρκεια της κάθε βελτιστοποίησης, επιβάλλονται περιορισμοι αστοχίας υλικών, λυγισμού, καθώς και εμφάνισης του φαινομένου πτερυγισμού.

Αρχικά, πραγματοποιείται μελέτη ελαχιστοποίησης της μάζας της πτέρυγας, χρησιμοποιώντας σύνθετα σταθερής δυσκαμψίας για την επιδερμίδα, ώστε να καθοριστεί η βέλτιστη συμβατική

δομή. Στην συνέχεια, εισάγονται στο πρόβλημα μεταβλητές για την καμπύλωση των ινών, οδηγώντας σε μείωση κατά 12.5% της μάζας σε σχέση με την βέλτιστη συμβατική δομή. Έπειτα, χρησιμοποιώντας την βέλτιστη συμβατική διάταξη και τις μεταβλήτες καμπύλωσης των ινών , η πτέρυγα βελτιστοποιείται, πρώτον ως προς τους δείκτες αστοχίας των πολύστρωτων της επιδερμίδας, και έπειτα ως προς την ταχύτητα πτερυγισμού. Σύμφωνα με τα αποτελέσματα, η χρήση σύνθετων υλικών μεταβλητής δυσκαμψίας οδήγησε σε μείωση του μέγιστου δείκτη αστοχίας κατά 47%, καθώς και αύξηση της ταχύτητας πτερυγισμού κατά 29% για τις εκάστοτε βελτιστοποιήσεις, χωρίς την προσθήκη επιπλέον μάζας. Τέλος, διενεργείται ταυτόχρονη βελτιστοποίηση της πτέρυγας ως προς την αντοχή της επιδερμίδας σε αστοχία, την αντοχή σε λυγισμό καθώς και την ταχύτητα πτερυγισμού. Ο σκοπός της παραπάνω μελέτης είναι η διερεύνηση των υφιστάμενων συμβιβασμών στον σχεδιασμό με χρήση συνθέτων μεταβλητής δυσκαμψίας.

Λέξεις Κλειδιά: Αεροελαστικότητα, Βελτιστοποίηση, NASA Common Research Model, Σύνθετα Υλικά Μεταβλητής Δυσκαμψίας, Πτερυγισμός

Abstract

Over the past several decades, composite materials have been used progressively more in the aerospace industry due to their superior performance, lightweight design and inherent ‘tailorability’. Traditionally, unidirectional composites stacked to form a laminated structure are used for high performance load carrying applications. However, with the introduction of automatic fiber placing machines, it is now possible to manufacture composites with fibers that follow curved paths inside the laminae. This process is known as *tow-steering* and the resulting composites as *variable stiffness composites*.

Tow-steered composites allow for even greater flexibility in the design of structures by taking advantage of the anisotropic behavior of fiber composites to enhance performance with no additional weight penalty compared to conventional composites. However, given the highly coupled nature of aeroelastic problems, it is not obvious nor intuitive to find the optimal fiber patterns, and sophisticated computational methods and optimization algorithms are required.

In this thesis, a structural optimization framework is developed for the aeroelastic benchmark uCRM-13.5 aircraft wing. More specifically, the wing structure is represented by a Finite Element model utilizing beam and shell elements, while a medium fidelity DLM model is incorporated to model the aerodynamics. A series of optimizations are performed to evaluate the benefits of using tow-steered composites over their conventional counterparts. Static strength, buckling and flutter constraints are imposed in all optimization runs.

Initially, a mass optimization is performed on the wing, using conventional composites for the skins, in order to determine the optimum unsteered configuration. Then, tow-steering variables are introduced into the problem, resulting in a 12.5% reduction in structural mass. Subsequently, two single-objective optimizations are performed on the optimum unsteered wing configuration using only the tow-steering design variables. First, the wing is optimized for stength, by maximizing the aggregated value (KS function) of the skin first-ply failure indices. A 47% reduction in the KS function was achieved in the tow-steered wing compared

to the optimum unsteered configuration. Next, the wing is optimized for flutter velocity, achieving an increase of 29% in the optimized tow-steered design. Finally, a multiobjective optimization run is conducted to explore the Pareto front between skin strength (KS function), buckling and flutter velocity in terms of the fiber steering patterns.

Keywords: Aeroelasticity, Optimization, Tow-Steering, Aeroelastic Tailoring, Variable Stiffness Composites, Flutter, u-CRM Wing

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Acronyms

ACO Ant Colony Optimization 22, 23

AFP Automated fiber placement 20, 21, 31, 157

AFRP Aramid Fiber Reinforced Polymer 38

AIC Aerodynamic Influence Coefficient 75, 84, 86, 87, 106

CAE Computational Aeroelasticity 27

CFD Computational Fluid Dynamics 27, 29, 31–34, 68, 94, 155, 156

CFRP Carbon Fiber Reinforced Polymer 34, 38, 117, 152

CLPT Classical Laminated Plate Theory 47, 55

CMCs Ceramic Matrix Composites 38

CNT Carbon Nanotube 30, 31

CRM Common Research Model 30, 32, 33, 94

CSM Computational Structural Mechanics 27

CTS Continuous Tow Shearing 21

DLM Doublet Lattice Method 28, 31, 32, 35, 68, 72, 74, 75, 77, 81, 84, 98–100, 152, 155

DoF Degrees of Freedom 56, 58–61, 63–66, 68, 75, 77, 81, 84–88, 92, 96, 152

DPM Doublet Point Method 28

EAS Equivalent Air Speed 106

ESL Equivalent Single-layer Theory 47

FEM Finite Element Method 25, 28, 53, 54, 61, 81, 100

FSDT First order Shear Deformation Theory 47, 50, 51, 58

FSI Fluid Structure Interaction 29

GFRP Glass Fiber Reinforced Polymer 38

KS Kreiseelmeier-Steinhauser 105, 106

LES Large Eddy Simulation 29

MIDACO Mixed Integer Distributed Ant Colony Optimization 22, 23, 31, 35, 36, 102, 107, 138, 141, 142, 149, 153, 154

MMCs Metal Matrix Composites 38

MTOW Maximum Take-off Weight 105

OML Outer Mold Line 96, 98, 99

PDEs Partial Differential Equations 53, 69, 85

RANS Reynolds-Averaged Navier Stokes 29, 31, 33, 155

TAS True Air Speed 106

TOGW Take-off Gross Weight 30, 31

TSD Transonic Small Disturbance 28

uCRM undeflected Common Research Model 20, 23, 31, 33, 34, 94, 152

UD Unidirectional 18, 26, 32, 43, 44

VAT Variable Angle Tow 18, 20

VLM Vortex Lattice Method 28, 74

VSL Variable Stiffness Laminates 18

Nomenclature

α	Speed of sound or angle of incidence
Δcp	Coefficient of Pressure Difference
ϵ_{ii}	Normal strain
γ	Ratio of specific heats
γ_{ii}	Shear strain
ν	Poisson's Ratio
ω	Circular Frequency
$\overline{Q}_{ii}$	Reduced Stiffness Coefficient
ψ_x, ψ_y	Rotations on Plate Theories
ρ	Density
σ_{ii}	Normal stress
τ_{ii}	Shear stress
C_L	Coefficient of Lift
D_{16}, D_{26}	Bending-Torsion Coupling Stiffness Terms
E	Young's Modulus
G	Shear Modulus
I	Bending Moment of Inertia

J Torsional Polar Moment of Inertia

k_x, k_y, k_{xy} Curvatures

p Pressure

q Modal participation factor

U_∞ Undisturbed Uniform Velocity

K Stiffness Matrix

k Reduced frequency

M Mach Number

Re Reynold's Number

u,v,w Displacements on x,y,z directions

W Dimensionless normal wash $\frac{w}{U_\infty}$

w Normalwash Velocity (In aerodynamics)

Chapter 1

Introduction

1.1 Overview

The Aerospace Industry has always been at the forefront of technological innovation, driven by the pursuit of safer, efficient and environmentally sustainable flight.
In the Aerospace Industry, weight reduction is of primary concern. For this reason, composite materials, due to their high specific strength and stiffness, i.e ratios of strength/stiffness over density, have been used progressively more. Most new aircraft are manufactured with a variety of composite materials to offer increased load carrying capabilities, increased stiffness, thermal shock resistance, impact resistance and more. For example, the Boeing 787 Dreamliner and the Airbus A350 XWB employ composite materials on more than 50% of their structures.
Aeroelastic Tailoring is the utilization of the anisotropic behavior of composite materials to 'tailor' it's stiffness characteristics in order to achieve the improvement of the response of the structure under aerodynamic loads.
In Unidirectional (UD) laminate structures, Aeroelastic Tailoring is traditionally performed by carefully designing the stacking sequence of constant angle plies and dropping or adding plies along the span of the wing.
However, over the last 3 decades, advancements in the field of composite manufacturing have introduced the concept of "Tow-Steering". Unlike conventional UD laminates where the angles are constant in each ply,tow-steered laminates allow the fibers of the plies to change orientation (i.e get steered) within the ply. These plies are commonly referred to as Variable Angle Tow (VAT) plies and the laminates as Variable Stiffness Laminates (VSL).
In the following subsections, we shall dive deeper into the fundamental concepts of Aeroe-

lasticity and Tow-steering.

1.1.1 Aeroelasticity

Aeroelasticity is the field that studies the interaction between elastic, aerodynamic and inertial forces acting on a flexible structure subjected to a fluid flow and the phenomena that they give rise to. This interaction can be well summarized by the classical *Collar's Aeroelastic Triangle*, shown in figure 1.1.

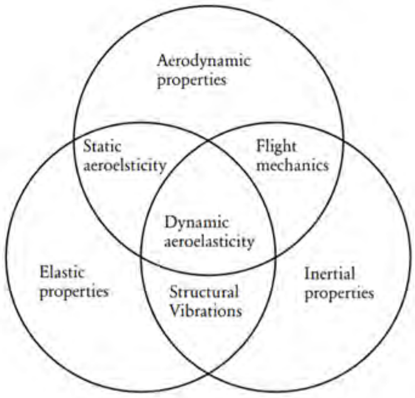


Figure 1.1: Collar's Aeroelastic Triangle

The aeroelastic phenomena can be classified as static or dynamic. *Static Aeroelasticity* considers the effects of the steady *n*on-oscillatory forces on the flexible structure. Because of this flexibility, the structure will deform under the influence of these forces. This deformation will then alter the aerodynamic forces because of the new shape, and then the new aerodynamic forces will again change the shape of the structure, until an equilibrium is reached between the elastic and aerodynamic forces, or the structure fails. The latter is called *Divergence* and it is the static phenomenon where the structure's restoring elastic forces fail to counteract the increasing aerodynamic forces.

The final deformed shape of an aircraft wing is also crucial because it determines the drag and lift coefficients, trim performance and stability of the aircraft in steady flight. The static aeroelastic effects can lead to a reduction in the *effectiveness* of the control surfaces, and eventually even to another phenomenon known as *Control Reversal*. In this phenomenon, the aileron, for example, produces the opposite rolling moment from that intended, because of the wing twist that accompanies the control rotation.

Dynamic Aeroelasticity is concerned with the effects of unsteady *oscillatory* aerodynamics

on the structure. Unsteady aerodynamics occur when the structure is oscillating within the flow, and so the aerodynamic forces also become oscillatory. In the core of dynamic aeroelasticity lies the potentially catastrophic phenomenon of *Flutter* . Flutter is a dynamic instability that features self-excited structural oscillations. These oscillations arise from the unfavourable coupling of aerodynamic, inertial and elastic forces. It usually involves two or more vibrational modes of motion that 'sync' at a certain frequency in a way that the structure can essentially extract energy from the air stream.
In addition to the phenomenon of flutter, which is self-excited, dynamic aeroelasticity also investigates the effects of *turbulence* and *gusts* on the structure, which is called *Dynamic Response*.

1.1.2 Tow-Steering

Since its dawn in the late 20th century, tow-steering and the benefits of VAT has been researched numerically and experimentally, as we will see in Section 1.3.2 that offers a Literature Review.
In a tow-steered composite, the fibers follow a prescribed curvilinear path. The production of VAT composites is mainly done with a technique called Automated fiber placement (AFP). In this process, robotic arms lay-down pre-impregnated tows in a prescribed reference path. The distance between 2 subsequent tows is known as shifting distance.
In an effort to validate the techniques used for modelling tow-steered composites and to showcase the use of tow-steering using AFP, a carbon fiber tow-steered wing which resembles the undeflected Common Research Model (uCRM) of aspect ratio 13.5 like the one that will be analyzed in this thesis, was designed and built by *Aurora Flight Sciences*. The optimal fiber paths were a result of full scale high fidelity aerostructural optimization of the wing with the goal of minimizing fuel burn, performed by the University of Michigan.
Subsequently, a scaled-down version of the design, constituting 27% of the full-scale model, was fabricated and subjected to static load tests at *Nasa Armstrong Research Center* during the months of September and October of 2018 *Jutte et al. [20]* . Figure 1.2 illustrates the manufacturing process of the fiber-steered wing utilizing the AFP technique on the left, while the right showcases the deflected wing during 2.5g static load testing.
During the AFP process, overlaps and gaps appear because of the non-parallel path course of the tows, that introduce non-uniform thickness and local discontinuities that are prone to failure [43]. These effects however won't be considered in this thesis.

Figure 1.2: Aurora Flight Sciences tow-steered carbon wing using AFP (*left*) 2.5g static load test at Nasa Armstrong Research Center (*right*)

Finally , another novel technique has emerged that produces fibers by means of shearing. In this technique, called Continuous Tow Shearing (CTS)*Kim et al. [27]*, unlike in AFP where the tows are placed in discrete patches, the tows are placed in a continuous manner. In the CTS method, fibers are steered smoothly and the gaps and overlaps are reduced.
A plethora of fiber paths have been proposed and used for optimization purposes in the literature, but the most widely employed scheme due to its simplicity and the one used in this thesis is that which describes a linear interpolation of the fiber angle between 2 control points. This one-dimensional variation, proposed originally by *Gurdal and Olmedo [16]* is described by the equation:

$$\theta(x) = \frac{(\theta_2 - \theta_1)x}{L} + \theta_1 \tag{1.1}$$

,where θ_1 and θ_2 are the specified fiber angles at the control points and L is a reference length. The variation of fiber angle is along the x-axis, and remains constant along the y-axis.

1.1.3 Optimization

Introduction

Optimization plays a crucial role in the field of engineering, among many others, as it enables the engineer to systematically refine designs and find optimal solutions that meet some specific objectives and constraints. Some objectives in the aerospace field include weight reduction, fuel efficiency, aerodynamic performance and structural integrity.
One major categorization of optimization algorithms is between gradient-based and gradient-

free ones. As the name implies, gradient-based algorithms utilize gradient information of the objectives and constraints functions in order to update the design variables. They can handle large design spaces and generally offer fast convergence to an optimal solution, albeit with the risk of being a local optimum. A major drawback of this type of algorithms is that the gradients of the objective and constraint functions are not always readily available and it can be a challenge to formulate the optimization problem.
Gradient-free algorithms on the other hand do not rely on gradient information and they are therefore more straightforward in formulating the optimization problem. They have the advantage of being able to efficiently search design spaces and find the global optimum, but they are usually slower than gradient-based algorithms.

Ant Colony Optimization

Ant Colony Optimization (ACO) algorithms are metaheuristic approaches inspired by the foraging behavior of ants. Ants use pheromone trails to guide them in their search for food. When an ant finds a source of food, it leaves a pheromone trail behind it. Shorter paths are favoured because the ants can traverse them in a shorter duration and therefore these paths will gain a larger concentration of pheromones. Eventually, the larger paths will evaporate and all of the ants will travel on the most efficient path.
ACO algorithms mimic the collective intelligence of the biological ants with artificial ants that traverse the design space and construct solutions. The artificial ants choose a path through probabilistic models that are updated based on the information gained on the search domain and the quality of the solutions.
ACO algorithms are effective for solving combinatorial optimization problems and strike a balance between exploration of the design space and exploitation of promising solutions. A main advantage of these algorithms is that they can be used for both single and multi-objective optimization problems and they can incorporate constraints.
Some drawbacks of the ACO algorithms are the relatively low convergence speed, the sensitivity to algorithm parameters and the computational cost for large design spaces.

MIDACO algorithm

Mixed Integer Distributed Ant Colony Optimization (MIDACO) is a numerical high performance software for solving single- and multi-objective optimization problems as a black-box optimizer, which means that the objective and constraint function need not be given in ex-

plicit mathematical form.
It is based on the ACO algorithm with an extension to mixed integer search domains. Constraints are handled within MIDACO by the *Oracle Penalty Method* which is an advanced and self-adaptive method for evolutionary algorithms to reach the global optimal solution. Finally, MIDACO implements a pseudo-gradient based backtracking line-search for fast local convergence, and is therefore classified as an evolutionary hybrid algorithm. More information about the MIDACO algorithm can be found in the website `http://www.midaco-solver.com/`.
The optimization problem considered by MIDACO can be formulated in the following way:

$$\text{Minimize:} \quad f(x) \qquad \text{(Objective Functions)}$$

$$\begin{aligned} \text{Subject to:} \quad & g(x) = 0 \quad \text{(Equality constraints)} \\ & g(x) \geq 0 \quad \text{(Inequality constraints)} \end{aligned}$$

The MIDACO is chosen for the structural optimization of the uCRM wing in this thesis, because of its efficiency to solve large design spaces and the ease of formulation of the optimization problem for both single- and multi-objective problems.

1.2 Historic Review

1.2.1 Aeroelasticity

On December 17 1903 the Wright Brothers made history with the first ever powered flight at Kitty Hawk. The Wright Flyer as it came to be known was a biplane.
Nine days before the first flight of the Wright Brothers, professor Samuel Langley attempted for the second time to launch his powered monoplane, however he failed because of a structural failure during the catapulted launch.
In 1913, Brewer attributed the failure of monoplane wings in excessive elastic twist, something that came to be known as the aeroelastic phenomenon of torsional divergence.
In professor Collar's words “It seems that but for aeroelasticity Langley might have displaced the Wright Brothers from their place in history”. Langley's monoplane failed because it lacked the torsional rigidity which the Wright flyer biplane had.
During World War I, the british Handley Page 0/400 biplane bomber experienced violent

antisymmetric oscillations of the fuselage and tail. scientist Lanchester, in his efforts of uncovering the causes of this incident of what came to be called flutter, concluded that the oscillations were self-excited and that an increase of torsional stiffness of the elevators could eliminate the problem.

Bairstow and Fage were the first to investigate flutter in a theoretical analysis. In their work, binary flutter of the fuselage torsion and elevator motion was used alongside quasi-steady aerodynamics in order to determine the aeroelastic equations of motion and investigate instability.

In the German side of World War I, the Albatros D-III biplane suffered by static divergence problems during high speed dives. Later near the end of the war, the Fokker D-VIII monoplane was introduced and rushed into production, however again wing failures occurred during dives. The cause of the failure was attributed later by Fokker to be the changes made on the production plane by the army compared to the prototype. The army, following regulations about the rear spar having proportional strength to the front spar, had strengthen the rear spar. However, this resulted in the elastic axis moving aft caused an increase on the angle of incidence which caused the wings to collapse. Ironically, while the plane had been made stronger, it was now prone to divergence.

In the later years, the british Frazer and Duncan performed wind tunnel tests to identify and study the aeroelastic phenomena published the so-called 'Flutter Bible' in 1929 which contained some design recommendations, however the aerodynamic basis of their work was not adequate. Wagner in 1925 studied the theoretical problem of 2D non-stationary flow over an harmonically oscillating airfoil which results by a sudden change of angle of attack, now known as indicial aerodynamics and introduced the Wagner function.

In the U.S, Theodorsen developed a 2D oscillating plate theory undergoing torsion, translation and aileron-type motions. The Theodorsen function $C(k)$ which is analogous to the reduced frequency k, establishes the lags between the airfoil motion and the change in forces and moments. In his approach sinusoidal frequency dependent aerodynamics are used, unlike Wagner's time dependent approach.

Garrick and Theodorsen during the 1930s studied the effects of parameters such as: center of mass, elastic axis, moments of inertia, mass ratio, bending/torsion frequency ratio and hinge location on aeroelastic phenomena. The basis for a general lifting surface theory for finite wings was developed by Küssner, in 1940 who used Prandtl's Acceleration Potential and the effect of a moving doublet to obtain an integral equation that relates the load distribution over the lifting surface to the normal velocity by a quantity known as the Kernel Function.

Note that the lifting surface theory and the later developed numerical method of solution will be shown in Chapter 2.3, as it is used in this Thesis.
With the rise of transonic and supersonic flight brought about by the jet engine after the World War II, new challenges arose within the field of aeroelasticity. Novel transonic flutter experimental methods were used and quickly enough transonic wind tunnels were created to tackle the study of these new phenomena. Later, the rise of computational power allowed for the introduction of numerical methods to solve the problem of aeroelasticity, including the Finite Element Method (FEM) for structural analysis and numerical panel and lifting surface methods for the aerodynamics.
The above offer merely a summary of some key points of the work of *Garrick and Reed [14]* who in their work presented a comprehensive review of the history of aeroelasticity and flutter analysis since its dawn.

1.2.2 Aeroelastic Tailoring

The definition of Aeroelastic Tailoring according to *Shirk et al. [35]* is the following:
Aeroelastic Tailoring is the embodiment of directional stiffness into an aircraft structural design to control aeroelastic deformation, static or dynamic, in such a fashion as to affect the aerodynamic and structural performance of that aircraft in a beneficial way.
Aeroelastic tailoring is therefore a form of passive control of the structure.
Professor Hill in as early as 1953 designed a wing, known as the aero-isoclinic wing, which was able to keep its angle of incidence constant along the span despite flexural distortion. Hill's design was coined to be one of the most important design developments of those years and was an excellent example of aeroelastic tailoring.
It wasn't until 1969 that the idea of passive control came again to the spotlight by a proposal of General Dynamics to the Air Force Flight Dynamics Laboratory(AFFDL) to apply advanced composite materials to the design of supercritical wings used in transonic flight. At General Dynamics, researchers discovered that the anisotropic nature of composite creates a coupling between bending and torsion deformation that can be used to produce the desired shape control on the supercritical wing. A contract between General Dynamics and AFFDL was formed to create a pilot computer program for the aeroelastic and strength optimization of aircraft lifting surfaces using advanced filamentary composites. The most significant result of this contract was the development of the TSO computer program in 1972, which soon became synonymous with aeroelastic tailoring.

In 1978, a remotely piloted research vehicle called HiMAT was developed by Rockwell under a contract to NASA, which incorporated aeroelastically tailored wing skins and canard. The flight test program of 1979 successfully demonstrated the benefits of aeroelastic shape control.

Figure 1.3: The X-29 Forward-Swept Research Plane

By that time, most aircraft used aft-swept wings, because of problems with aeroelastic divergence found in forward-swept wings. The re-introduction of the forward-swept wing concept came partly by Krone, who in 1974 showed that tailored composites can be used to avoid divergence in forward-swept wings with little to no weight penalty. Grumman, a civil and military aircraft producer of the time, used Krone's data and investigated forward sweep designs. Later, Grumman was chosen by the Defense Advanced Research Projects Agency (DARPA) of the U.S to design and build a research forward-swept plane named X-29, shown here in figure 1.3. The X-29 successfully completed its first flight in 1984.
In the years to come, the introduction of mathematical programming and optimization tools established the concept of aeroelastic tailoring. Aeroelastic tailoring has been used for to maximize the lift-to-drag ratio, expand the flight envelope, improving aircraft controllability and more. For the aeroelastic tailoring of a lifting surface, many different approaches are possible, such as changing the stacking sequence, finding the optimal fiber orientation of UD plies, changing the thickness along the wing by dropping plies, or as is the topic of this

Thesis, by making use of Tow-Steering.

1.3 State of the Art

1.3.1 Aeroelastic Analysis

Owing to the ever increasing computing power, it is now possible to perform high fidelity aeroelastic Aaalysis by means coupling of Computational Fluid Dynamics (CFD) models with Computational Structural Mechanics (CSM), which is called Computational Aeroelasticity (CAE).
Traditionally in the last decades, low and medium fidelity tools have been widely used, because of their lower computational cost.
In modern days, preliminary design is still being predominantly done by the use of low to medium fidelity analysis tools in order to be able to perform multiple calculations in a short period of time. However, in higher stages, an increase in fidelity of the models is often needed to capture additional effects not modelled by lower fidelity tools *Afonso et al. [1]*.
The aerodynamic and structural models used in aeroelastic analysis are shown in figures 1.4 and 1.5 taken by *Afonso et al. [1]*.

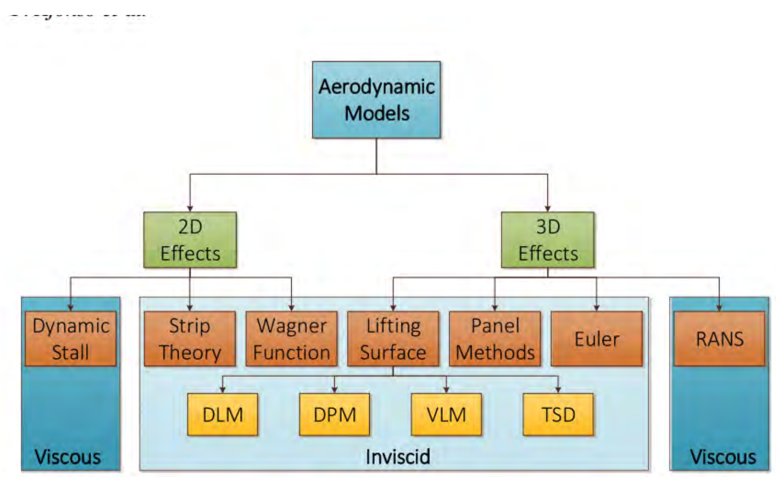


Figure 1.4: Aerodynamic Models used in Aeroelastic Analysis *Afonso et al. [1]*

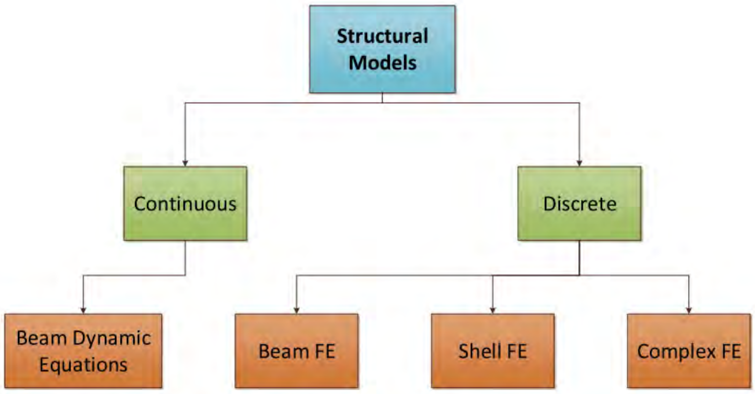


Figure 1.5: Structural Models used in Aeroelastic Analysis *Afonso et al. [1]*

Low Fidelity Models

In low fidelity models the beam dynamic equations of motion can be used for structural analysis by making simplifying assumptions, coupled with simple 2D aerodynamic analytical theories such as Strip Theory or Wagner/Theodorsen 2D airfoil theory. The 2D aerodynamic theories used here do not include compressibility and viscous effects along with effects of the 3D flow over a finite body.

One advantage of these methods is the direct coupling of the aerodynamic and structural model in a set of analytical aeroelastic equations of motion. The beam dynamic equations are not classified as low fidelity models due to being inaccurate but rather due to the lack of capability to be used in complex geometries and structures, where the numerical methods such as FEM take the upper hand.

Medium Fidelity Models

Moving to the medium fidelity Models, these have and are used extensively in the industry. They include the coupling of Finite Element models with 2D and 3D Potential Flow aerodynamic theories.

The Finite Elements are usually the less computationally expensive to model. Beam and shell elements are often used and more rarely 3D and more complex elements might be incorporated if needed.

Lifting Surface include the Doublet Lattice Method (DLM), Vortex Lattice Method (VLM) and Doublet Point Method (DPM) for incompressible or low compressible inviscid flow where the thickness effects are neglected. Then the Transonic Small Disturbance (TSD) theory de-

veloped by $ZAERO^{TM}$ can be used in transonic flows along with potential corrections.
Finally there are 3D Panel Methdos that include the effects of thickness. This improves mainly the calculation of induced drag, and is not that important for flutter calculations where lift plays a more significant role.
Unsteady flow can be considered in the above methods by considering small non-planar harmonic oscillations, which poses a limitation on the accuracy when the amplitude of oscillations becomes larger.
Unlike low fidelity models, where there was a direct coupling of structural and aerodynamic models in the aeroelastic equations, here the two models are used separately and therefore the loads and displacements have to be transferred successfully from one model to the other by means of an Fluid Structure Interaction (FSI) algorithm.

High Fidelity Models

High fidelity aerodynamic models include inviscid Euler or Navier Stokes CFD solvers. Both of them can capture shocks in the transonic regime but only the second can capture viscous effects such as boundary layer, flow separation etc.
Euler methods can be linearized around an equilibrium state and be used in flutter calculations, with less accuracy but great computational savings.
The last step before the solution of the full Navier-Stokes equations are the models that use the Reynolds-Averaged Navier Stokes (RANS) equations, where viscous effects are included however the turbulence quantities are time-averaged with the use of semi-empirical formulae.
An improvement on the accuracy of turbulence modelling can be incorporated through the use of Large Eddy Simulation (LES) where the larger eddies are calculated directly while the smaller ones use the turbulence models. However the computational cost increase can often not be justified by the resulting accuracy *Afonso et al. [1]*.
Last but not least, there are models that solve the full Navier-Stokes equations, however as of today these are limited to simpler models for academic research rather than in industry.
As the computational power keeps increasing, the shift from lower to higher fidelity models will be greater and it is possible that high fidelity models will be incorporated even for early stage preliminary design calculations in the future.

1.3.2 Literature Review

In this section a brief review of some papers on aeroelastic tailoring and tow-steering will be presented which however is by no means an exhaustive list of aeroelastic tailoring work. Studies can be found that have used aeroelastic tailoring for different objectives such as drag reduction, weight reduction, gust response, flutter and divergence speed increase, maximum load carrying capacity increase and combinations there-of.

Aerolastic Tailoring of Metallic and Composite Configurations

- In 2003, *Guo et al. [15]* conducted a study on the optimization of a laminated composite wing structure to achieve maximum flutter speed and minimum weight without a strength penalty and achieved a 18% increase in flutter speed and a 13% decrease in weight. He found that in comparison to symmetric layups, the assymetric is favorable for aeroelastic optimization because of bending-torsion coupling. He also found that the torsional rigidity plays a more dominant role in aeroelastic tailoring compared to the bending rigidity.

- In 2012, *Kennedy and Martins [23]* employed a gradient-based aerostructural design optimization framework that used a medium fidelity aerodynamic panel method to compare metallic and composite aircraft wings. A weighted combination of the fuel burn and Take-off Gross Weight (TOGW) were minimized in order to find a Pareto-front. The structural model considered strength and buckling constraints. Both geometric and structural sizing design variables were used, including detailed parametrization of the laminate stacking sequence. In their work, they found that composite wings up to 40% lighter than metallic ones, leading to 8% fuel burn savings and 11% reductions in take-off weight.

- In 2014, *Martins et al. [32]* solved a series of medium- and high-fidelity aerostructural optimization problems on a wing geometry based on the jig shape of the Common Research Model (CRM). Three types of materials were investigated : aluminum, conventional carbon-reinforced composites and a hypothetical Carbon Nanotube (CNT) based composite. The Pareto fronts between fuel burn and TOGW were investigated considering both structural sizing and aerodynamic shape design variables. The use of conventional composite materials resulted in a 7.7% reduction in fuel burn and a 8.4% reduction in TOGW compared to the all-metallic design. The corresponding values for the CNT based composite versus the conventional composite were 5.2% and 5.6% respectively. The optimal wing trends showed that the wings with minimum fuel burn were longer, heavier, thinner and more flexible than the ones with minimum TOGW. Additionally, the wing span of the optimal designs increased as the material stiffness increased, from metal to CNT composite.

- In 2023, *Kilimtzidis and Kostopoulos [26]* developed a sequential multi-fidelity optimization framework for high aspect ratio composite aircraft wings, focusing on an all-composite design of the uCRM-13.5 geometry, using the MIDACO algorithm. Structural representation utilized a 3D shell and 1D beam Finite Element model, while for the aerodynamics models tools of various fidelities were employed, including the DLM, a 3D panel-method and RANS-based CFD. Initially, a comparison between the different aerodynamic fidelities highlighted significant differences in the aerodynamic loadings between the different tools. The optimization framework aimed to minimize the wing mass while meeting strength, local panel buckling and flutter constraints. Lower-fidelity tools were used in the initial runs to guide the exploration of the design space, while higher-fidelity CFD runs were used for refinement in later stages. It was found that the resulting structure was mainly driven by panel buckling constraints, with increased thickness near the Yehudi break.

Variable Stiffness Laminates and Tow-steering

- In 2010, *Alhajahmad et al. [3]* investigated the effect of tow-steering on the maximum strength of fuselage panels including buckling considerations and obtained optimal designs that substantially increased the maximum load carrying capacity and showed better buckling performance compared to traditional straight fiber designs.

- In 2011, *IJsselmuiden [18]* developed a multi-step variable stiffness composite design optimization framework aimed at maximizing the structural performance in simple plate or shell-like structures under static load. Initially, an optimal design was found using lamination parameters. Subsequently, the optimal fiber angle distribution was obtained in order to satisfy manufacturing constraints and retain the same structural performance. Finally, this fiber angle distribution was translated into continuous tow paths suitable for manufacturing through the AFP process. Through practical design applications, significant improvements were observed in both buckling and strength performance for the optimized variable stiffness structures compared to their constant stiffness counterparts.

- In 2014, *Dillinger et al. [13]* extended the use of the optimization framework developed by *IJsselmuiden [18]* to wing structures under aeroelastic constraints. He focused on the first step of the framework, i.e defining the optimum lamination parameters, and optimized the wing for mass and aileron control effectiveness. Interestingly, he found that unbalanced laminates performed better compared to balanced ones. Furthermore, the same author, in his PHD thesis (*Dillinger [12]*) developed practical corrections for the DLM based on the wing camber and twist that will be used later on, along with possible CFD corrections.

- In 2013, *Stodieck et al. [39]* used tow-steering principles to investigate the effects of spanwise fiber variation on the flexural axis position, natural frequencies, divergence and flutter speeds and gust response of a rectangular plate. In subsequent work, *Stodieck et al. [40]*, working on the same plate, included higher order fiber angle variations in two dimensions and achieved an increase of 7% on flutter speed and a 52% reduction in peak wing root gust loads compared to optimized UD laminates.

- In 2017, *Stodieck et al. [41]* extended her work to a realistic wing geometry, the CRM, and performed a mass optimization study comparing 1D and 2D variable distributions for steering and thickness. Several constraints were considered, including strains and buckling under 2.5g/-1g maneuver and gust loads, along with flutter instability and control effectiveness, and their effect on the optimization was evaluated by optimizing the wing separately for each individual constraint, where the 2.5g maneuver was deemed to be the most critical. The tow-steered configurations were found to outperform the UD ones by 7.3% and 10.5% in mass reduction for the 1D and 2D optimizations respectively.

- In 2014, *Stanford and Jutte [36]* performed aeroelastic optimization on both a cantilever plate and the CRM wing using tow-steering principles and an evolutionary algorithm. For the plate, the tow-steered design showed significant improvement in peak Tsai-wu stresses, while for the flutter speed only a slight increase was achieved. Regarding the wing, initially a multi-objective optimization between mass, flutter and peak stresses was performed without steering. A strong trade-off between flutter and wing mass was observed. On the other hand, the heaviest design didn't necessarily have the lowest stresses. Then, while keeping the laminate fixed, the effect of steering of the skins on flutter speed and peak stresses was investigated. Both the peak stresses and to a lesser extent the flutter speed were shown to be benefited by tow-steering.

- In 2016, *Stanford et al. [38]* continued his work on aeroelastic tailoring of the CRM and tested several methods to minimize the structural mass and compared the results with an all metallic baseline design. First, he tried a spatially detailed metallic design, where each Finite Element has its own thickness variable. Then, he tested the use of functionally graded metals(FGMs), where a single structural member is composed of a distribution of two metals. Subsequently, composite material was considered for the skins, while the rest of the components remained metallic. Balanced and unbalanced laminates of both straight and tow-steered were investigated. Finally, a different approach of using a distribution of tailored control surfaces along the trailing edge was explored, while returning to the all metallic configuration. In his work, a drop of 6.01% was achieved using tow-steering compared to the straight fiber design when balanced laminates were used, but only a 1.61% drop in the case of unbalanced laminates. Interestingly, the use of unbalanced over balanced laminates resulted in a drop of 4.56% in mass without steering. It's worth noting that in this work, buckling constraints were also included and were in fact the main drivers of the mass minimization.

- In 2017, *Stanford and Jutte [37]* extended his research on the higher aspect ratio derivative of the CRM, the uCRM-13.5, which is also used in this thesis. More specifically, the study investigated two approaches to reduce the wing's mass: tow-steering of the composite skins and the implementation of curvilinear stiffeners. The use of tow-steering resulted in a mass reduction of 8.8% compared to the unsteered wing. Interestingly, the use of curvilinear stiffeners along with straight fibers yielded a similar result of 8.5% reduction in mass. If these 2 methods are used simultaneously, an 11.4% drop in mass is achieved compared to the unsteered wing, however comparably the benefit of using these methods independently is larger.

- In 2019, *Brooks et al. [9]* conducted gradient-based aerostructural optimization of the uCRM geometry and compared the performance of conventional and tow-steered composite configurations. Unlike previous studies, he used a higher fidelity model based on RANS CFD to model the aerodynamics of the wing. Additionally, cross-sectional shape design variables were included in the optimization in conjunction with thickness and steering variables. The objective of the optimization was to minimize the fuel burn for a given range under normal cruise conditions. Also 2.5g and -1g maneuvers were included as critical conditions for buckling and conventional strength constraints, along with several manufacturing and geometric constraints to ensure a meaningful design. A series of studies were conducted in different aspect ratio derivatives of the uCRM ranging from 7.5 to 13.5. For the uCRM-9, tow-steering allowed for a reduction of 2.3% in the aircraft fuel burn compared to the conventional composite design, owing to a 24% reduction in wing structural mass. For the uCRM-13.5, accordingly, a decrease of 1.5% was achieved in fuel burn by a 14% drop in wing weight. Interestingly, a trend of diminishing returns of tow-steering was identified as the aspect ratio increased, because of the less additional load alleviation compared to the conventional design.

- In 2020, *Brooks et al. [10]* continued his work on aeroelastic tailoring of the uCRM using high-fidelity CFD by exploring the trade-offs between aircraft fuel burn performance and structural weight across three material technologies, aluminum alloy, conventional Carbon Fiber Reinforced Polymer (CFRP) composites and tow-steered composites. Like previous work, both structural sizing and geometric shape design variables are used, however in this case additional constraints are imposed for buffet onset and 1-g cruise condition at the Mach crossover altitude. Similar trends emerged across all materials in the trade-off curves, with larger wing spans favored for fuel burn optimization and smaller spans for structural weight reduction. Transitioning from aluminum to conventional composite and then to tow-steered composite uniformly improves performance of all Pareto-optimal designs. For the designs optimized for fuel-burn, switching from aluminum to conventional composites resulted in a 6.3% decrease in fuel burn, with an additional 1.5% achieved with the use of tow-steered composites. Similarly, for the designs optimized for structural weight the use of conventional composites saved 3.7% in structural weight compared to aluminum, and tow-steering resulted in an additional 1.6% decrease.

1.4 Thesis Outline

Chapter 1 gives an overview of aeroelasticity and provides and introduction tow-steering and optimization via an Ant-Colony algorithm, the MIDACO. Then, we take a trip back in time to discover the roots of aeroelasticity and how it emerged as a field, followed by the rise of aeroelastic tailoring as a term some decades later. Subsequently, the state of the art for aeroelastic analysis using modern computational tools is discussed. The chapter wraps up with a literature review of aeroelastic tailoring and tow-steering.

Chapter 2 is a detailed exploration of the theories and methods used in this thesis. We begin by explaining the basics of composite mechanics, followed by an introduction to Finite Element analysis. Next, we explore the derivation and formulation of the aerodynamic DLM based on potential aerodynamics theory. We then examine the critical aeroelastic phenomena of divergence and flutter, along with the fundamental methods and equations for static and dynamic aeroelastic analysis using Nastran. Finally, this chapter provides an overview of linear buckling analysis with Nastran.

Chapter 3 presents the computational framework used throughout this work. Initially, the geometric details of the reference wing geometry (the uCRM-13.5) are given along with the internal configuration. Then, the structural and aerodynamic models of the wing are developed. For the structural representation, a Finite Element model is employed, while the aerodynamics are modeled with a DLM panel mesh. The chapter concludes with the coupling of the two models.

Chapter 4 deals with the development of the optimization framework for the minimization of mass of the wing using aluminum for the internal configuration and conventional composites for the skin. The optimization problem is formulated and the design variables, problem constraints and optimization parameters are defined. Finally, the results of the structural optimization are presented and discussed in detail.

Chapter 5 dives into the topic of aeroelastic tailoring with the use of tow-steered composites for the skins. A series of single-objective optimizations is conducted to evaluate the advantages of tow-steering over conventional designs. Initially, a mass optimization similar to the one of the previous Chapter is performed, albeit with the introduction of additional design variables for tow-steering. Subsequently, the thickness variables are eliminated from the problem, and the optimal unsteered configuration is used for the wing optimization using only tow-steering design variables. Firstly, the wing is optimized for skin strength based on first-ply composite failure, and subsequently for flutter performance. Each optimization's parameters and results are presented and discussed.

Chapter 6 explores the trade-offs between skin strength, flutter and buckling performance of the wing from a tow-steering perspective. A multi-objective optimization is set-up between the aforementioned three objectives and with only the tow-steering variables active. The entire Pareto front is generated using MIDACO, and the results are subsequently presented and analyzed.

Chapter 7 serves as an overview of the findings and conclusions drawn from this work. Additionally, it provides suggestions for improving the current study and discusses avenues for future research.

Chapter 2

Theoretical Background

2.1 Composite Material Mechanics

In this section, we will deal with the mechanics of composite materials and laminates.

2.1.1 Introduction to Composites and Laminates

A composite is a material that consists of two or more constituents that are not soluble in each other. One of these phases is the *reinforcing phase* and the other is the *matrix* phase. The matrix phase is usually made of polymers such as epoxy, polyester or vinyl ester resins and thermoplastics. The matrix phase is generally continuous and offers support to the reinforcement material. The loads are transferred from the matrix to the reinforcement phase, which can take the form of fibers, particles or flakes. The most used polymer matrix, and the one featuring in this thesis, is the epoxy resin because of its high strength and low viscosity, which is important for good wetting of fibers and load transfer.
For fiber reinforced composites, there could be a distinction made about short fiber composites and long fiber composites. Short fiber composites, albeit being cost effective, are rarely used in applications where high mechanical properties are needed because the discontinuities in the fibers are detrimental to their performance. Long fiber composites on the other hand offer very high mechanical strength and stiffness.

CFRP and Glass Fiber Reinforced Polymer (GFRP) composites are used quite extensively in the industry for structural application because of their high specific mechanical properties. CFRPs are generally stronger and stiffer but at the expense of a higher cost compared to GFRPs. Aramid Fiber Reinforced Polymer (AFRP) composites have high impact resistance and are used in applications that impact energy needs to be dissipated.
Metal Matrix Composites (MMCs) have good mechanical properties along with good electrical and thermal conductivity and increased ductility. They are used in applications that involve higher temperatures or structural applications. Ceramic Matrix Composites (CMCs) are used extensively in applications that involve very high temperatures because of their excellent heat resistance properties and chemical inertness.
A *lamina* (or ply) is a thin layer of composite material, like for example a CFRP. A laminated structure consists of multiple plies stacked on top of each other. The plies can generally have different thickness and be made of different materials. A laminate composite is shown in figure 2.1.

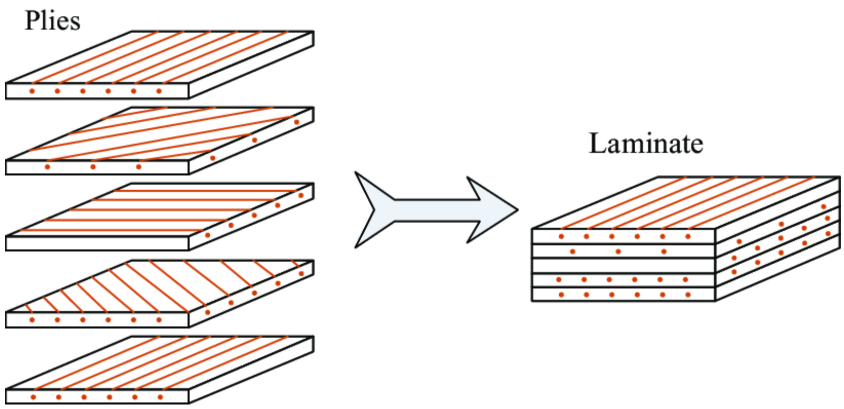


Figure 2.1: Structure of a Laminated Composite

2.1.2 Review of Basic Mechanics Principles

Let us consider a 3-dimensional structure in equilibrium which is acted upon by external forces. These forces can be surface forces like a uniform load on a beam or body forces like the weight of the body. The result of these forces is that internal forces on the body. If a cross section is cut in the structure, the internal forces must be in equilibrium with the external loads.

Stress is defined as the intensity of force per unit area. In a structure, the internal loads result in stresses. Knowledge of the stress distribution in a structure is crucial to determine its safety. If the stresses induced by the loads surpass the strength of the structure, the structure will fail catastrophically.
If we consider a cross-section surface of the structure in the y-z plane, then a force ΔP is acting on an area ΔA, as shown in figure 2.2. In general the internal force vector has a component normal to the surface ΔP_x and 2 components parallel to the surface ΔP_y and ΔP_z The stresses in the infinitesimal area ΔA are given by :

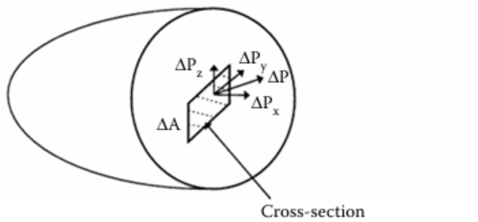


Figure 2.2: Stresses on an infinitesimal area of a cross section

$$\sigma_x = \lim_{\Delta A \to 0} \frac{\Delta P_x}{\Delta A} \tag{2.1}$$

$$\tau_{xy} = \lim_{\Delta A \to 0} \frac{\Delta P_y}{\Delta A} \tag{2.2}$$

$$\sigma_{xz} = \lim_{\Delta A \to 0} \frac{\Delta P_z}{\Delta A} \tag{2.3}$$

The component normal to the surface is called *normal* stress and the components parallel to the surface are called *shear* stresses. Note that stress is defined at a point and remains unchanged, however if one was to cut a different cross section, the components of stress would change.
The same can be shown if we cut cross-sections in the x-y and x-z planes. An infinitesimal cuboid under stress is shown in figure 2.3. A total of 9 different stresses acts on a point in this infinitesimal cuboid. However, by imposing equilibrium of moments at the cuboid, we have three relations between the shear stresses, and the total number of independent stresses reduces to 6.

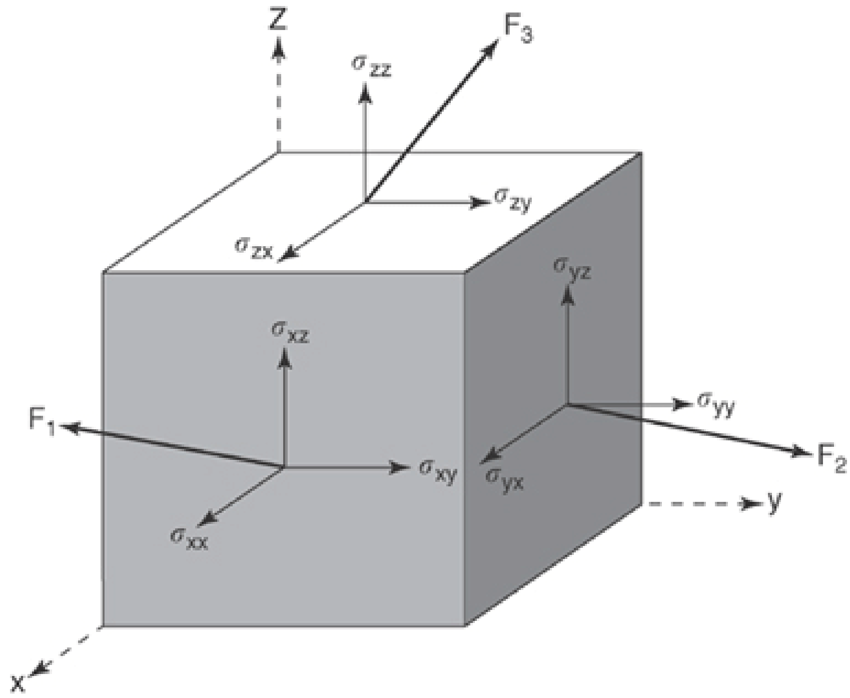


Figure 2.3: Stresses on an infinitesimal cuboid

These relations are :

$$\tau_{xy} = \tau_{yx} \tag{2.4}$$

$$\tau_{yz} = \tau_{zy} \tag{2.5}$$

$$\tau_{zx} = \tau_{xz} \tag{2.6}$$

Tensile normal stresses are positive and compressive normal stresses are negative. As for the shear stresses, a shear stress is positive if its direction and the direction normal to the face on which it is acting are both positive or negative, otherwise it is negative.

An elastic structure under mechanical or thermal loads also develops deformations. The knowledge of this deformations is also very important, because excessive deformations in a structure, although not necessarily catastrophic, can be detrimental to the function of the structure.

Deformations are specified in terms of *strains*, that is the relative change of the size and shape of the body, and are thus dimensionless. As with the case of stresses, strains are also divided in shear and normal strains. Normal strains, denoted by the letter ϵ are responsible for the change of size of the body , whereas shear strains, denoted by the letter γ, are responsible for the change of shape of the body.

Let us consider an infinitesimal area in the x-y plane under deformation as shown in figure 2.4.

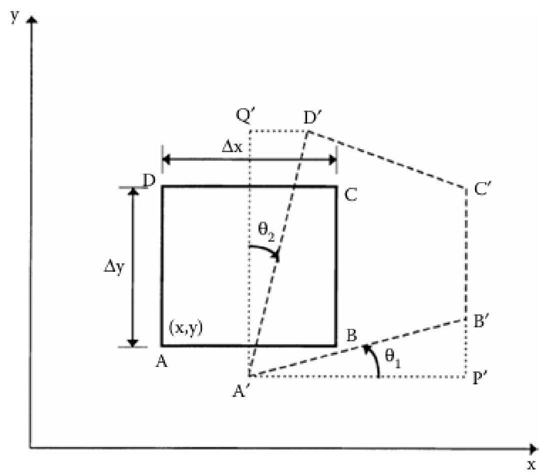


Figure 2.4: Strains on an infinitesimal area on the x-y plane

The normal strain in the x-y direction is then defined as the change of length in the line AB per unit length, i.e :

$$\epsilon_x = \lim_{AB \to 0} \frac{A'B' - AB}{AB} \tag{2.7}$$

The shearing strain in the x-y plane is defined as the change in the angle between the sides AB and AD from 90°, i.e :

$$\gamma_{xy} = \theta_1 + \theta_2 \tag{2.8}$$

If we consider a point in the body that is under deformation, then the displacement of the point relative to its initial position in the (x,y,z) coordinate system can be defined by its components :

$$u(x, y, z) = \text{displacement in x-direction}$$

$$v(x, y, z) = \text{displacement in y-direction}$$

$$w(x, y, z) = \text{displacement in z-direction}$$

The relationships between displacements and strains, under the assumptions of **small displacements**, are the following :

$$\epsilon_x = \frac{\partial u}{\partial x} \tag{2.9}$$

$$\epsilon_y = \frac{\partial v}{\partial y} \tag{2.10}$$

$$\epsilon_z = \frac{\partial w}{\partial z} \tag{2.11}$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \tag{2.12}$$

$$\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \tag{2.13}$$

$$\gamma_{xz} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \tag{2.14}$$

The above equations are known as the *compatibility equations*.
If the body is said to be linear elastic, which makes therefore again the assumption of small deformations, then the stresses and strains at a point are related through 6 simultaneous linear equations called *Hooke's Law*. For the simplest case of a linear isotropic material, these equations take the form :

$$\begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{bmatrix} = \begin{bmatrix} \frac{1}{E} & -\frac{\nu}{E} & -\frac{\nu}{E} & 0 & 0 & 0 \\ -\frac{\nu}{E} & \frac{1}{E} & -\frac{\nu}{E} & 0 & 0 & 0 \\ -\frac{\nu}{E} & -\frac{\nu}{E} & \frac{1}{E} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G} \end{bmatrix} \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{zx} \end{bmatrix} \tag{2.15}$$

,where E is the Young's modulus, G is the shear modulus and ν is the Poisson's ratio of the material. These constants are known as engineering constants because they are found by experiments. The above equations are also known as *constitutive equations* of the material. The 6x6 matrix in 2.15 is known as the *compliance matrix* and relates strains with stresses. The inverse of the compliance matrix is the *stiffness matrix*, which instead relates the stresses with the strains.
Note that both the stiffness and the compliance matrices are symmetric.
In the isotropic case, only 2 constants need to be known for the calculation of the compliance

and stiffness matrices. However if the material is anisotropic, this is no longer the case. For a generally anisotropic (also known as triclinic) linear elastic material, 21 constants need to be known instead. The stiffness matrix of a triclinic material is shown in equation 2.16. It can be seen that in a triclinic material, unlike an isotropic material, normal stresses and strains are related to shear stresses and strains. This is called *shear coupling*.
Luckily, a triclinic material is very rare, and materials tend to have some degree of symmetry in their structure.

$$[C] = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} \\ C_{21} & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} \\ C_{31} & C_{32} & C_{33} & C_{34} & C_{35} & C_{36} \\ C_{41} & C_{42} & C_{43} & C_{44} & C_{45} & C_{46} \\ C_{51} & C_{52} & C_{53} & C_{54} & C_{55} & C_{56} \\ C_{61} & C_{62} & C_{63} & C_{64} & C_{65} & C_{66} \end{bmatrix} \tag{2.16}$$

In the case of an *Orthotropic material*, which is widely used as a model for laminae as we shall see in the next section, the independent constants that need to be defined are reduced to 9. The compliance matrix of an orthotropic material in the *material principle axes* has the form of equation 2.17. Note that an orthotropic material in its material axes doesn't have shear coupling, however if the coordinate system is rotated, then shear coupling terms also appear in the stiffness and compliance matrices.

$$.[S] = \begin{bmatrix} 1/E_1 & -\nu_{12}/E_1 & -\nu_{13}/E_1 & 0 & 0 & 0 \\ -\nu_{21}/E_2 & 1/E_2 & -\nu_{23}/E_2 & 0 & 0 & 0 \\ -\nu_{31}/E_3 & -\nu_{32}/E_3 & 1/E_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G_{23} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G_{31} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G_{12} \end{bmatrix} \tag{2.17}$$

The Young's moduli, shear moduli and Poisson's ratios are found experimentally for a specific material that displays orthotropic behavior. After this brief introduction, we are ready to take a deeper look at the topic of mechanics of composite Materials.

2.1.3 Macromechanical Analysis of a Lamina

A UD fiber-reinforced lamina with its material coordinate axes is shown in 2.5. A thin plate with no out-of-plane loads is said to be under *plane stress*. In this case, the upper and lower

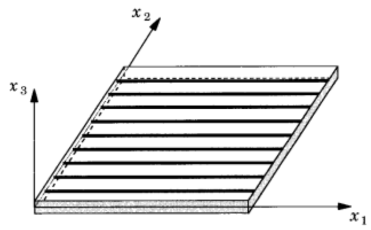


Figure 2.5: Unidirectional fiber-reinforced Lamina

surfaces of the plate are free of external loads and thus free of stress, meaning that $\sigma_3 = 0$, $\tau_{31} = 0$ and $\tau_{23} = 0$. Because the plate is thin, these stresses are assumed to also be 0 within the plate.
A lamina is usually very thin, and it is common to model it as being under plane stress. This assumption then reduces the stress-strain equations to two dimensions.
In the *macromechanical analysis* of the lamina, the UD lamina is assumed to be a homogeneous and orthotropic material. Fiber-matrix interactions in the inner structure of the lamina are not considered in this approach. These effects can be included with the*micromechanical* approach, which considers the volume fractions of fibers and matrix, bonding of the materials etc. Here only the macromechanical approach will be utilized for the mechanics of a lamina.
In two dimensions, the compliance matrix for a *2-D orthotropic* material is written as :

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & 0 \\ S_{12} & S_{22} & 0 \\ 0 & 0 & S_{66} \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} \tag{2.18}$$

Note that $\gamma_{23} = \gamma_{31} = 0$. However ϵ_3 is not zero, but it can be ommitted from the above equation because it is dependent on strains ϵ_1 and ϵ_2. Therefore a total of 4 independent quantities are present in the compliance matrix in two dimensions.

The stiffness matrix, shown in equation 2.19, is the inverse of the compliance matrix in equation 2.18, and its members are usually denoted with Q_{ij} and are called *reduced stiffness coefficients*.

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix} \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} \tag{2.19}$$

In the case of a unidirectional lamina where the material axes coincide with the global axes, there is no shear coupling in the stiffness matrix. However this is not the case for an angle lamina, where the material axes 1-2 are rotated from the global axes x-y by an angle θ, as shown in figure 2.6.

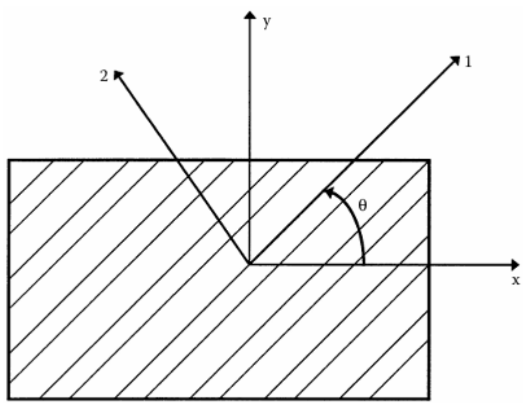


Figure 2.6: Material and Global Axes of an angle Lamina

In this case, it can be shown that the stresses in the global coordinate system are related to the stresses in the material system by the *transformation matrix* $[T]$ as shown in equation 2.20.

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = [T]^{-1} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} \tag{2.20}$$

The elements of the transformation matrix are the sines and cosines of the rotation. Using $s = cos(\theta)$ and $c = cos(\theta)$, it takes the form :

$$[T] = \begin{bmatrix} c^2 & s^2 & 2sc \\ s^2 & c^2 & -2sc \\ -sc & sc & c^2 - s^2 \end{bmatrix} \tag{2.21}$$

The same is true for the global strains and the local strains, with the addition of the Reuter matrix $[R]$, as shown in equation 2.22.

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} = [R][T][R]^{-1} \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix} \tag{2.22}$$

The Reuter matrix has the form :

$$[R] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \tag{2.23}$$

By combining equations 2.20 and 2.22 and carrying the multiplications we get the stress-strain equation on the global coordinate system in equation 2.24.

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix} \tag{2.24}$$

where $\bar{Q}_{ij}$ are the transformed reduced stiffness matrix elements. The same can be done for the transformed compliance matrix.

With the mechanics of a single lamina covered, we can now move on to the mechanics of a laminate structure.

2.1.4 Macromechanical Analysis of Laminate Structures

A laminate is made by stacking laminae on top of each other and bonding them. A laminate is identified by its laminate code, that contains the information about how its layout.

For example, a laminate with the code $[0/-45/90/-45/0]$ has 5 layers, with the orientation of each layer from top to bottom being $0°$,$-45°$,$90°$,$-45°$ and $0°$ respectively.

If a subscript of 2 is used in the code, as in $[-45_2/0]$, it means that there are 2 adjacent layers with $-45°$.

If the subscript s is used outside the brackets, as in $[0/90_2]_s$, it means that the laminate is symmetric about the midplane and consisting of double the layers, in this case 6 symmetric layers.
If instead an odd number of plies is used, then the middle ply is symbolized with an overbar. For example $[0/45/\bar{60}]_s$ denotes a 5 layer symmetric laminate with a middle ply of 60°.
The most classical way of analysing laminate structures is by using the Equivalent Single-layer Theory (ESL), that states that the heterogeneous 3-D laminated plate is treated as a statically equivalent 2-D single-layer with complex constitutive behavior. Bonding of the laminates is assumed to be perfect.
The 2 main theories developed with ESL are the Classical Laminated Plate Theory (CLPT) and the First order Shear Deformation Theory (FSDT). The difference between the 2 theories, is that the first assumes perfect bending situation with $\gamma_{xz} = \gamma_{yz} = 0$ and is a good approximation for structures like slender beams where transverse shear is negligible, while the second allows for shear to be present, which is more accurate for shorter beams and plates, where shear is significant. Both theories assume that the layer material is linear elastic and that small deformation occur.
We will start with the CLPT theory which is simpler and later we will introduce the additional shear predicted by the FSDT.
The laminate of thickness t is assumed to be in plane stress in the x-y plane. The origin of the laminate plate is lying on its midplane, where z=0. Consider now a cross section of the plate under load in the x-z plane. Under the kinematic assumptions of CLPT, which are also called Kirchoff's assumptions, the plate is deformed as shown in figure 2.7.

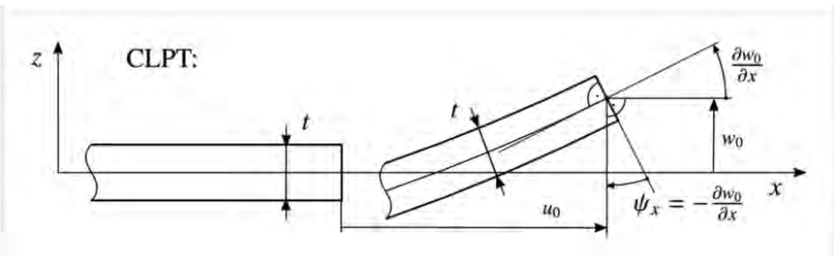


Figure 2.7: Kinematic Assumptions of CLPT Theory

The displacements at the midplane of the plate are denoted with the subscript 0 as u_0 ,v_0 and w_0. With the above kinematic assumptions, the displacement in the x direction in any point of the plate is :

$$u = u_0 - z\frac{\partial w_0}{\partial x} \tag{2.25}$$

If similarly we cut a cross section in the y-z plane, then the displacement in the y- direction is given by :

$$v = v_0 - z\frac{\partial w_0}{\partial y} \tag{2.26}$$

Using the above equations in the definitions of strains given by equations 2.9, 2.10 and 2.12, we get :

$$\epsilon_x = \frac{\partial u}{\partial x} = \frac{\partial u_0}{\partial x} - z\frac{\partial^2 w_0}{\partial x^2} \tag{2.27}$$

$$\epsilon_y = \frac{\partial v}{\partial y} = \frac{\partial v_0}{\partial y} - z\frac{\partial^2 w_0}{\partial y^2} \tag{2.28}$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = \frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x} - 2z\frac{\partial^2 w_0}{\partial x \partial y} \tag{2.29}$$

The first part of the above equations are the midplane strains denoted by ϵ_x^0, ϵ_y^0 and γ_{xy}^0. The second part are the midplane curvatures denoted by k_x, k_y and k_{xy}. If we write equations 2.27-2.29 in matrix form, we get:

$$\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} \epsilon_x^0 \\ \epsilon_y^0 \\ \gamma_{xy}^0 \end{bmatrix} + z\begin{bmatrix} k_x \\ k_y \\ k_{xy} \end{bmatrix} \tag{2.30}$$

If we substitute equation 2.30 into 2.24, then the stresses can be calculated for each ply by means of the following equation :

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{bmatrix} \epsilon_x^0 \\ \epsilon_y^0 \\ \gamma_{xy}^0 \end{bmatrix} + z\begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{bmatrix} k_x \\ k_y \\ k_{xy} \end{bmatrix} \tag{2.31}$$

Note that strains vary linearly through the thickness of the laminate, however stresses vary linearly only inside of each lamina, and may jump from lamina to lamina because of the differences in the reduced stiffness matrix $[\bar{Q}]$.
The only thing left now is to determine the midplane strains and curvatures. If we integrate

the stresses through the thickness of the laminate, we will get the resultant moments and forces, which are known. Then we can associate these moments and forces with the midplane strains and curvatures through equation 2.31.

Let us now denote the thickness of each layer of the laminate as h_k, with k=0,1...,n. Then the resulting forces and moments in the laminate in matrix form are given by :

$$\begin{bmatrix} N_x \\ N_y \\ N_{xy} \end{bmatrix} = \sum_{k=1}^{n} \int_{h_k-1}^{h_k} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}_k dz \tag{2.32}$$

$$\begin{bmatrix} M_x \\ M_y \\ M_{xy} \end{bmatrix} = \sum_{k=1}^{n} \int_{h_k-1}^{h_k} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}_k zdz \tag{2.33}$$

In the above equations, N_x and N_y are normal forces, while N_{xy} resembles the in-plane shear force. As for the moments, M_x and M_y are bending moments and M_{xy} is twisting moment. Note that all quantities are per unit length.

If we substitute equation 2.31 into the above equations and carry out the integrations, we will end with:

$$\begin{bmatrix} N_x \\ N_y \\ N_{xy} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} \\ A_{12} & A_{22} & A_{26} \\ A_{16} & A_{26} & A_{66} \end{bmatrix} \begin{bmatrix} \epsilon_x^0 \\ \epsilon_y^0 \\ \gamma_{xy}^0 \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} & B_{16} \\ B_{12} & B_{22} & B_{26} \\ B_{16} & B_{26} & B_{66} \end{bmatrix} \begin{bmatrix} k_x \\ k_y \\ k_{xy} \end{bmatrix} \tag{2.34}$$

$$\begin{bmatrix} M_x \\ M_y \\ M_{xy} \end{bmatrix} = \begin{bmatrix} B_{11} & B_{12} & B_{16} \\ B_{12} & B_{22} & B_{26} \\ B_{16} & B_{26} & B_{66} \end{bmatrix} \begin{bmatrix} \epsilon_x^0 \\ \epsilon_y^0 \\ \gamma_{xy}^0 \end{bmatrix} + \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{bmatrix} k_x \\ k_y \\ k_{xy} \end{bmatrix} \tag{2.35}$$

where

$$A_{ij} = \sum_{k=1}^{n} [\bar{Q_{ij}}]_k (h_k - h_{k-1}) \qquad \text{i,j=1,2,6} \tag{2.36}$$

$$B_{ij} = \frac{1}{2} \sum_{k=1}^{n} [\bar{Q_{ij}}]_k (h_k^2 - h_{k-1}^2) \qquad \text{i,j=1,2,6} \tag{2.37}$$

$$D_{ij} = \frac{1}{3} \sum_{k=1}^{n} [\bar{Q_{ij}}]_k (h_k^3 - h_{k-1}^3) \qquad \text{i,j=1,2,6} \tag{2.38}$$

The $[A]$,$[B]$ and $[D]$ matrices are called the *extensional, coupling* and *bending stiffness matrices* respectively.
Equations 2.34 and 2.35 offer a set of six simultaneous linear equations and 6 unknowns. The $[A]$ matrix relates in-plane forces to in-plane strains, and the $[D]$ matrix relates bending and twisting moments to curvatures. The $[B]$ matrix however relates forces and moment terms to both strains and curvatures, therefore 'coupling' them. This means that if someone where to apply a moment to the plate, the response would not only be bending and torsion, but also include extension.
We will return to this conversation in the end of this section, but now we will briefly introduce the FSDT theory. In the FSDT theory, all assumptions remain the same, except for the assumptions of zero transverse shear. Therefore the cross-section no longer remains perpendicular to the axis of the midplane like in the case of perfect bending. This is demonstrated in figure 2.8.

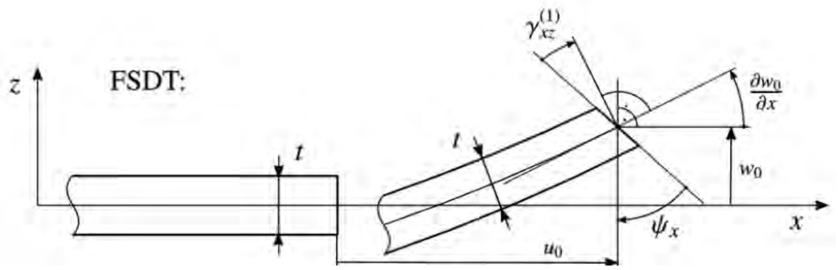


Figure 2.8: Kinematic Assumptions for FSDT theory

The difference is, that unlike before where $\psi_x = -\frac{\partial w_0}{\partial x}$ (and the same for ψ_y), here this is not true, because an additional rotation has been added due to the transverse shear deformation. The new equations for the displacements are then :

$$u = u_0 + z\psi_x \tag{2.39}$$

$$v = v_0 + z\psi_y \tag{2.40}$$

$$w = w_0 \tag{2.41}$$

The normal and in-plane shear strains remain the same as in equation 2.30, but now the curvatures are partial derivatives of the angles ψ_x and ψ_y:

$$k_x = \psi_{x,x} \qquad k_y = \psi_{y,y} \qquad k_{xy} = \psi_{x,y} + \psi_{y,x} \tag{2.42}$$

where the subscript after the comma denotes partial differentiation, for example $\psi_{x,x} = \frac{\partial \psi_x}{\partial x}$. However, transverse shear strains are now not 0, but rather constant, and will be denoted as γ^0_{xz} and γ^0_{yz}. They can be calculated by :

$$\begin{bmatrix} \gamma^0_{yz} \\ \gamma^0_{xz} \end{bmatrix} = \begin{bmatrix} w^0_{,y} + \psi_y \\ w^0_{,x} + \psi_x \end{bmatrix} \tag{2.43}$$

The transverse shear stresses are related to the transverse shear deformations via the constitutive relation :

$$\begin{bmatrix} \sigma_{yz} \\ \sigma_{xz} \end{bmatrix} = \begin{bmatrix} \bar{Q}_{44} & \bar{Q}_{45} \\ \bar{Q}_{45} & \bar{Q}_{5} \end{bmatrix} \begin{bmatrix} \gamma^0_{yz} \\ \gamma^0_{xz} \end{bmatrix} \tag{2.44}$$

The transverse shear stresses produce transverse shear forces, that can be calculated by integration. Then by using equation 2.44 we can relate the transverse shear forces to the shear deformations. This is shown in equation 2.45.

$$\begin{bmatrix} Q_y \\ Q_x \end{bmatrix} = \sum_{k=1}^{n} \int_{h_{k-1}}^{h_k} \begin{bmatrix} \sigma_{yz} \\ \sigma_{xz} \end{bmatrix} dz = \begin{bmatrix} A_{44} & A_{45} \\ A_{45} & A_{55} \end{bmatrix} \begin{bmatrix} \gamma^0_{yz} \\ \gamma^0_{xz} \end{bmatrix} \tag{2.45}$$

With this, the brief discussion of the FSDT comes to an end.

Note that in practice to involve transverse shear in the 2-D orthotropic lamina, the transverse shear moduli G_{yz} and G_{xz} need to be known from experiments, adding 2 extra variables to a total of 6. In this thesis as we will see later, transverse shear effects will be neglected in the lamina modelling.
Before we move on to the next section, we shall revert out attention again to the [A],[B] and [D] matrices from equations 2.34 and 2.35, and inspect the form that they take for some specific cases of laminates.
In the case of a *symmetric* laminate, the [B] matrix can be proved to be $[B] = 0$, and thus uncoupling the curvatures and strains to the forces and moments respectively. In practice most laminates are chosen to be symmetric, and indeed in this thesis this is also the case.
Another case is this of a *cross-ply* Laminate. A cross-ply laminate is made of only 0° and 90° plies, like for example a $[0/90_2/0] laminate$. It can be seen that for cross-ply laminates :

$$A_{16} = A_{26} = B_{16} = B_{26} = D_{16} = D_{26} = 0$$

In this case we have uncoupling of shear and normal forces, as well as between bending and twisting moments. Note here that *bending-torsion coupling* is crucial for the aeroelastic behavior of the structure, as we will see in subsequent chapters.
A *balanced* laminate has layers that come only in pairs of $+\theta$ and $-\theta$. The pairs also need to have the same thickness. An example balanced laminate is a $[40/-30/60/-40/-60/30]$. In this case, the terms A_{16} and A_{26} are zero.
If in addition to being balanced, the laminate is *antisymmetric* with same plies above and below the midplane, then also the terms D_{16} and D_{26} are zero. An example of an antisymmetric is $[45/30/-30/-45]$.
The basics of mechanics of laminates have now been established, and we are ready to move on to the next section, where the Finite Element Method will be introduced.

2.2 Finite Element Method

The finite element method is a powerful tool to solve multiple problems numerically, such as heat transfer, electromagnetic and, of course, structural problems. Analytical methods, such as the one shown in the previous section, usually cannot or are too hard to be solved for complex geometries. In this thesis, the FEM will be used for the structural analysis of the wing.

The core concept of the FEM is the *discretization* of the structure into a finite set of elements, in order to transform the analytical differential equations into algebraic equations, which can be solved by a computer easily. This set of elements is called a *mesh*. Each element has a specific number of nodes, depending on its topology. For example, a 2-D surface element could have 4 nodes in each of its edges, as we will see in section 2.2.2

In the discreticized problem, displacements(or rotations) are only known in the nodes of the elements. For the rest of the element, the displacements are interpolated by *shape functions*, which are symbolized by the letter N. The continuity of the displacements is satisfied by ensuring that adjacent elements have common nodes. Forces and boundary conditions in the FEM must also be applied in nodes.

To solve a structural problem, the *stiffness matrix* [K], in addition to the *mass matrix* [M] in the case of a dynamic problem, is calculated by using the structural properties of the element and its topology, i.e the type of discretization used. These matrices, in the FEM are calculated approximately by means of energy theorems, as it will be shown in the next sections.

2.2.1 General Formulation

The exact equations of motion for any point in a 3D body can be formulated by considering a cuboid such as the one in figure 2.3. The resulting Partial Differential Equations (PDEs) in the Cartesian coordinate system using the Newton's second law are :

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} = \rho \ddot{u} \tag{2.46}$$

$$\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{y}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} = \rho \ddot{v} \tag{2.47}$$

$$\frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{yz}}{\partial y} + \frac{\partial \sigma_{z}}{\partial z} = \rho \ddot{w} \tag{2.48}$$

The above exact equations are used to formulate new approximate solutions to the equations of motion using what is known as *Variational Formulations* of the equations of motion.
There are several methods to arrive to variational formulations, but the one commonly used in the FEM comes from the *Galerkin Method* by using arbitrary variations of the displacements as the trial functions. The actual method won't be shown here, but the interested reader can consult the literature. Note that in this method, the equations of motion are satisfied over the whole volume of the body instead of any arbitrary point. For that reason they are also called *Weak Formulations*.
By using the aforementioned method for a static problem, that is the second derivatives of the displacements with time are equal to 0, one can end up with the Variational Formulation of the Equations of Motion of equation 2.49.

$$-\int_V \boldsymbol{\delta\epsilon}^T \boldsymbol{\sigma} dV + \int_V \boldsymbol{\delta u}^T \boldsymbol{b} dV + \oint_{\Gamma_\sigma} \boldsymbol{\delta\bar{u}} \boldsymbol{f}^{\boldsymbol{\Gamma_\sigma}} d\Gamma = 0 \tag{2.49}$$

Where $\boldsymbol{\epsilon}$ and $\boldsymbol{\sigma}$ are the strain and stress vectors, $\boldsymbol{b}$ is the vector of body forces per unit volume, $\boldsymbol{\delta u}$ is a virtual displacement around equilibrium, $\boldsymbol{f}^{\boldsymbol{\Gamma_\sigma}}$ is the vector of surface forces acting on the surface Γ_σ and $\boldsymbol{\delta\bar{u}}$ is a virtual displacement on the surface.
Note that the surface Γ_σ is the part of the whole surface Γ that doesn't include boundary conditions. This is because in boundary conditions the displacements are known, and $\boldsymbol{\delta\bar{u}} = 0$.
To shed some light on the meaning of the above equation, we can see that each part has units of energy. In fact, for any arbitrary admissible virtual displacement $\bar{\delta u}$, the first integral expresses the elastic strain energy stored in the body, while the second and third integrals express the work done by external body and surface forces respectively. In essence equation 2.49 expresses that the work done by external forces in the body must equal the work done by internal forces, i.e *The Principle of Virtual Displacements*.
It can be also seen that the above equation is equivalent to the *Principle of Minimum Potential Energy*. The total potential energy of an elastic body is defined as the sum of the elastic strain energy and the work potential, i.e : $\Pi = U + WP$.

According to the principle, if the body is to be in stable equilibrium, then the total potential energy has to be minimum. This can be expressed mathematically as :

$$\delta\Pi = -\int_V \boldsymbol{\delta\epsilon}^T \boldsymbol{\sigma} dV + \int_V \boldsymbol{\delta u}^T \boldsymbol{b} dV + \oint_{\Gamma_\sigma} \boldsymbol{\delta \bar{u}} \boldsymbol{f}^{\boldsymbol{\Gamma}_\sigma} d\Gamma = 0 \tag{2.50}$$

which is in fact the same equation as equation 2.49.
In the next sections, we will use this equation to calculate the stiffness matrix [K] for 1D and 2D elements.

2.2.2 One-Dimensional Euler-Bernouli Beam Element

We will start with the simpler 1D elements and work our way to the 2D elements that are actually used in this thesis. 3D elements won't be shown here but they follow the same pattern.
One-dimensional elements are used to model either rods and truss elements that are assumed to only take axial forces, or even slender beams under torsion and bending in 1 axis.
Here we will calculate the stiffness matrix for a beam element subjected to bending using the *perfect bending condition*. This means, as we showed in the previous section, that the beam won't be subjected to transverse shear stresses and the cross-section will remain perpendicular to its centerline axis. This type of beam is called *Bernouli-Euler beam*. By using the same kinematic assumptions used in CLPT theory, but now for one dimension , we can find the strain ϵ_x to be :

$$\epsilon_x = k_x z \text{ , where } k_x = -\frac{\partial^2 w^0}{\partial x^2} \text{ is the curvature.} \tag{2.51}$$

Note that the rest of the strains can be found to be 0.
Now let us assume that the beam has a thickness of h and is made from linear isotropic material. Then according to Hooke's law :

$$\sigma_x = E\epsilon_x = Ek_x z \tag{2.52}$$

From the above equations we can clearly see that the strain and stress developed in the beam are linear in variation through the thickness. This results in zero net force N_x, as expected, but results in a bending moment, that can be calculated by integration to be :

$$M_x = \int_{-h/2}^{h/2} \sigma_x z dz = \int_{-h/2}^{h/2} E k_x z^2 dz = EIk_x \tag{2.53}$$

where I is the *area moment of inertia* of the beam cross section per unit length.
Now let us use the above equation to equation 2.49. Note that the only admissible displacement is δw^0.

$$-\int_0^L \int_{-h/2}^{h/2} \delta\epsilon_x \sigma_x dx dz + \int_0^L \int_{-h/2}^{h/2} \delta w^0 b_z dx dz + \int_0^L \delta w^0 \tau_z dx = 0 \tag{2.54}$$

where b_z are the transverse body forces and τ_z are the transverse surface forces. Then by using equations 2.51 and 2.53 and assuming that the forces are constant for simplicity, we can rewrite the equation of equilibrium for the beam as :

$$-\int_0^L \delta k_x M_x dx + \int_0^L \delta w^0 b_z h dx + \int_0^L \delta w^0 \tau_z dx = 0 \tag{2.55}$$

Now we are ready to discreticize the problem with finite elements. We will use the common two-node element for this, that is shown in figure 2.9. Note that three-node or higher one-dimensional elements exist as well.

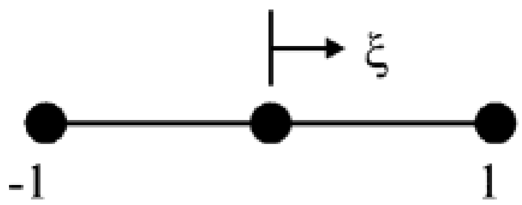


Figure 2.9: Two-node Element with its Natural coordinate system

The shape functions for the 2-node 1D element are usually linear interpolation, however for this problem because we want to calculate the curvature that contains the second derivative of w^0, we will have to use more complex shape functions, called *Hermetian polynomials.*
The node i Degrees of Freedom (DoF) are the transverse displacement w_i^0 and its derivative θ_i that expresses the slope. The only independent variable in this problem is w^0. It can be

expressed as a function of the shape functions in the form :

$$w^0(\xi) = H_1(\xi)w^1 + H_3(\xi)w^2 + \frac{L_e}{2}\left(H_2(\xi)\theta^1 + H_4(\xi)\theta^2\right) \tag{2.56}$$

where H_1, H_2, H_3 and H_4 are the Hermitian polynomials as a function of ξ that takes values from -1 to 1 inside the element. The Hermitian polynomials shape functions can be visualized in figure 2.10. Note that continuity of both w and θ is achieved with these shape functions. By differentiating equation 2.56 two times we can find the curvature to be:

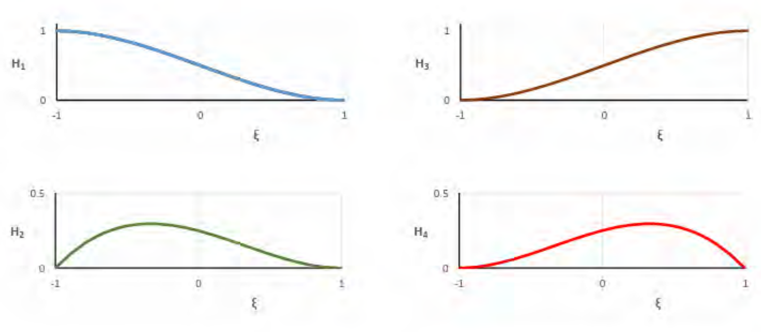


Figure 2.10: The Hermitian Polynomials as Shape Functions

$$k_x = -\frac{\partial^2 w^0}{\partial x^2} = [R^1(\xi)]\begin{bmatrix} w^1 \\ \theta^1 \end{bmatrix} + [R^2(\xi)]\begin{bmatrix} w^2 \\ \theta^2 \end{bmatrix} \tag{2.57}$$

where :

$$[R^1(\xi)] =[-\frac{4}{L_e^2}H_{1,\xi\xi}(\xi), -\frac{2}{L_e}H_{2,\xi\xi}(\xi)] \tag{2.58}$$

$$[R^2(\xi)] =[-\frac{4}{L_e^2}H_{3,\xi\xi}(\xi), -\frac{2}{L_e}H_{4,\xi\xi}(\xi)] \tag{2.59}$$

We are finally ready to calculate the element stiffness matrix. By using equation 2.57 into the elastic energy found in equation 2.55, the 2x2 submatrix between nodes i and j is given by the integral:

$$\underset{2x2}{[K_e^{ij}]} = \int_0^L [R^i(\xi)]^T EI[R^j\xi]dx \tag{2.60}$$

The integrals are usually calculated by means of *numerical integration*. The resulting element stiffness matrix for the 2-node Euler-Bernouli beam element under bending takes the final form :

$$[K_e] = \frac{EI}{L} \begin{bmatrix} \frac{12}{L^2} & \frac{6}{L} & \frac{-12}{L^2} & \frac{6}{L} \\ \frac{6}{L} & 4 & -\frac{6}{L} & 2 \\ -\frac{12}{L^2} & -\frac{6}{L} & \frac{12}{L^2} & -\frac{6}{L} \\ \frac{6}{L} & 2 & -\frac{6}{L} & 4 \end{bmatrix} \tag{2.61}$$

Note that we can improve the developed beam element, by adding transverse shear strain ϵ_{xz} to the kinematic assumptions. The resulting beam model is called *Timoshenko beam*, and it gives significantly more accurate results for shorter beams, where shear becomes significant. In addition, if a Timoshenko beam model were to be used, we would have 2 independent variables per node, and therefore we could use normal linear interpolation instead of the Hermitian polynomials used above. We will see this in the next section where we will develop the plate element including transverse shear.
Axial stiffness and torsion stiffness can be included in the 2-node element, if the problem includes axial and torsion loads in addition to bending loads, by adding additional DoF to the nodes.
The loads can be calculated from the integrals in the same way, by using the shape functions. This way any load results on equivalent nodal forces.

2.2.3 Two-Dimensional Plate Element

The element topology that we will use here is the four-node quadrilateral(QUAD4), that offers linear interpolation. The element with its natural coordinate system is shown in figure 2.11. Higher order elements can be used as well like the QUAD8 or QUAD9 elements that offer quadratic interpolation, but won't be considered here.
We will now start from the FSDT to construct the stiffness matrix of the plate element. For simplicity, we will begin by ignoring the terms that have to do with membrane stiffness(in-plane). Therefore, we will assume that the terms ϵ_x^0, ϵ_y^0 and γ_{xy}^0 from equations 2.34 and 2.35 are zero, and we will initially only consider the [D] matrix. Later, we will discuss how membrane stiffness can be added.
Since the FSDT theory was shown in the previous section, it won't be repeated here, and we move on to the discretization. The shape functions of the QUAD4 elements can be written in the compact form :

$$N_i = \frac{1}{4}(1 + \xi\xi_i)(1 + \eta\eta_i) \tag{2.62}$$

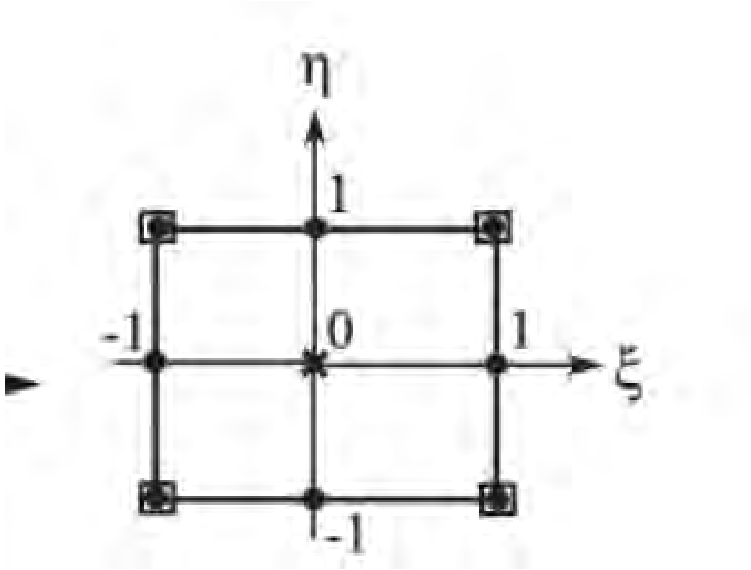


Figure 2.11: 4-node Quadrilateral Element with its Natural Coordinate System

where (ξ_i, η_i) are the coordinates of the node i in the element natural coordinate system.
If we ignore the in-plane stresses as we discussed, then each node has 3 DoF. Those are the transverse displacement w^0 and the 2 rotations ψ_x and ψ_y.
The interpolation inside the element using the shape functions then gives:

$$\begin{bmatrix} w^0 \\ \psi_x \\ \psi_y \end{bmatrix} = \sum_{i=1}^{4} \begin{bmatrix} N^i & 0 & 0 \\ 0 & N^i & 0 \\ 0 & 0 & N^i \end{bmatrix} \begin{bmatrix} w^i \\ \psi_x^i \\ \psi_y^i \end{bmatrix} \tag{2.63}$$

The curvatures then can then be written as a function of the node DoF by using the shape functions :

$$\begin{bmatrix} k_x \\ k_y \\ k_{xy} \end{bmatrix} = \begin{bmatrix} \psi_{x,x} \\ \psi_{y,y} \\ \psi_{x,y} + \psi_{y,x} \end{bmatrix} = \begin{bmatrix} N^i_{,x} & 0 \\ 0 & N^i_{,y} \\ N^i_{,y} & N^i_{,x} \end{bmatrix} \begin{bmatrix} \psi_x^i \\ \psi_y^i \end{bmatrix} = \underset{3x2}{[R_b^i]} \begin{bmatrix} \psi_x^i \\ \psi_y^i \end{bmatrix} \tag{2.64}$$

Doing the same for the transverse shear strains we get :

$$\begin{bmatrix} \epsilon^0_{yz} \\ \epsilon^0_{xz} \end{bmatrix} = \begin{bmatrix} w^0_{,y} + \psi_y \\ w^0_{,x} + \psi_x \end{bmatrix} = \begin{bmatrix} N^i_{,y} & 0 & N^i \\ N^i_{,x} & N^i & 0 \end{bmatrix} \begin{bmatrix} w^i \\ \psi_x^i \\ \psi_y^i \end{bmatrix} = \underset{2x3}{[R_s^i]} \begin{bmatrix} w^i \\ \psi_x^i \\ \psi_y^i \end{bmatrix} \tag{2.65}$$

In the above equations the subscript b comes from bending and the subscript s from shear. The elastic strain energy term of the variational formulation of equilibrium of equation 2.49 then takes the form:

$$\int_A \boldsymbol{\delta k}^T [D] \boldsymbol{k} dA + \int_A \boldsymbol{\delta \epsilon_s^0}^T [A_s] \boldsymbol{\epsilon_s^0} dA \tag{2.66}$$

The first integral expresses the bending and torsion strain energy, while the second integral expresses the transverse shear strain energy. Note that the matrices $[D]$ and $[A_s]$ depend on the material constitutive model. So in the case of an orthotropic material, they take the form of equations 2.35 and 2.45 respectively. The vectors $\boldsymbol{k}$ and $\boldsymbol{\epsilon_s^0}$ are the curvatures and transverse strains.

The stiffness submatrix of the element between the nodes i and j then is a 3x3 matrix and is calculated by the following integrals:

$$[K_e]^{ij} = \int_{A_e} [R_b^i]^T [D] [R_b^j] dA + \int_{Ae} [R_s^i]^T [A_s] [R_s^i] dA \tag{2.67}$$

The total DoF of the element are $3x4 = 12$ so the resulting element stiffness matrix $[K_e]$ will be a $12x12$ matrix.

Let us now also try to include the membrane stiffness from in-plane strains and stresses. The in-plane strains are the normal strains ϵ_x^0 and ϵ_y^0 and the in-plane shear strain γ_{xy}^0.

In case we have an isotropic material for example, the coupling matrix $[B]$ found in equations 2.34 and 2.35 is zero, and therefore bending and extension are uncoupled. In this specific case then, we can write the forces as :

$$\begin{bmatrix} N_x \\ N_y \\ N_{xy} \end{bmatrix} = \underset{3x3}{[A]} \begin{bmatrix} \epsilon_x^0 \\ \epsilon_y^0 \\ \epsilon_{xy}^0 \end{bmatrix} \tag{2.68}$$

where $[A]$ is the Extensional Matrix.

If we denote as u_0 and v_0 the displacements because of the in-plane forces, then the in-plane strains are calculated as:

$$\begin{bmatrix} \epsilon_x^0 \\ \epsilon_y^0 \\ \epsilon_{xy}^0 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} \tag{2.69}$$

We discretize the displacements with the linear QUAD4 shape functions:

$$\begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = \sum_{i=1}^{4} \begin{bmatrix} N^i & 0 \\ 0 & N^i \end{bmatrix} \begin{bmatrix} u_0^i \\ v_0^i \end{bmatrix} \tag{2.70}$$

Then equation 2.69 is written as:

$$\begin{bmatrix} \epsilon_x^0 \\ \epsilon_y^0 \\ \epsilon_{xy}^0 \end{bmatrix} = \sum_{i=1}^{4} \begin{bmatrix} N^i_{,x} & 0 \\ 0 & N^i_{,y} \\ N^i_{,y} & N^i_{,x} \end{bmatrix} \begin{bmatrix} u_0^i \\ v_0^i \end{bmatrix} = [R_M^i] \begin{bmatrix} u_0^i \\ v_0^i \end{bmatrix} \tag{2.71}$$

The in-plane strains contribute to the Elastic Strain Energy by the integral :

$$\int_A \boldsymbol{\delta\epsilon^0}^T [A] \boldsymbol{\epsilon^0} dA \tag{2.72}$$

The submatrix of the membrane stiffness then is a 2x2 Matrix that can be calculated between the nodes i and j as:

$$[K_e^{ij}] = \int_{Ae} [R_M^i]^T [A] [R_M^i] dA \tag{2.73}$$

The total DoF of each node of the plate element are now 5. In the case of an isotropic material as we have seen, the DoF that are associated with bending and torsion are the $[w^i, \psi_x^i, \psi_y^i]$, and those associated with in-plane extension and shear are the $[u_0^i, v_0^i]$. However, in the case of an orthotropic material, this is no longer true, because the $[B]$ matrix is in general not 0, so there will be coupling through the $[B]$ matrix between the bending and extension DoF.
In this thesis, the laminate that will be analyzed will be symmetric, which means that the $[B]$ Matrix will be 0, and no such coupling will be present, as we discussed in the end of section 2.1.4.

2.2.4 Isoparametric Transformation

For simple geometries like a rectangular plate, we can create a FEM model with simple QUAD elements such as the one shown in figure 2.11. However in more complex geometries with curvatures and non-uniform mesh , like the wing geometry that we will analyze in this thesis, we need to use different shapes to properly cover the whole body. In order to do that, we use a transformation from the undistorted element, or so called *parent* element.
The transformation that is commonly used because of its simplicity and good convergence,

is the one called *isoparametric transformation*. In this type of transformation, the shape functions that are used to interpolate the variables of the problem, are also used to interpolate the coordinates of the distorted element, as a function of its nodal coordinates. For a 4-node quadrilateral element, this is expressed as :

$$x = N^1(\xi,\eta)x_1 + N^2(\xi,\eta)x_2 + N^3(\xi,\eta)x_3 + N^4(\xi,\eta)x_4 \tag{2.74}$$

$$y = N^1(\xi,\eta)y_1 + N^2(\xi,\eta)y_2 + N^3(\xi,\eta)y_3 + N^4(\xi,\eta)y_4 \tag{2.75}$$

In the isoparametric transformation, the number of points used for the interpolation is the same as the number of nodes in the element. Note that the interpolation of the element is of the same order as the shape functions used for the problem. A typical isoparametric element with its parent element is shown in figure 2.12. This element uses linear shape functions and therefore its transformed shape is also linear.

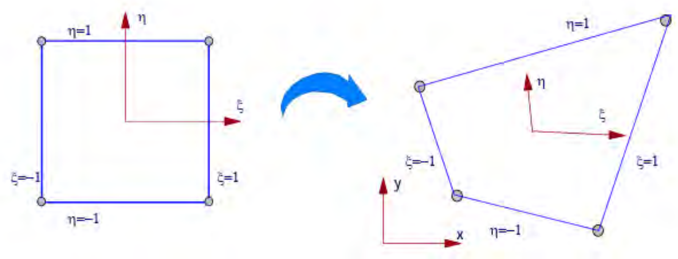


Figure 2.12: Isoparametric Element and its Parent Element

In order to calculate the stresses and strains in this transformed element, we need the derivatives of the shape functions in the transformed x-y system. These can be calculated using the chain rule as:

$$\begin{bmatrix} N^i_{,\xi}(\xi,\eta) \\ N^i_{,\eta}(\xi,\eta) \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} \begin{bmatrix} N^i_{,x}(\xi,\eta) \\ N^i_{,y}(\xi,\eta) \end{bmatrix} = [J] \begin{bmatrix} N^i_{,x}(\xi,\eta) \\ N^i_{,y}(\xi,\eta) \end{bmatrix} \tag{2.76}$$

The matrix [J] is called the Jacobian of the transformation. To find the derivatives in the x-y system we use the inverse of the Jacobian $[J]^{-1}$. In order for it to be invertible, its determinant must not be 0.
Finally, for integration in the transformed system carry out the integration in the element natural coordinate system (ξ,η) and multiply by the determinant of the Jacobian of the

transformation. For example :

$$\int_A F(x,y)dxdy = \int_{-1}^{1}\int_{-1}^{1} F(\xi,\eta)|J|d\xi d\eta \tag{2.77}$$

2.2.5 Structural Analysis with Finite Elements

Static Analysis

For static analysis, the resulting equation takes the form :

$$[K]\boldsymbol{u} = \boldsymbol{F} \tag{2.78}$$

where $[K]$ is the global stiffness matrix of size NXN, $\boldsymbol{u}$ is the vector of the N nodal DoF (displacements or rotations) and $\boldsymbol{F}$ is the vector of the equivalent N nodal forces(or moments).

The global stiffness matrix is calculated by adding all the element stiffness matrices , according to the connectivity of the elements in the structure, such that the correct stiffness term is associated with its corresponding nodal DoF.

The above system of equations is then partitioned to the free DoF system and the constrained DoF system, that satisfies the boundary conditions and has known displacements or rotations.

If we use the subscript f for the free system and the subscript c for the constrained system, then the equations take the form:

$$\begin{bmatrix} K_{cc} & K_{cf} \\ K_{fc} & K_{ff} \end{bmatrix} \begin{bmatrix} \boldsymbol{u_c} \\ \boldsymbol{u_f} \end{bmatrix} = \begin{bmatrix} \boldsymbol{R_c} \\ \boldsymbol{F_f} \end{bmatrix} \tag{2.79}$$

where $\boldsymbol{R_c}$ is the vector of the support reactions. Then the equation of the unknown displacements takes the form:

$$K_{fc}\boldsymbol{u_c} + K_{ff}\boldsymbol{u_f} = \boldsymbol{F_f} \tag{2.80}$$

The above equation can be solved for the unknown DoF. Finally, in the special case where the constrained DoF are zero (like in the cantilever boundary condition of this thesis), then the system to be solved takes the simpler form :

$$\boldsymbol{u_f} = [K_{ff}]^{-1}\boldsymbol{F_f} \tag{2.81}$$

Then, from the resulting displacements and rotations, we can calculate the element strains and stresses, as we have shown. Note that by using the remaining equations of the constrained DoF, we can also find the reaction forces acting on the constraint.

Dynamic Analysis

So far in the previous sections we have ignored the terms of accelerations that exist in Newton's second law of equations 2.46-2.48 and essentially dealt with static problems. Now we will include these terms, in order to analyze dynamic problems. Note that in dynamic problems, variables are also time-dependent.

To include the acceleration terms in equation 2.49, we will use *D'Alambert's principle*, which states that the acceleration terms of Newton's second law can be treated as an extra set of forces, called *inertial forces*. Then, we can treat this problem as a problem of equilibrium in the form:

$$\Sigma F_{external} - \Sigma F_{inertial} = 0 \qquad ,\text{where } \Sigma F_{inertial} = \rho\ddot{u} \tag{2.82}$$

Note that in the above equation the forces are per unit volume.

Inertial forces are body forces, so in order to include them to equation 2.49, we will express the vector $\boldsymbol{b}$ as the sum of the external body forces, denoted now by $\boldsymbol{b^*}$, like gravity, and the inertial forces.

$$\boldsymbol{b(t)} = \boldsymbol{b^*(t)} - \rho\ddot{\boldsymbol{u}} \tag{2.83}$$

where $\ddot{\boldsymbol{u}}$ is the vector of DoF and could also contain rotations. The *mass matrix* of an element is then calculated by employing the shape functions of the discretization. In general then, it has the form:

$$[M_e] = \int_{V_e} \rho [N]^T [N] dV \tag{2.84}$$

where $[N]$ is a diagonal matrix containing the shape functions and has the size of nxn, where n is the total number of shape functions.

The mass matrix calculated by the equation 2.84 is called *consistent mass* representation and is more accurate, however it is usually fully populated.

An alternative representation is the *lumped mass* representation, where the mass matrix is made diagonal, while the total mass is preserved. For example for a 2-node element, half the mass will be 'lumped' at each node. In this way, the computational cost is reduced, at the expense of a less accurate solution. By involving a sufficient amount of elements however, the error from using a lumped mass representation will be small.

The equations of motion for a dynamic Finite Element model take the general form:

$$[M]\ddot{\boldsymbol{u}} + [C]\dot{\boldsymbol{u}} + [K]\boldsymbol{u} = \boldsymbol{F} \tag{2.85}$$

where the $[C]$ matrix is the *viscous damping* matrix. If $[C] = 0$ the system is called *conservative.*

In this thesis, we will often deal with the subject of *normal modes.* A normal mode is the pattern at which an undamped linear system oscillates, when it vibrates freely ($\vec{\boldsymbol{F}} = 0$) at a frequency equal to the natural frequency corresponding to that mode. Note that the natural frequencies of a system are dependent only on its stiffness and mass, along with the boundary conditions imposed.

To understand the concept of normal modes let us consider a cantilever plate, that is vibrating freely and without damping around its equilibrium point. Because the plate is a continuum, it has an infinite number of natural frequencies and corresponding modes. However when a Finite Element model is employed to describe the problem, then the problem reduces to a multiple DoF discrete system.

The equations of motion for a conservative discrete system without excitation take the form:

$$[M]\ddot{\boldsymbol{u}} + [K]\boldsymbol{u} = 0 \tag{2.86}$$

The first 4 mode shapes of an arbitrary cantilever plate are shown in figure 2.13.

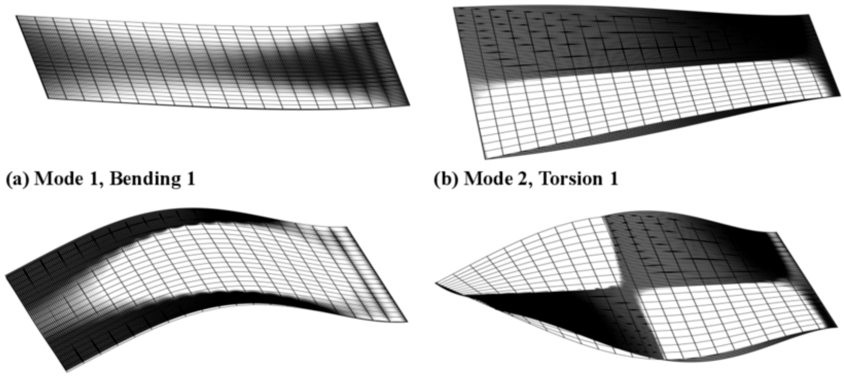


Figure 2.13: First 4 Normal Modes of a Cantilever Plate

The pattern and frequency at which the plate will vibrate, depends on the initial conditions. If the initial conditions are such that the structure vibrates at exactly one of its natural frequencies, then the pattern of oscillation will be exactly that of its corresponding mode. However, in general, the motion of the structure will involve a combination of its natural modes.

Using then the principle of Linear Superposition, we can express this complex motion as a superposition of its individual normal modes. Let us write the DoF vector $\boldsymbol{u}$ in the following form:

$$\boldsymbol{u(t)} = [\Phi]\boldsymbol{q(t)} \tag{2.87}$$

where $[\Phi]$ is the modal matrix, and $\boldsymbol{q(t)}$ is the mode participation vector, that expresses the contribution of each mode shape to the final motion. The $[\Phi]$ matrix is a NxN matrix for a N-DoF discrete system.

It can be proven, that modes are orthogonal with each other. We can use equation 2.87 in order to transform the equation of motion to the *modal space*. In the modal space, the mass and stiffness matrices are diagonal, and so the multiple DoF system can be solved as a combination of single DoF systems because they are uncoupled. Equation 2.86 in the modal space takes the form :

$$[\Phi]^T[M][\Phi]\boldsymbol{\ddot{q}(t)} + [\Phi]^T[K][\Phi]\boldsymbol{q(t)} = 0 \tag{2.88}$$

One additional advantage of the modal space representation, is the ability to perform model reduction,by not using all the normal modes in the solution, with great saves in computational load. It is often enough to only consider the first several modes of a structure to get

accurate results for its vibrational behavior, since these modes have the highest contribution and the largest energy content.
In order to perform a normal mode analysis of a Finite Element model, we use equation 2.86. Normal modes refer to the natural modes of the structure.
To solve the equation, we assume solutions of the form:

$$\boldsymbol{u} = \mathbf{u}\mathrm{e}^{i\omega t} \tag{2.89}$$

where ω is the angular frequency, i is the imaginary number. This form of solutions is very convenient because it can turn differential equations into algebraic equations. Indeed if we differentiate the above solution 2 times over time we get:

$$\ddot{\boldsymbol{u}} = -\omega^2 \mathbf{u}\mathrm{e}^{i\omega t} \tag{2.90}$$

Using the assumed solution to equation 2.86 yields"

$$\left([K] - \omega^2 [M]\right) \boldsymbol{u} = 0 \tag{2.91}$$

The above equation is now an eigenvalue problem, with the eigenvalues being the eigenfrequencies ω_i. Eigenvalue problems are computationally efficient and are often used. To determine the non-trivial solution of equation 2.91 we set the determinant equal to zero :

$$det\left([K] - \omega^2 [M]\right) = 0 \tag{2.92}$$

The eigenvectors are the normal modes. Each mode is found by solving equation 2.91 with the corresponding eigenfrequency. Note that because the determinant is set to 0, the absolute values of the vector $\boldsymbol{u}$ cannot be determined, but rather the relative values of each DoF. The modes therefore describe only the shape of the final motion, rather than the actual amplitude of vibration.

2.3 Doublet Lattice Method

The DLM, is a numerical aerodynamic panel method, designed to model compressible subsonic unsteady flows. Its theoretical basis is *inviscid* flow and *potential aerodynamics*. In addition, the DLM is designed for flat surfaces, and cannot account for camber or thickness of the body.
The DLM is used extensively for flutter analysis where oscillatory aerodynamics occur, because it can give give satisfactory results without the need to use more complex and computationally expensive CFD models. Note however, that because the DLM is based on potential aerodynamics, it is a linear theory, and therefore it cannot account for non-linear effects.

2.3.1 Euler's Equations for Inviscid and Adiabatic Fluid Flow

The most basic assumption, that is often used in aerodynamic models to relieve some of the complexity, is the assumption of inviscid flow, that is, that there are no phenomena of friction between the fluid and the body surface. The frictionless fluid doesn't exist in nature, however it is often used to facilitate important principles that can be approximately true for viscous fluids.
Some of the limitations of inviscid flow include :

- Without the effects of viscosity, the boundary layer effects cannot be accounted for. Boundary layer is a thin region near the surface of the body where the viscous effects dominate. Note that on the surface, the velocity of the flow is zero, and gradually increases to the flow speed by moving away from the surface.
- Without viscosity, surface friction drag cannot be predicted. Other phenomena such as heat conduction and mass diffusion also cannot be modelled.
- Viscosity plays a crucial role in the transition of the fluid from the laminar to the turbulent regime. In fact, the regime of the flow can be approximately predicted by

the *Reynold's number*, a non-dimensional number that considers the ratio of inertial forces to viscous forces. The formula for the Reynolds number is :

$$Re = \frac{\rho u l}{\mu}$$

where μ is the dynamic viscosity of the fluid, u is the relative velocity of the flow with respect to the body, l is a characteristic length and ρ is the density of the fluid. Turbulent flow generally contains more energy, higher mixing of the flow and considerable higher friction forces.

- Inviscid flow cannot account for the phenomenon of flow separation, that occurs when the fluid flow finds severe adverse pressure gradient, and the low speed and energy fluid particles near the surface slow down to a halt and even reverse in direction of motion, causing the fluid to separate from the object. Note that this effect is more pronounced in laminar flow, and therefore in real life applications it is often desired to induce turbulence in the flow, because turbulent flow contains more energy in the boundary layer, in order for the flow to remain attached to the surface.

The PDEs that describe the equations of an inviscid and adiabatic flow are the *Euler's equations*. These are 5 equations that can be solved for the 5 state variables. The 5 state variables include the pressure $p(x, y, z, t)$, the density $\rho(x, y, z, t)$, and the 3 cartesian components of the flow velocity, $u(x, y, z, t), v(x, y, z, t)$ and $w(x, y, z, t)$ (not to be confused with the displacements used in structural analysis).
Let us consider a control volume that includes the flow over a body with the reference coordinate system attached to the body. Then the Euler's equations take the form:

1. Continuity equation :

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{V}) = 0 \tag{2.93}$$

   where $\boldsymbol{V}$ is the vector of velocity and $\nabla\cdot$ is the divergence operator. The continuity equation expresses the *conservation of mass*.

2. Momentum equation
   The Momentum equation can be divided in 3 equations expressing the *conservation of*

momentum in each of the 3 cartesian directions :

$$\frac{\partial \rho u}{\partial t} + \nabla \cdot (\rho u \boldsymbol{V}) = -\frac{\partial p}{\partial x} + \rho f_x \tag{2.94}$$
$$\frac{\partial \rho v}{\partial t} + \nabla \cdot (\rho v \boldsymbol{V}) = -\frac{\partial p}{\partial x} + \rho f_y \tag{2.95}$$
$$\frac{\partial \rho w}{\partial t} + \nabla \cdot (\rho w \boldsymbol{V}) = -\frac{\partial p}{\partial z} + \rho f_z \tag{2.96}$$

where here f_i are body forces per unit mass in the volume of the fluid (such as gravity and acceleration inertial forces).

3. Isentropic Relation:

$$p = \rho^{\gamma} \qquad \text{,where } \gamma = \frac{c_p}{c_v} \text{ is the ratio of specific heats.} \tag{2.97}$$

Note that an isentropic flow must both inviscid and adiabatic, like Euler's equations assume.

Finally, we will also need the speed of sound, defined by the formula :

$$\alpha^2 = \frac{dp}{d\rho} \qquad \text{,or if air is assumed to be a perfect gas:} \quad a = \sqrt{\gamma R T} \tag{2.98}$$

2.3.2 Potential Aerodynamics Theory

It can be shown that inviscid flow is irrotational, i.e the vorticity of the flow is zero at any point. The condition of irrotationality is mathematically expressed as: $\nabla \times \boldsymbol{V} = 0$.
These conditions allows or the definition of a scalar function Φ, such that the velocity is given by the gradient of Φ. This function is called the *velocity potential.* In cartesian coordinates, the 3 components of velocity can be written:

$$u = \frac{\partial \Phi}{\partial x} \qquad v = \frac{\partial \Phi}{\partial y} \qquad w = \frac{\partial \Phi}{\partial z} \tag{2.99}$$

Note that the above equation can be written in the more compact form $\boldsymbol{V} = \nabla\Phi$, where ∇ is the gradient operator.
We can manipulate Euler's equations in order to reach to an equation that describes the velocity field of the flow and the only variable is the velocity potential Φ. If we assume small disturbances around an undisturbed uniform velocity and far field pressure, then by

dropping out the non-linear terms we can reach to the *Linear Small Disturbance Velocity Potential equation*

$$(1 - M_\infty^2)\tilde{\Phi}_{xx} + \tilde{\Phi}_{yy} + \tilde{\Phi}_{zz} - \frac{1}{\alpha_0^2}\left(2U_\infty\tilde{\Phi}_{xt} + \tilde{\Phi}_{tt}\right) = 0 \tag{2.100}$$

where U_∞ is the free stream velocity, α_0 is the constant value of speed of sound in the far field and $M_\infty = \frac{V_{00}}{\alpha}$ is the Mach number of the undisturbed flow. The tilde on top of the letters expresses that they are small perturbations around mean values.
The above equation is used to model small disturbances in the flow caused by the motion and existence of the lifting surface. Those disturbances travel with the speed of sound.
In potential aerodynamics, the principle of Linear Superposition is used to superimpose the flows induced by a combination of elementary solutions of the aerodynamic potential equation in order to model a more complex flow. These solutions must satisfy some boundary conditions.
For a flow over the surface of a body, the first condition that must be met, is that the fluid cannot penetrate the body surface. This is called the *zero normal flow condition*, to ensure that the flow is always tangential to the body. Note however that in the case of a moving body, the normal velocity of the fluid must equal the normal velocity of the body surface in order for this condition to be satisfied. Mathematically, for a time variant surface given by the equation $F(x, y, z, t)$, this condition is described by :

$$\frac{\partial F}{\partial t} + \boldsymbol{V} \cdot \nabla F = 0 \tag{2.101}$$

For a thin plate or wing where the undeformed midplane lies at $z = 0$, the surface can be described by:

$$F(x, y, z, t) = z - h(x, y, t) \tag{2.102}$$

Then by using the normal flow boundary condition we get :

$$w = \frac{\partial h}{\partial t} + U_\infty \frac{\partial h}{\partial x} \tag{2.103}$$

Thus, the downwash velocity w needs to be equal with the *substantial derivative* of the deflection of the midplane in the streamwise direction.
The second condition is that all elementary solutions must satisfy the *far field boundary condition*, i.e that in far distances the velocity field takes the values of the undisturbed uniform

flow and all influence dies out.
The last condition that needs to be met, is the *Kutta condition.* For incompressible steady flow, this requires a smooth flow off the trailing edge of the body, i.e the velocity isn't allowed to deflect as the flow passes through the sharp trailing edge. However for linearized unsteady compressible flow, it has been shown that the flow off the trailing edge is not tangential. So in this case, a zero pressure difference condition is imposed instead.
In potential aerodynamics, a point source is a localized source of fluid emanating radially from its origin in all directions. The opposite of a source, is a point sink, with the fluid moving towards the origin.
A point doublet is the limiting condition where a point source and a point sink of equal but opposite strengths are located at the same point.
In the case of a steady incompressible flow, the aerodynamic potential equation takes the form of the Laplace equation: $\nabla^2\Phi = 0$, and it can be shown that a stationary point source and a stationary point doublet are both elementary solutions. However, in the case of equation 2.100, this is no longer the case, because of the time dependent derivatives.
A moving point source solution can be derived as an elementary solution of the time-dependent velocity potential equation. It can be also shown that a moving point doublet found by differentiation with respect to z from the point source solution, is also an elementary solution. The derivation of these solutions is out of the scope of this background section and the interested reader is directed to *Blair [6]*.
Now, we are really interested in the pressure rather than the velocity potential. So instead of calculating the velocity potential and then solving for the pressure, we manipulate equation 2.100 and express it only in terms of pressure.

$$(1 - M_\infty^2)p_{xx} + p_{yy} + p_{zz} - \frac{1}{\alpha_0^2}\left(2U_\infty p_{xt} + p_{tt}\right) = 0 \tag{2.104}$$

This new equation, called the *Pressure Potential equation*, is mathematically the same equation, and therefore has the same elementary solutions. Only the physical interpretation is different.
A pressure source and a pressure doublet are elementary solutions of equation 2.104. A point source is symmetrical with respect to the midplane of the thin plate, and therefore a pressure source cannot model a pressure difference. This is why the pressure doublet is used, which is antisymmetric and therefore capable of producing a pressure difference.
In the DLM, we further restrict the time dependence to harmonics in time, and therefore to

constant frequencies. This results in great computational savings, and is consistent with the approach used in flutter analysis.

Instead of the pressure potential, we will use a new quantity called *Acceleration Potential* ψ. The acceleration potential is in fact the pressure doublet solution, divided by the density ρ_0. We write this solution the acceleration potential in the form $\psi = \overline{\psi} e^{i\omega t}$. By substituting the acceleration potential doublet in the potential equation 2.100 in the frequency domain we get:

$$(1 - M_{\infty}^2)\frac{\partial^2 \overline{\psi}}{\partial x^2} + \frac{\partial^2 \overline{\psi}}{\partial y^2} + \frac{\partial^2 \overline{\psi}}{\partial z^2} - \frac{1}{\alpha_0^2}\left(2i\omega \frac{\partial \overline{\psi}}{\partial x} - \omega^2 \overline{\psi}\right) = 0 \tag{2.105}$$

The pressure that arises with the acceleration potential is simply $p = \rho_0 \psi$.

The aim of panel method is to determine a distribution of the superimposed elementary flow Solutions on the lifting surface that properly models the flow over a body, satisfying the potential equation and the boundary conditions imposed.

For this reason, we extend the concept of a single pressure doublet to a doublet sheet. A doublet sheet has a continuous distribution of doublet strength along the surface. Each infinitesimally small area dS has strength AdS. It can be shown, that the amplitude of the doublet strength is related to the jump in pressure (in amplitude because it is now defined as a complex number to account for oscillations) across the doublet sheet by the following formula:

$$\Delta \overline{p}(x, y) = 4\pi\rho A \tag{2.106}$$

Finally, by considering the contributions of the whole doublet sheet and imposing the surface boundary condition, the oscillating lifting surface theory provides a linear relationship between the downwash velocity component induced at point w and the pressure jump across the surface.

Instead of the pressure difference, we will use the dimensionless coefficient of pressure defined as: $\Delta\overline{cp} = \frac{\overline{p}}{\frac{1}{2}\rho U_{\infty}^2}$ and instead of the downwash velocity w we use the dimensionless downwash $W = \frac{w}{U_{\infty}}$. Then the final equation takes the form:

$$\overline{W}(x, s) = \frac{1}{8\pi}\int_S K(x, \xi, s, \sigma, \omega, M)\Delta\overline{cp}(\xi, \sigma) d\xi d\eta \tag{2.107}$$

where the (x,s) system is an orthogonal system that lies on the surface such as the x-direction is the direction of the undisturbed flow velocity.

The K function inside the integral is called the *Kernel* function and it relates the downwash to the change in the coefficient of pressure.

2.3.3 Doublet Lattice Method Formulation

With the rise of computational power, the implementation of numerical solutions became possible. In the case of numerical solutions, the lifting surface is divided into a number of trapezoidal elements, arranged in columns parallel to the free stream.
The DLM is an approximation of the theory of oscillating lifting surface and it offers a way to calculate the Kernel function numerically.
In the DLM, each box contains a line distribution of acceleration potential doublets of uniform but unknown strength. This line is positioned at the 1/4 chord length of each box. For the Kutta condition to be met, the control point where the downwash will be calculated for each element, is located at the 3/4 chord length of the box. This location in the DLM is empirical and has been derived from numerical experimentation rather than rigorous mathematical proof.
In the case of steady flow, the doublet line of the DLM is equivalent to a horseshoe vortex of the steady VLM at the 1/4 chord line. The VLM uses vortices instead of doublets in a similar way to satisfy the boundary conditions and creates lift based on the *Kutta-Joukowski Theorem*. The VLM however won't be shown here.
A typical aerodynamic mesh of a lifting surface with a sweep angle is shown in figure 2.14.

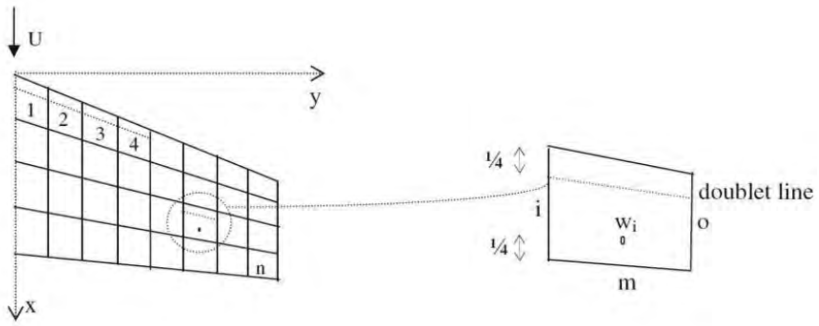


Figure 2.14: Typical DLM Element

Now let us consider a general swept lifting surface like the above that is discretized in n boxes. The *constant* force per unit length acting on the j_{th} box at the 1/4 chord line is f_j. Then the amplitude of the j_{th} doublet line strength is constant and given by :

$$A = \frac{f_j}{4\pi\rho} \int_{l_j} d\mu \tag{2.108}$$

where l_j is the length of the line and $d\mu$ is the incremental length. Then the total normalized downwash induced at a point (x_i, s_i) by all the doublet lines is given by:

$$W(x_i, s_i) = \sum_{j=1}^{n} \frac{f_j}{4\pi\rho U_\infty} \int_{l_j} K[x_i, s_i, x_j(\mu), s_j(\mu), \omega, M]d\mu \tag{2.109}$$

The above equation is the discretized 'version' of equation 2.107. The input to this equation is the downwash of each point which is generally known by the deflection of the surface (see equation 2.103, and the output is the force f_j acting on the j_{th} element. Then we can find the constant pressure acting on the element by dividing with the box area :

$$P_j = \frac{f_j l_j}{\Delta x_j cos(\lambda_j)} \tag{2.110}$$

where λ_j is the sweep angle of the doublet line and Δx_j is the average chord of the j_{th} box. Then if instead of the pressure we use the pressure coefficient we get the similar to equation 2.107 expression:

$$\frac{f_j}{4\pi\rho U_\infty} = \frac{1}{8}\Delta\overline{cp}_j \Delta x_j cos(\lambda_j) \tag{2.111}$$

that expresses the pressure coefficient in the j_{th} box.

For aeroelastic analyses by coupling the aerodynamic model with the Finite Element model, as we shall see in more depth in section 2.5, the pressures calculated on the aerodynamic elements by the DLM are transformed to equivalent forces on the structural nodes. The resulting matrix that relates the aerodynamic forces to the structural DoF is called Aerodynamic Influence Coefficient (AIC) matrix. The AIC matrix of the unsteady case is dependent upon the frequency and Mach number.

For the DLM, it is suggested by *Albano and Rodden [2]* that the boxes have aspect ratios of order unity, i.e not much larger or much smaller than 1.

Also, it is concluded that the DLM is accurate for reduced frequency of order unity even for high subsonic Mach numbers. The reduced frequency is a dimensionless quantity defined as $k = \frac{\omega b}{U_\infty}$,where b is the semi-chord length, that is used in aeroelastic analysis because it is more convenient.

2.4 Introduction to Aeroelasticity

The purpose of this section is to describe the fundamental and potentially catastrophic aeroelastic phenomena of divergence and flutter and provide the general equations of static and dynamic aeroelasticity.

2.4.1 Divergence

Divergence is a static aeroelasticity phenomenon. The most common type of divergence, is that of torsional divergence, where the angle of attack because of elastic twist increases excessively up until a speed is reached, the divergence speed, where the structure's elastic forces are unable to counter-act the increasing aerodynamic forces. In order to better understand the phenomenon of divergence, we will use a simple example problem.
Let us consider a flexible rectangular plate of span s and chord length c, fixed at its root, as shown in figure 2.15. This wing has an initial angle of incidence of θ_0 and torsional rigidity GJ. The plate is subjected to air flow of speed V and is assumed to only twist without bending. The elastic twist of the plate, for simplicity, will be assumed to have a linear relationship with the spanwise distance from the root, i.e $\theta = \frac{y}{s}\theta_T$, where θ_T is the elastic twist at the tip.

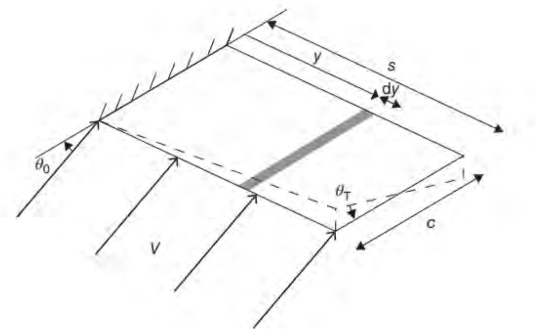


Figure 2.15: Flexible Rectangular Cantilever Plate under twisting *Wright and Cooper [45]*

Let us assume that the elastic axis of the plate lies at a distance *ec* after the aerodynamic center. It is important to pause here and give some explanation of the above terms before we move on with the problem..
The *aerodynamic center* lies at approximately the 1/4 chord point of an airfoil section(or in this case the plate) and it is the point where the lift and drag forces are taken to act, along

with a pitching moment. The use of the aerodynamic center instead of the center of pressure (where only forces act) is convenient because its location remains constant irrespective of changes in the angle of attack and airfoil shape. In addition at this point the pitching moment is not a function of the lift.
The *shear center* is defined as the point in a $2-D$ section, where if a transverse load is applied it produces zero twist. The *elastic axis* is then the locus of the shear centers along the span of the structure. The elastic axis is commonly used in the literature, however it shouldn't be confused with the *flexural axis*, which refers to the locus of the *flexural centers* with zero twist relative to the wing root. For homogeneous unswept wings the 2 axes coincide, but in general they do not.
This problem is taken directly from the classical textbook of *Wright and Cooper [45]*. Strip theory is used for the aerodynamic modelling. Strip theory is the simplest theory that divides the plate in chordwise strips of length dy, upon which the incremental lift dL acts at its aerodynamic center at 1/4 chord. The lift on one strip has no influence upon another, unlike panel methods such as the DLM that we saw above.
If the lift-curve slope is $a_1 = \frac{\partial C_L}{\partial a}$ where a is the total angle of incidence given by $a = \theta_0 + \theta$, then the equations of motion are found to be:

$$\left(\frac{GJ}{s} - \frac{1}{6}\rho V^2 ec^2 a_1 s\right)\theta_T = \frac{1}{4}ec^2 a_1 s\theta_0 \tag{2.112}$$

From the above equation it can be seen by the equation inside the parenthesis, that the effective torsional stiffness is reduced by the aerodynamic term as the velocity is increased. In fact, in static cases like this, we can treat the aerodynamic term as *aerodynamic stiffness*, and write the *Static Aeroelasticity equation* of motion in the general form :

$$\left(\rho V_\infty^2 [C] + [E]\right) \boldsymbol{q} = \mathbf{0} \tag{2.113}$$

where $[C]$ is the aerodynamic stiffness matrix, $[E]$ is the structural stiffness matrix and the vector $\boldsymbol{q}$ contains generalized coordinates(DoF).
Returning to the current problem, the divergence velocity is the velocity at which the effective torsional stiffness becomes zero, and thus the elastic twist becomes infinite. The divergence speed is then :

$$V_{div} = \sqrt{\frac{6GJ}{\rho ec^2 s^2 a_1}} \tag{2.114}$$

In a real structure, the elastic twist can obviously not become infinite, and thus the structure fails.

Factors that affect the Divergence Speed

From this simple example we can deduct some design rules to increase the divergence speed :

- The greater the torsional stiffness, here GJ, the higher the divergence speed becomes.
- The closer the elastic axis is to the aerodynamic center, the divergence speed also increases. In the limiting case that they coincide, divergence does not occur at all.
- In case the elastic axis moves before the aerodynamic center, then the aerodynamic moment becomes negative resulting in nose down twist and divergence does not occur.

The last two scenarios are not generally possible in practical structures and wings, therefore adequate torsional stiffness is crucial along with designs with the elastic axis as close to the aerodynamic center as possible.
The effect of sweep is also important in divergence speed, as effective angle of incidence changes when the wing bends. In order to visualize, let us look at figure 2.16.

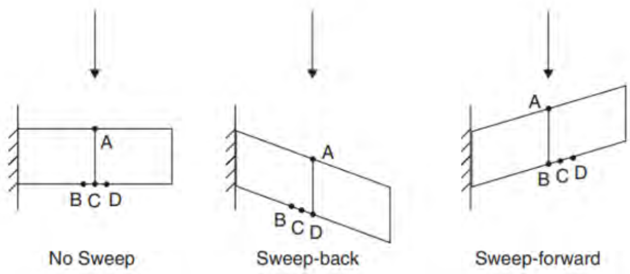


Figure 2.16: Effect of Sweep angle on Divergence speed *Wright and Cooper [45]*

Considering upwards bending, the following occur in each case :

1. In the unswept case, incidence is unchanged due to bending.
2. When the wing is swept back, the streamwise incidence changes to the line (AD) of the figure. Because point D is further from the root than point A, it bends upwards more, therefore causing a decrease in the effective angle of incidence.

3. For the swept forward wing, the effective incidence changes to the line (AB), and as a result the opposite is true, increasing the angle of incidence.

As a consequence of the above, it can be deducted that backward swept wings have generally higher divergence speeds and they are usually out of the range of interest and flutter, which will be considered next, is usually the critical mode of failure of the two. On the other hand, for forward swept wings divergence is a lot more critical.
Finally, in the case that we have a composite plate, there is another factor that comes into play. Let us revisit equation 2.35 and recall that the D_{16} and D_{26} terms of the $[D]$ matrix represent bending-torsion coupling. If these terms are positive, it means that positive bending will also result in positive twist and therefore has divergence tendency. This will inevitably decrease the divergence speed. The opposite is true for negative bending-torsion coupling, where the upwards bending will result in a decrease in the angle of attack, and therefore divergence will be increased. This is demonstrated in figure 2.17 taken by *Stanford and Jutte [36]*, where no coupling, negative coupling (wash-out) and positive coupling (wash-in) are shown from left to right.

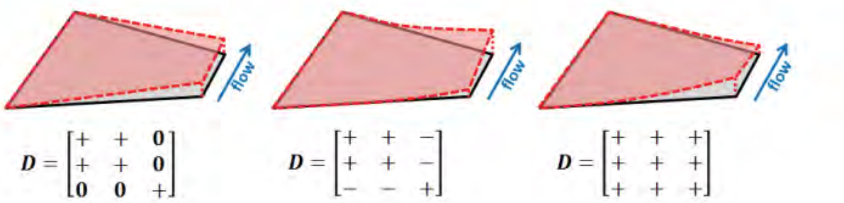


Figure 2.17: Bending-Torsion Coupling Effect on Wing Deformation *Stanford and Jutte [36]*

Note that in essence, a positive bending-torsion coupling has a so called *wash-in* effect that behaves like a forward swept wing, while a negative bending-torsion coupling has a *wash-out* effect that behaves like a backward swept wing. We will see in the next section that flutter speed is benefited by wash-in behavior, unlike divergence.

2.4.2 Flutter

Flutter is arguably the most important aeroelastic phenomenon and usually the most critical. It is a dynamic instability that occurs in bodies subjected to large aerodynamic loads.

It features self-excited oscillations of the structure caused by unfavorable coupling of modes of vibration that result in motion such that the structure is able to extract energy from the air(*Bisplinghoff et al. [5]*).

The flutter speed is defined as the lowest speed at which the structure sustains harmonic oscillations of constant amplitude, and the frequency of these oscillations is the flutter frequency. For speeds above the flutter speed, these oscillations get larger in amplitude and therefore become unstable.

Flutter analysis is done in the frequency domain by assuming harmonic motion of the structure. In return this results in harmonic frequency dependent aerodynamic forces. Note that unlike the steady aerodynamics, unsteady aerodynamics feature phase lag between the motion of the structure and the resulting lift, because the flow needs time to adjust to the new flow conditions. Also, as one can imagine, unsteady aerodynamic forces not only depend on the angle of attack and displacement of the surface, but also on their derivatives, that are now not zero.

The oscillations that may occur in speeds under the flutter speed will be quickly damped by the aerodynamic forces, even if no structural damping is considered, which is usually the case for conservative results. However when the speed reaches and surpasses this limit, even the smallest of perturbations will cause the structure to destabilize.

The methods of flutter analysis by coupling the DLM aerodynamic model to the FEM structural model will be discussed in the next section. However, we will show here that the *Dynamic Aeroelasticity Equations* without external excitation can be written in the general form:

$$[A]\ddot{\boldsymbol{q}} + (\rho V_\infty [B] + [D])\,\dot{\boldsymbol{q}} + \left(\rho V_\infty^2 [C] + [E]\right)\boldsymbol{q} = 0 \tag{2.115}$$

where $[A]$ is the structural inertia matrix, $[B]$ is the aerodynamic damping matrix, $[C]$ is the aerodynamic stiffness matrix, $[D]$ is the structural damping matrix and finally $[E]$ is the structural stiffness matrix. The aerodynamic damping matrix contains the unsteady aerodynamic forces that are dependent upon the derivatives of the DoF.

Note that both the $[B]$ and $[C]$ aerodynamic matrices in the unsteady case are time dependent. In flutter analysis, as it has been discussed, these unsteady terms are assumed to be harmonic and they become frequency-dependent instead, and thus the above aeroelastic equations are defined at specific frequencies(or rather reduced frequencies). This poses some difficulty in flutter analysis, and iterative methods are used, as the frequency of flutter is not known before-hand.

Let us now revert our attention to the physical aspects of the phenomenon of flutter. We will discuss the phenomenon of classical binary flutter, that involves coupling of a bending and a torsion mode.

First we will discuss about what happens during the flutter condition, and why the structure is able to extract energy from the air with a very simplistic example. Let us imagine an airfoil that is oscillating in bending and torsion under the quasi-steady assumption, meaning that no phase lags occurs in aerodynamic terms, as shown in figure 2.18. Also for this simplified example imagine that the two motions happen with the same frequency.

During bending oscillation, positive lift is produced as the airfoil is moving downwards because of effective angle of attack increasing, and negative lift is produced as it is moving upwards. On the other hand during torsion oscillation, positive lift is produced when the twist is nose up and negative lift is produced when the twist is nose-down.

If the two motions are in-phase, meaning that they pass from their equilibrium and extreme points at the same time, then the lift force extremes will be at 90° out of phase, and no flutter occurs.

However if we consider the 2 motions being at 90° phase difference with each other then the lift produced by the two motions will be in-phase, and therefore the structure is able to extract energy in a similar fashion as the phenomenon of resonance in classical vibration analysis.

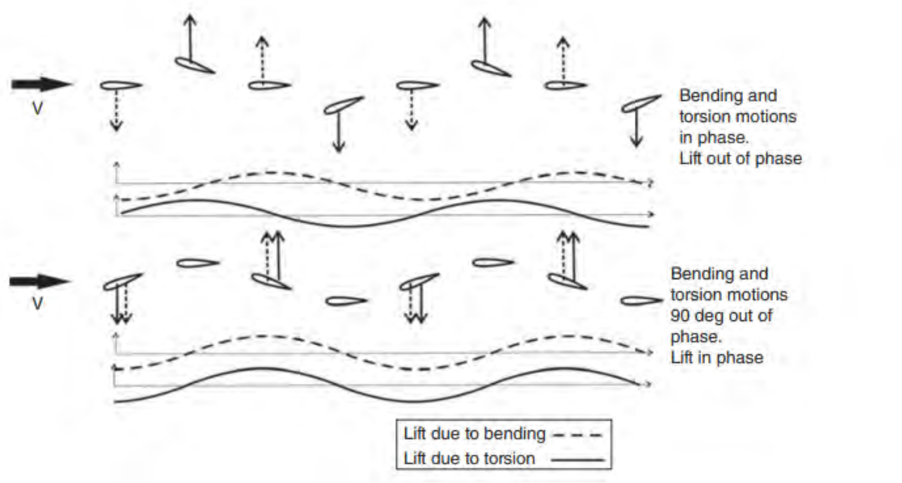


Figure 2.18: Binary Flutter Mode Phasing *Wright and Cooper [45]*

Note that even if the two modes are oscillating in the same frequency as in this example, if the unsteady aerodynamics are let to include phase lag, the phasing of the critical motion will no longer be exactly 90°. It is important to understand that in reality neither do the two motions happen in the same frequency, nor does the flutter occur by only those two modes. In fact, each mode has a different frequency and the movement of the structure includes a linear combination of all the modes,as it was seen in Section 2.2.5. However it is indeed the case that the critical modes' frequencies get closer in order for flutter to occur.
The frequency of the mode for the steady case is simply the natural frequency provided by a normal mode analysis. However, as the velocity increases, because of the aerodynamic damping and stiffness terms, the eigenfrequencies that occur from the dynamic aeroelastic equation 2.115, become different, and some modes can even become non-oscillatory. In this case those modes don't contribute in the flutter motion. Note however that a non-oscillatory mode can also be unstable and this would be the case of divergence that was seen in the previous section.

Factors that affect the Flutter Speed

There are several parameters that affect the flutter speed.

- Moving the *mass axis* more forward closer to the aerodynamic center, increases the effective mass and reduces the inertial-aerodynamic coupling, thereby increasing the flutter Speed.

- Moving the elastic axis closer to the aerodynamic center is also beneficial for flutter, since the aerodynamic forces have a smaller lever arm. However note that even if the elastic axis coincides with the aerodynamic center, flutter will still occur at some speed due to aerodynamic coupling.

- Perhaps the most important parameter that strongly effects flutter speed is the spacing between the natural frequencies of the modes. The further apart the modes are, the more the flutter mechanism is delayed to higher speeds.

- For composite structures that exhibit bending torsion coupling, it is found that positive bending-torsion coupling is beneficial for flutter speed. Remember however that a positive bending torsion coupling has divergent tendencies in its static response, so a trade-off has to be made. Generally positive bending-torsion coupling results in higher effectiveness of the aerodynamic loads, and therefore it can also improve control effectiveness, but it also increases Induced drag because of the higher lift. The influence of bending-torsion coupling is to move the *primary stiffness axis*, which refers to the locus of points that exhibit the most resistance to bending deformation *Weisshaar et al. [44]*, forward (positive coupling) or aft (negative coupling) of the elastic axis. Depending on if the bending torsion coupling is positive or negative, the resulting effects on the wing performance are nicely summarized in figure 2.19 by *Shirk et al. [35]*.

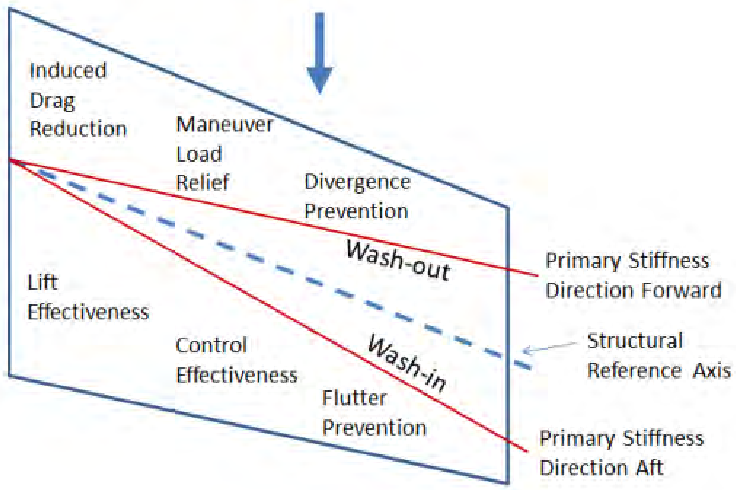


Figure 2.19: Effects of Bending-Torsion Coupling on Wing Performance *Shirk et al. [35]*

2.5 Aeroelastic Analysis with Nastran

In this section we will introduce how aeroelastic analysis is done in Msc Nastran *Nastran [34]*.

2.5.1 The Doublet Lattice Method in Msc Nastran

The theory of the DLM was presented in section 2.3. In order to determine the AIC, 3 basic matrix equations are needed :

$$\boldsymbol{w_j} = [A_{jj}]\frac{\boldsymbol{f_j}}{q} \tag{2.116}$$

which is the basic equation of the DLM that relates the dimensionless downwash w_j with the pressure of the lifting surface.

$$\boldsymbol{w_j} = [D^1_{jk} + ikD^2_{jk}]\boldsymbol{u_k} + \boldsymbol{w^g_j} \tag{2.117}$$

which is the substantial differentiation matrix of the aerodynamic grid point deflections to provide the downwash.

$$\boldsymbol{P_k} = [S_{kj}]\boldsymbol{f_j} \tag{2.118}$$

which is the integration of the pressure in order to obtain the forces on the aerodynamic grid points.

In the above equations, q is the dynamic pressure,$\boldsymbol{f_j}$ is the lifting pressure, k is the reduced frequency, $[S_{kg}$ is called the integration matrix, $[D^1_{jk}, D^2_{jk}]$ are the real and imaginary parts of the substantial differentiation matrix, $A_{jj}(M, k)$ is the influence coefficient matrix that is a function of the Mach number and the reduced frequency and P_k and u_k are the forces and DoF of the aerodynamic grid points.

Finally note that the term $\boldsymbol{w^g_j}$ is a correction of the downwash that can be implemented to introduce the effects of initial angle of attack, camber and twist. This correction will be implemented in Chapter 3 to correct the coefficient of lift (C_l) of the wing, when modeled with flat surfaces, based on the actual camber and twist of the wing surface.

Now we can combine the above equations to obtain the aerodynamic influence coefficient (AIC) matrix that relates the forces acting on the aerodynamic grid points to the displacements of the aerodynamic grid points:

$$[Q_{kk}] = [S_{kj}][A_{jj}]^{-1}[D^1_{jk} + ikD^2_{jk}] \tag{2.119}$$

2.5.2 Coupling of the Finite Element Model with the Aerodynamic Model

In general, the aerodynamic grid points don't coincide with the structural grid points. However, as we have seen in previous sections, the forces that are produced from the aerodynamic model must be applied in structural nodes, while also the downwash that is calculated by structural movement must be known in the aerodynamic element control point.
Therefore, two transformations are necessary between the two models, the first one to relate the the structural DoF to the aerodynamic DoF, and the second one to transform the aerodynamic pressure on the elements to equivalent structural node forces.
In Nastran this is called *splining*. *Splining* is an interpolation that is based on a model. It is used to solve for the deflections in the whole domain based on some discrete points, by using a theory. For example for surface splines, like the one used in this Thesis, the finite plate splining method finds for the distribution $w(x, y)$ of the deflections in the surface by using the nodal DoF by solving the respective PDEs from finite plate theory.
For the first transformation, the structural grid points are the independent ones and the aerodynamic grid points are the dependent. Splining then leads to the interpolation matrix $[G_{kg}]$ that relates the two sets:

$$\boldsymbol{u_k} = [G_{kg}]\boldsymbol{u_g} \tag{2.120}$$

where $\boldsymbol{u_k}$ are the aerodynamic DoF and $\boldsymbol{u_g}$ are the structural DoF. The transformation of the forces between the two systems is found based on the requirement that the two sets of forces are *structurally equivalent* rather than *statically equivalent*. What this means is that the two sets of forces deflect the structure equally, something that isn't necessarily true for a statically equivalent system. This is because in aerolasticity the deformations are of main interest.
Mathematically, the above conditions requires that the two sets of forces do the same virtual work on their respective deflection modes:

$$\delta W = \boldsymbol{\delta u_k}^T \boldsymbol{F_k} = \boldsymbol{\delta u_g}^T \boldsymbol{F_g} \tag{2.121}$$

Then by using equation 2.120 and dropping the virtual deflections because of their arbitrariness we obtain:

$$\boldsymbol{F_g} = [G_{kg}]^T \boldsymbol{F_k} \tag{2.122}$$

Therefore the same transpose of the interpolation matrix used in equation 2.120 is needed for the force transformation. Note however that it is possible to use different spline methods for the two transformations.

2.5.3 Static Aeroelastic Analysis

For the static aeroelasticity case, only the real part of the substantial derivative matrix D_{jk} is used from equation 2.117, as no frequency dependent motion is considered. In the same manner, the $[A_{jj}]$ is no longer frequency dependent.
If the static aeroelastic behavior of the structure under the aerodynamic loads is to be determined, as is the case during steady flight of an aircraft, then Nastran allows for the introduction of additional degrees of freedom that correspond to control surfaces and rigid body motions.
Therefore the more general equation for the downwash takes the form:

$$\boldsymbol{w_j} = [D_{jk}]\boldsymbol{u_k} + [D_{jx}]\boldsymbol{u_x} + \boldsymbol{w_j^g} \tag{2.123}$$

where $\boldsymbol{u_x}$ corresponds to extra DoF of control surface and rigid body motions, and $[D_{jx}]$ is the substantial derivative matrix of those extra points.
Let us denote with the subscript a the set of the structural DoF and with the subscript k the aerodynamic DoF.
Now let us combine the AIC Matrix obtained for the aerodynamic grid points in equation 2.119 with the transformations from the splining method found in the previous section. This results to the matrix $[Q_{aa}]$ that relates forces acting on the structural grid points to the structural deflections.

$$[Q_{aa}] = [G_{ka}]^T[S_{kj}][A_{jj}]^{-1}[D_{jk}][G_{ka}] \tag{2.124}$$

We can also include the matrix $[Q_{ax}]$ that provides the forces on the structural grid points as a result of the extra aerodynamic points:

$$[Q_{aa}] = [G_{ka}]^T[S_{kj}][A_{jj}]^{-1}[D_{jx}] \tag{2.125}$$

Finally, in order for gravity and inertial forces to be included, we will include the mass matrix and acceleration terms. Recall however that the DoF are not time-dependent in this static analysis.

The *Equations of Motion* for the static case then take the form:

$$[K_{aa} - qQ_{aa}]\boldsymbol{u_a} + [M_{aa}]\boldsymbol{\ddot{U}_a} = q[Q_{ax}]\boldsymbol{u_x} + \boldsymbol{P_a} \tag{2.126}$$

where the vector $\boldsymbol{P_a}$ contains extra external structural loads, like thermal or mechanical loads.
In the case that the structure is an aircraft, its *free-flying* nature has to be taken into account. Therefore the rigid body motions will be included. In that case we can further partition the above equation into two sets, one that contains the rigid body DoF of the *free-flying aircraft*, and a second one that contains the flexible DoF.

Divergence Analysis

The divergence speed can be also found by performing flutter analysis, as we will discuss in the next section. However if only the divergence speed is to be found, it would be more computationally effective to only solve the static problem.
Denoting with the subscript l only the flexible DoF (disregarding the possible rigid body motions), the divergence speed can be found by the following equation:

$$[K_{ll} - qQ_{ll}]\boldsymbol{u_l} = 0 \tag{2.127}$$

The above equation is transformed to classic eigenvalue problem, where the eigenvalues λ are the dynamic pressures of divergence.

$$[K_{ll} - \lambda Q_{ll}]\boldsymbol{u_l} = 0 \tag{2.128}$$

Obviously from all the eigenvalues, only the lowest is important since it is the most critical.

2.5.4 Flutter Analysis

For flutter analysis we use the full AIC matrix introduced in equation 2.119 for the aerodynamic model. In addition to the transformation of splining, we will also use an additional transformation for this case and transform the system to modal coordinates, as it was shown in section 2.2.5. Then the AIC that relates the modal coordinates to the equivalent modal forces in the structural system takes the form:

$$[Q_{ii} = [\Phi_{ai}]^T [G_{ka}][Q_{kk}][G_{ka}][\Phi_{ai}] \tag{2.129}$$

The a-set refers to the structural node DoF, and the k-set to the aerodynamic DoF. The i-set contains the normal mode vectors, and it depends on the number of modes considered. This is called *modal reduction*. Basically, not all modes are considered in the motion of the structure, because that would be extremely computationally expensive. Therefore only the first few modes are usually calculated and reserved for the dynamic analysis.
Note that empirical correction factors can also be introduced here in Nastran, however they will not be considered here.
The above matrix is now dependent upon the reduced frequency and Mach number. In practice it is calculated for a discrete number of reduced frequencies and Mach numbers that are of interest for the specific problem. The rest values that may occur during the flutter analysis are then interpolated or extrapolated based on those values.
In order for this interpolation to be accurate, the exact calculated matrices should contain the range of frequencies of interest, which are dependent upon the normal modes used, and also the velocities and Mach numbers of interest.
In order to solve perform a flutter analysis, traditionally there are two methods, that were developed independently in around the same period. Theodorsen(1935) in America in order to solve the flutter problem, introduced the aerodynamics as complex inertial terms and transformed the analysis into a classical vibration analysis but with complex arithmetic. Because flutter assumes sustained oscillations, he also introduced the artificial complex structural damping g term. This damping has no real physical meaning, it is an imaginary damping term that is necessary for sustained oscillations. Therefore, g takes negative terms when the system is damped, and becomes zero at the flutter speed. In what came to be known as the *K-Method*, flutter analysis is a double eigenvalue problem in frequency and velocity that is solved by iterating through the reduced frequency of the assumed harmonic motion until no artificial damping is required ($g = 0$).
Frazer and Duncan(1928) in England tried a different approach. They introduced the aerodynamic terms into the equations of motion as frequency dependent stiffness and damping terms. Note that those terms are slowly-varying functions of the reduced frequency, unlike in the K-Method that as mass terms they are highly sensitive to differences in reduced frequency. In this method some iteration is still needed to align the eigenvalue solution fre-

quency to the reduced frequency in each mode. This method is known as the *British Method* of flutter analysis. In Nastran, a variation of the British Method that was introduced by Hassig(1971) is used. Hassig called this method the *P-K Method* and Nastran adopted his terminology.

Here the P-K Method will be used. Note that in the P-K Method, a linear interpolation scheme is used by Nastran in order to interpolate the values of Q_{ii}.

The *fundamental equation of Flutter Analysis* with the P-K Method in Nastran is:

$$\left([M_{ii}]p^2 + ([B_{ii}] - \frac{1}{4k}\rho c V_\infty [Q^I_{ii}])p + (K_{ii} - \frac{1}{2}\rho V^2_\infty [Q^R_{ii}])\right) \boldsymbol{u_i} = 0 \tag{2.130}$$

where :

- M_{ii}, B_{ii} and K_{ii} are the modal mass,damping and stiffness matrices respectively.
- c is the reference length
- k=$\frac{\omega c}{2V_\infty}$ is the reduced frequency
- $\boldsymbol{u_i}$ is the vector of the modal participation factors
- Q^I_{ii} and Q^I_{ii} are the modal aerodynamic damping (imaginary part) and modal aerodynamic stiffness (real part) matrices respectively, that are function of Mach number and reduced frequency.
- $p = \omega(\gamma \pm i)$ is the complex eigenvalue
- $\gamma = 2g$ is the transient decay coefficient that is the two times the artificial structural damping used in the K-Method.

All the matrices are real in the above equation. Note that the reduced frequency is in fact the imaginary part of the complex eigenvalue p, multiplied by the factor $\frac{c}{2V_\infty}$:

$$k = \frac{c}{2V_\infty} Im(p) \tag{2.131}$$

Equation 2.130 for the P-K methods is then rewritten in the state space form of twice the order :

$$([A] - p[I])\,\overline{\boldsymbol{u_i}} = 0 \tag{2.132}$$

where $[I]$ is the identity matrix, $\overline{\boldsymbol{u_i}}$ now includes both the modal displacements and their derivatives and $[A]$ is given by:

$$[A] = \begin{bmatrix} 0 & I \\ -[M_{ii}]^{-1}\left([K_{ii}] - \frac{1}{2}\rho V^2_\infty [Q^R_{ii}]\right) & -[M_{ii}]^{-1}\left([B_{ii}] - \frac{1}{4k}\rho c V_\infty [Q^I_{ii}]\right) \end{bmatrix} \tag{2.133}$$

The eigenvalues of the real $[A]$ matrix are either real or complex conjugate pairs. Real roots are non-oscillatory and indicate either convergence or divergence. Therefore we see that the divergence speed can be identified by a flutter analysis as well, as discussed in the end of section 2.5.3.

The damping is expressed by the decay rate coefficient, that represents the distance travelled in chord lengths to half (or double for positive real part) the amplitude. Note that the damping expressed as $g = 2\gamma$ is a more realistic estimate of the physical damping than the parameter g used in the K-method. The solution process for the P-K Method is the following:

1. Calculate the generalized aerodynamic matrix ($[Q_{ii}]$ for k=0 and for a specified Mach number and velocity. Solve the eigenvalue problem of equation 2.132 to obtain the eigenvalues p. The non-oscillatory solutions immediately satisfy equation 2.131 and no iteration is needed for the corresponding modes. These roots indicate Divergence if they are positive.

2. For the oscillatory roots found in the previous step, an iterative process begins as follows in the next steps. Take the first oscillatory root and obtain the new reduced frequency for this iteration as :
$$k_i^j = \omega_i^j \left(\frac{c}{2V_\infty}\right) \tag{2.134}$$
where i is the mode under investigation and j is the iteration.

3. Calculate the generalized aerodynamic matrix ($[Q_{ii}]$ and with that the aerodynamic stiffness and damping matrices for the specified reduced frequency and Mach Number and solve the eigenvalue problem of equation 2.132

4. Take the reduced frequency solution closest to the initial guess (based on equation 2.131) and repeat the process until the reduced frequency converges (which is usually a few iterations) and note the corresponding damping value $g = 2\gamma_i$. The reduced frequency has converged when it doesn't change more than a specified threshold (usually 0.001) compared to the last estimate.

5. Considered the next mode of interest and repeat for all modes.

6. Then consider for the next air speed and Mach number of interest and repeat the process.

The above procedure results in a set of frequency and damping for specified air velocity and Mach number values. The flutter velocity and frequency can be determined by the first mode where the damping becomes zero.

2.6 Linear Buckling Analysis with Nastran

Buckling is a phenomenon where the structure, when under compressive load, becomes unstable and suddenly deflects excessively. In order to understand this behaviour, we consider the higher order terms of the strain-displacement relationships on the static equilibrium equations. This results in what is known as the *differential stiffness matrix* K_d. The total potential energy of the system with the addition of the differential stiffness matrix becomes:

$$U = 0.5\boldsymbol{u}^T[K_a]\boldsymbol{u} + 0.5\boldsymbol{u}^T[K_d]\boldsymbol{u} = 0 \tag{2.135}$$

In equation 2.6, $\boldsymbol{u}$ is the vector of all the DoF of the Finite Element model and K_a is the linear stiffness matrix.
For static equilibrium, the total potential must be at an extremum (note that in this case it is not necessarily a minimum since buckling is an instability). Then the following equation must be satisfied:

$$\frac{\partial U}{\partial u_i} = [K_a]\boldsymbol{u} + [K_d]\boldsymbol{u} = 0 \tag{2.136}$$

The differential stiffness matrix is a function of the internal loads on each element, and therefore a function of the applied loads. Therefore, equation 2.6 can be written in the form:

$$\Big([K_a] + P_a\overline{[K_d]}\Big)\,\boldsymbol{u} = 0 \tag{2.137}$$

where P_a is the applied load and $[K_d] = P_a\overline{[K_d]}$.
The differential stiffness matrix is also known as incremental stiffness, because it adds or subtracts from the linear stiffness of the structure, depending on if the load is tensile or compressive respectively. Linear buckling is the limiting case when the sum of the linear stiffness and the incremental stiffness is equal to zero, and so the structure 'buckles'.

Equation 2.6 is an eigenvalue problem, where the critical values of the applied loads obtainable is equal to the number of the DoF of the model. We can formulate the critical buckling loads in the form $P_{cr_i} = \lambda_i P_a$ and so for equation 2.6 to have a non-trivial solution the determinant must be zero:

$$|[K_a] + \lambda_i[K_d]| = 0 \tag{2.138}$$

The buckling eigenvalues λ_i resemble the factor by which the applied loads must be increased for the structure to become unstable. Note that only the first eigenvalue has any practical

meaning because the structure will fail prior to reaching higher eigenvalues.
In Nastran, SOL 105 is used the solve the problem of Linear Buckling Analysis and extract the eigenvalues. The critical buckling loads can then be calculated by taking the product of the applied loads with the buckling eigenvalue.
Linear buckling assumes small deflections and linear elastic element stresses. If large deflections are to occur, the non-linearity may cause the actual buckling load to be different than the linear buckling load obtained via SOL 105 and a non-linear buckling approach might be needed.

Chapter 3

Computational Framework

In this section, the model of the reference wing geometry, the uCRM 13.5 will be developed and presented.

3.1 Wing Geometry

The reference wing geometry that is chosen for the optimization studies of this thesis is the uCRM geometry, which is based on NASA's aerodynamic benchmark CRM resembling the wing shape of contemporary and future aircraft designs (*Vassberg et al. [42]*). Originally developed for CFD studies, the CRM depicts the wing in its deformed state. Brooks et al. [8] extended this model for aeroelastic analysis through an inverse design process, resulting in two variants of the uCRM. The first variant, the uCRM-9, maintains the original CRM planform, while the uCRM-13.5, which will be used throughout this thesis, serves as a higher aspect ratio derivative of the uCRM-9. The geometric details of the uCRM-13.5 are summarized in table 3.1.

The internal configuration of the wing comprises two spars, 41 evenly distributed ribs and a pair of 7 upper and lower skin stringers. Spar caps are also included in the model. The leading edge spar is located at 10 % of the local chord at the root and 35 % at the tip, while the trailing edge spar located at 60 % chord. Both the spar caps and the stringers are modeled with rectangular cross sections.

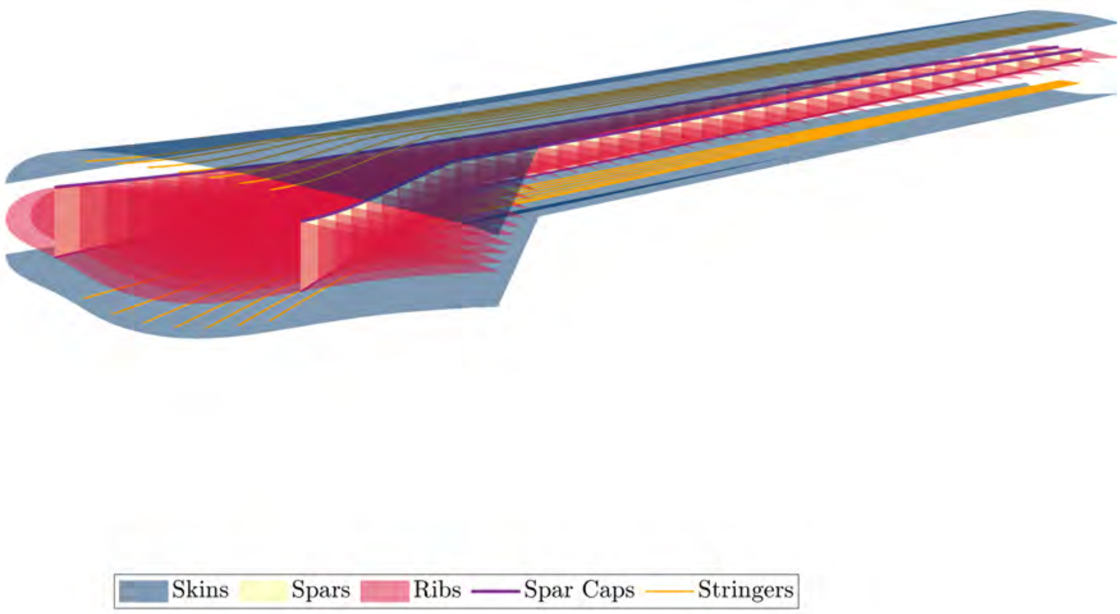
It's worth noting that the number of stringers in this design is relatively low compared to other reference configurations. This deliberate choice aims to reduce the support against buckling for the composite skins. Such a design consideration could potentially offer advantages from an aeroelastic tailoring perspective in subsequent stages.

Table 3.1: uCRM-13.5 Geometry Data

Entity	Value
Wingspan	72 m
Root Chord	11.02 m
Tip Chord	2.06 m
Wimpress Reference Area	384.05 m^2
Aspect Ratio	13.5
Taper Ratio	0.25
Quarter Chord Sweep	35°
Yehudi Chord	7.56 m
MAC	5.36 m

An exploded view of the wing geometry is shown in figure 3.1, where the spar caps and stringers are illustrated as curved lines.

Figure 3.1: uCRM-13.5 exploded view of internal and external geometry



3.2 Finite Element Model

The Finite Element model of the wing is constructed using the Outer Mold Line (OML) of the uCRM-13.5 combined with the previously discussed internal configuration. More specifically, the skins, ribs and spars are modeled with 4-noded quadrilateral shell elements (CQUAD4 in Nastran), that are proven to provide relatively good accuracy, while being computationally cost-effective compared to 3D volume elements.
On the other hand, the stringers and spar caps are modeled via 1D beam elements (CBEAM in Nastran) with rectangular cross sections, which is common practice due to their relatively small cross-sectional area compared to their length that allows the simplification of the model without significant drop in the accuracy.
The model comprises a total of 16,355 elements, resulting in 84,222 DoF. Table 3.2 summarizes the number of elements used for each wing component.
The wing structure is assumed to be fixed at its root. To achieve this, all relevant DoF in the Finite Element model are constrained in both translational and rotational motions.

Table 3.2: Number of elements for each major component

Component	No. of Elements	Type of Element
Skins	9480	CQUAD4
Ribs	4109	CQUAD4
Spars	606	CQUAD4
Stringers	1680	CBEAM
Spar Caps	480	CBEAM

The internal structure of the wing is considered to be made of Aluminum 7075 while the skins are assumed to be manufactured of the Hexcel IM7/8552 UD composite material *Marlett [31]*, with a ply thickness of 0.1828 mm. The elastic and strength properties of these materials are detailed in table 3.3.
In order for the structural representation of the wing to be more complete, non-structural external masses of the engine and fuel have been included to the model via concentrated mass elements (CONM in Nastran). The engine mass corresponds to that of a GE90 engine, weighting approximately 7500 kg, while the fuel load considered is 55 000 kg, consistent with the masses used in *Brooks [7]*.

Table 3.3: Material Elastic and Strength Properties

Material	Property	Symbol	Unit	Value
Aluminum 7075	Young's Modulus	E	GPa	71.7
	Shear Modulus	G	GPa	27
	Poisson's Ratio	ν_{12}	–	0.33
	Density	ρ	kg/m^3	2810
	Yield Strength	σ_y	MPa	500
Hexcel IM7/8552 Ply	Young's Modulus (1-axis)	E_{11}	GPa	158.51
	Young's Modulus (2-axis)	E_{22}	GPa	8.96
	Poisson's Ratio	ν_{12}	–	0.315
	Shear Modulus (1-2 plane)	G_{12}	GPa	4.68
	Shear Modulus (1-3 plane)	G_{13}	GPa	4.68
	Density	ρ	kg/m^3	1590
	Strength (Tension, 1-axis)	X_t	MPa	2500
	Strength (Compression, 1-axis)	X_c	MPa	1531
	Strength (Tension, 2-axis)	Y_t	MPa	640.5
	Strength (Compression, 2-axis)	Y_c	MPa	285.7
	Shear Strength	S	MPa	53.5

The concentrated mass elements connection to the structure is achieved with rigid body elements (RBE3 in Nastran), that allow for the weighted connection of the mass to multiple nodes. The engine mass is divided into 13 concentrated masses of 577 kg connected with with the element at the center of mass of the engine via RBE2 links. The positions of these masses are taken directly from Brooks work for the uCRM-13.5. The concentrated mass on the center of mass of the engine is then connected to the wing structure at 8 nodes via an RBE3 connector.
Regarding the fuel mass, each concentrated mass element is bounded by two ribs and positioned between the front and rear spars. The connection to the structure is made using eight RBE3 links at rib-spar junction nodes. The weight of each concentrated mass is calculated to be analogous to the volume inside of each rib bay and to correspond to the total fuel weight of 55 000 kg. Finally, the fuel mass is considered to be stored up until roughly 70% of the span of the wing, which corresponds to a total of 27 concentrated mass elements. Figure 3.2 illustrates the model of the wing including the masses of the engine and fuel and their connections to the structure. The size of the masses in the plot is proportional to the weight of each concentrated mass.

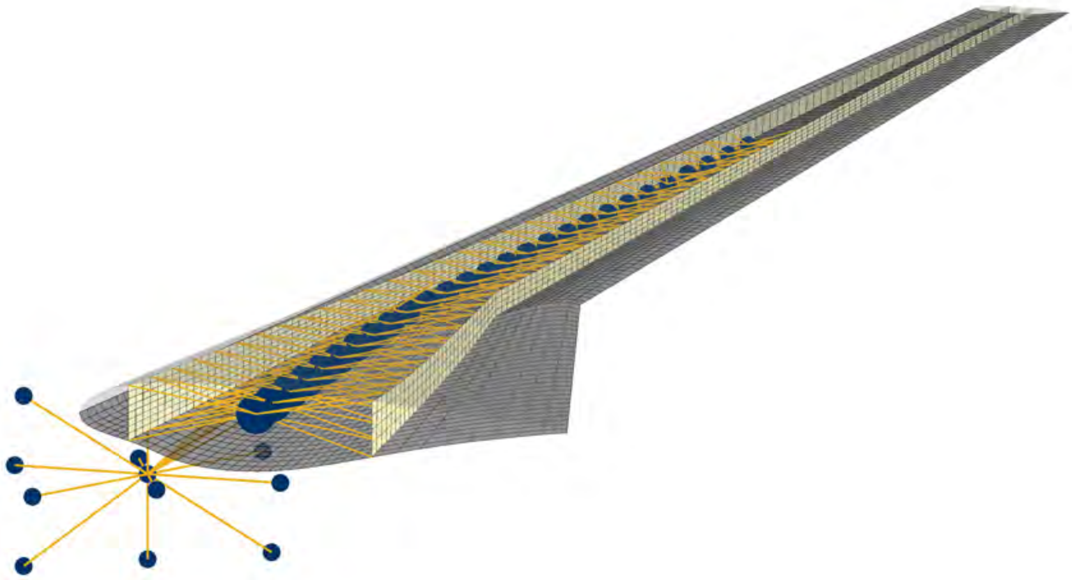


Figure 3.2: uCRM-13.5 with fuel and engine concentrated masses

3.3 DLM Aerodynamic Model

In order to model the aerodynamics of the wing with Nastran, the generation of a flat lifting surface is necessary. The numerical method employed to solve the lifting surface problem, as discussed in Chapter 2, is the DLM, which is suitable for both static and flutter aeroelastic calculations.

The wing OML is approximated with 2 rectangular surfaces, one from the root and until the yehudi break, and the other from the yehudi break to the wing tip. The first surface is modeled via a 20×20 element aerodynamic mesh, while the second with a 60×20 mesh. This results in a total of 1,600 box elements covering the entire surface, with 80 elements in the spanwise direction and 20 in the chordwise direction. Figure 3.3 provides a visual representation of the aerodynamic surface, along with the wing's OML for reference.

To ensure the aerodynamic model accurately represents the wing, it's essential to account for the wing surface's camber and twist. As discussed in Chapter 2, in Nastran this can be done by implementing the W2GJ correction matrix for the local angle of attack of each box of the DLM mesh. This correction method has been also used in *Dillinger [12]*.

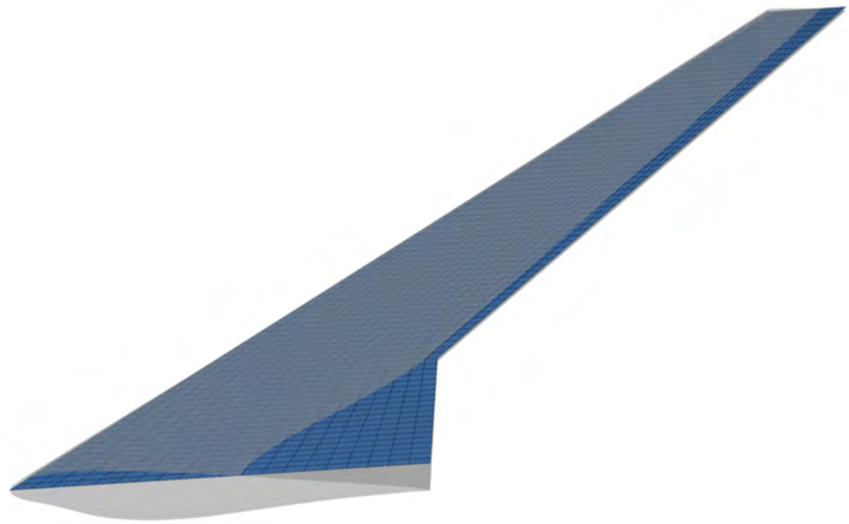


Figure 3.3: Aerodynamic surface for DLM model of the uCRM-13.5

For the purpose of calculating the distribution of twist and camber, the airfoil section data of the OML are extracted at 82 points along the span of the wing. At each airfoil section, the chord line and twist angle are calculated . Subsequently, the twist is removed from the section and the camber line is formed based on the curvature of the upper and lower lines of the untwisted section. An example untwisted section along with the calculated camber line is shown in figure 3.4.

The total local angle of attack is then determined along each airfoil section as the sum of the inclination due to camber and that due to twist. Recalling from Chapter 2 that the W2GJ downwash correction is applied at the 3/4 point of each box, a linear interpolation scheme is used to calculate the local angle of attack at the desired point based on the data points available. The resulting downwash correction of the DLM mesh is illustrated in figure 3.5.

Figure 3.4: Example airfoil section of the uCRM-13.5

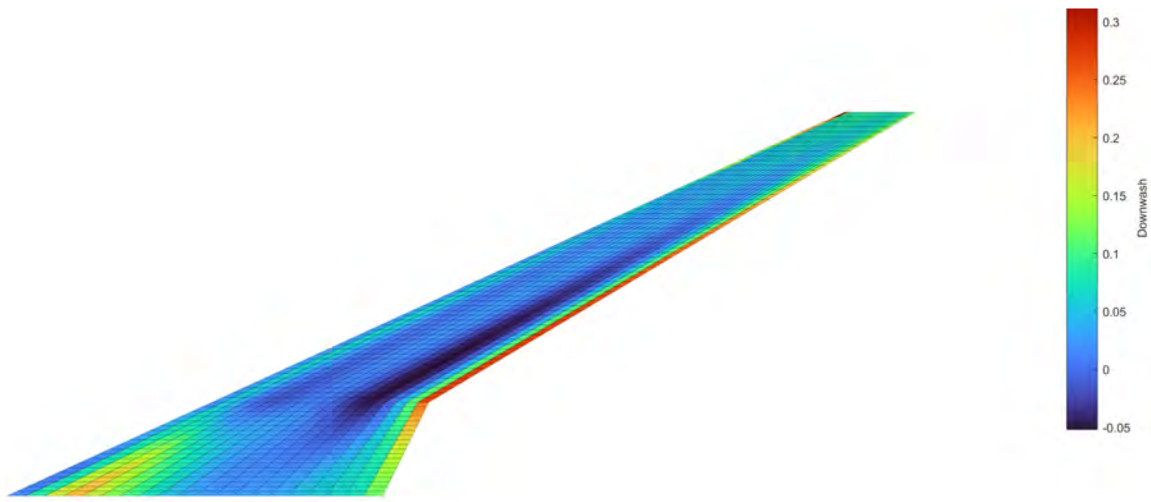


Figure 3.5: W2GJ local angle of attack correction for the uCRM-13.5 wing

3.4 Coupling

The final step for the wing model to be complete is to couple the FEM mesh to the DLM mesh. As discussed in Chapter 2, this is done with the use of splines.
An infinite surface spline is used to transfer the loads acting on the aerodynamic nodes to structurally equivalent forces on a set of structural nodes. Then, the information for

the displacements of the structural nodes is transferred back to the aerodynamic mesh in a similar manner.

The choice of structural nodes that are to be used for the surface spline plays a crucial role for the accuracy and quality of the coupling. In order to allow for better load distribution on the structure, as is common practice, structural nodes located at the spars and ribs are chosen for the surface spline. On the other hand, all aerodynamic grid points are used for the spline. Finally, the structural nodes used for the spline are presented in figure 3.6.

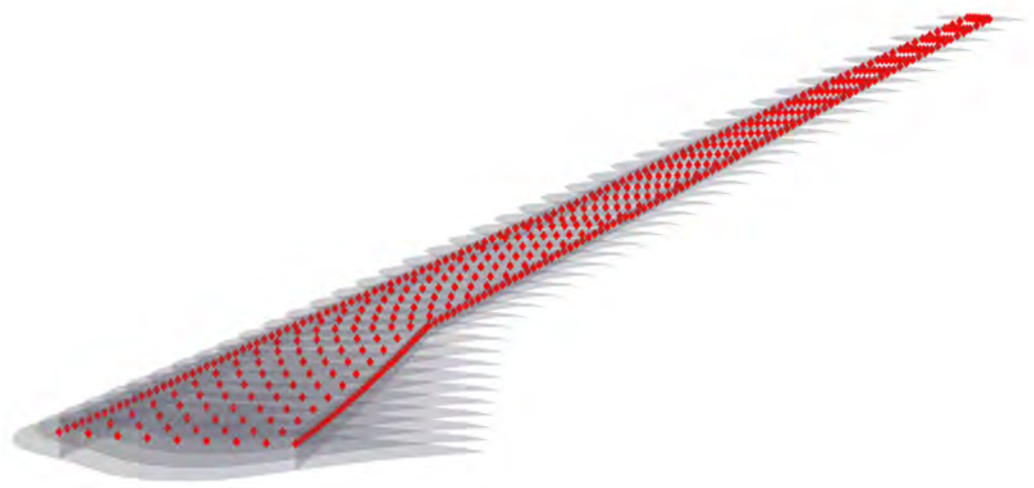


Figure 3.6: Structural nodes used for splining in Nastran

Chapter 4

Structural Optimization of the uCRM-13.5

This chapter deals with the structural optimization of the uCRM-13.5 wing in order to minimize the weight. The optimization algorithm that will be used is the MIDACO algorithm that was introduced in Chapter 1. First, the development of the optimization framework will be addressed in detail. Subsequently, the resulting structural sizing of the wing will be presented and the aeroelastic performance of the wing will be discussed.

4.1 Design Variables Definition

The design variables play a crucial role in the optimization process. The design space needs to be sufficiently large to ensure meaningful results, but without being too computationally expensive.

In the study of this chapter, the thicknesses of the major components of the wing structure constitute the design variables of the weight minimization problem. For the aluminum spars and ribs, one design variable for the thickness is sufficient, while for the aluminum spar caps and stringers, the width as well as the thickness of the cross section needs to be specified. In an effort to keep the design space manageable, the width of the bars is kept constant during the optimization process and therefore it doesn't constitute a design variable for the optimizer.

As for the composite skins, the thickness is determined by the number of $0°$, $90°$, $45°$ and $-45°$ plies that make up the laminate. Note that a $0°$ ply is defined to be parallel to the leading edge of the wing. In a laminate structure, both the thickness as well as the stacking

sequence are important, however assigning a design variable to every ply would make the design space too large to be solved with gradient-free optimization techniques.
For this reason, a symmetric and balanced layup of $[(45_{n_{45}}/0_{n_0}/-45_{n_{-45}}/90_{n_{90}})_S]_S$ is used for every laminate considered, which reduces the design space to 3 per laminate. These are the number of 0°, 90° and ±45°, assuming that for every 45° ply, a −45° ply is also added for balancing.
The wing structure is further divided into 6 regions along the span, in order to provide a reasonably large design space for the thickness distribution of the wing. The regional division of the wing is visualized in figure 4.1. In addition, the composite skins are allowed to have different laminates for the upper and lower part, due to notable thickness variations observed between the two.
Note that initially the wing had been divided into 4 regions, however it was deemed beneficial to increase this number to 6. The original division had the boundary of the first with the second region laying in the Yehudi break, which features the highest stresses of the structure. This was problematic because the abrupt change of thickness in that region was amplifying the already high stresses due to the break, effectively necessitating the use of more material. This adjustment not only resolves this issue, but also facilitates a smoother thickness distribution along the wing overall.

Figure 4.1: Division regions for the optimization of the uCRM-13.5

The total number of design variables is 60, from which 36 are of type integer while the rest are continuous. The design variables along with their upper and lower bounds during the optimization process are summarized in table 4.1.

Table 4.1: Design Variables

Design Variable	Lower Bound	Upper bound	Type	Total in Model
Spars thickness	1 mm	25 mm	Continuous	6
Ribs thickness	1 mm	35 mm	Continuous	6
Stringers thickness	1 mm	20 mm	Continuous	6
Spar caps thickness	1 mm	20 mm	Continuous	6
No. of $0°$ plies	0	10	Integer	12
No. of $90°$ plies	0	10	Integer	12
No. of $\pm45°$ plies	1	10	Integer	12

Finally, the widths of the rectangular sections for the spar caps and stringers are tapered along the span of the wing and depicted for each region in table 4.2. These values have been determined after some intuition had been gained for the problem from initial runs.

Table 4.2: Width Values for Each Region

	Region 1	Region 2	Region 3	Region 4	Region 5	Region 6
Spar caps width	12 cm	12 cm	9 cm	7 cm	4 cm	1 cm
Stringers width	13 cm	13 cm	9 cm	7 cm	4 cm	1 cm

4.2 Objective Function and Constraint Definition

As discussed, the focus of this chapter is on the structural sizing of the wing components. Consequently, the objective function of the optimization problem in this chapter is to minimize the structural mass.

To ensure that mass minimization is feasible without compromising safety or structural integrity, constraints need to be imposed in the optimization problem. Throughout this study, these constraints include the static strength and buckling safety of the wing under a critical aerodynamic load equivalent to a 2.5g pull-up maneuver, as well as ensuring a sufficiently high flutter velocity. To verify that the constraints are being met for a specific design, a total of three analyses are conducted during each optimization cycle..

Static Strength Constraint

First, a static aeroelastic analysis is performed using SOL 144 of Nastran at the critical maneuver condition, which is summarized in table 4.3. The Maximum Take-off Weight (MTOW) of the aircraft is estimated at around 297 tonnes as per *Brooks [7]*. The lift generated by the wings will need to be equal to 2.5 times the MTOW for the 2.5g maneuver condition.

In order to achieve sufficient lift, an angle of attack of 4.5° was found to be required. Although this angle may need adjustment due to passive load alleviation from wing deformation, iteratively finding the correct angle of attack for each individual run would significantly increase the computational cost. Consequently, the angle of attack will be kept constant during the optimization, introducing a minor error that is expected to be mitigated with the use of a safety factor.

Table 4.3: 2.5g Maneuver Condition

Condition	Lift Constraint	Mach Number	Altitude (ft)
2.5 Maneuver	2.5 MTOW	0.64	0

After the analysis is complete, the stresses and composite first-ply failure indices are extracted for all elements of the model. Regarding the aluminum components, the stresses are used to form failure indices similar to those of the composite skins , by using the formula Failure index= $\frac{\sigma_{max}}{S_{yield}}$.

Rather than taking the most critical value, stress values are aggregated using the Kreiseelmeier-Steinhauser (KS) function *Kreisselmeier and Steinhauser [28]*. The KS function uses all failure index values of the model and compresses them into one single scalar value. The KS function has the following form :

$$KS(g_j) = g_{max} + \frac{1}{\rho} ln[\sum_{j}^{N} exp(\rho(g_j - g_{max}))] \leq 1 \tag{4.1}$$

A value greater than 1 would indicate that an element of the model has failed elastically. The elements that exhibit the highest stresses will have greater weight than others, while

the relevant weight can be controlled via the tuning parameter ρ.
Lower values of this parameter (< 80) will indicate that individual highly stressed elements will have less weight, while higher values (> 100) will result in the KS value to converge to the most critical value.
Here, due to some stress concentrations from regional divisions that are a result of the modeling rather than the geometry, an intermediate value of 50 is chosen for the tuning to strike a balance between the two.
Finally, a safety factor of 1.5 is used for all stress values of the model prior to the KS aggregation, meaning that a KS value of 1 in reality would indicate a failure index has exceeded the value of 0.67.

Buckling Constraint

After the static aeroelastic analysis has been completed, the loads acting on the structure are extracted and used as an input for a buckling analysis with SOL 105 of Nastran.
The first buckling eigenvalue less than 1 would indicate that buckling has occurred in the structure. A safety factor of 1.5 is again added and the first buckling eigenvalue is constrained to be above 1.5 for a buckling-free structure.

Flutter Constraint

Finally, the wing dynamic aeroelastic instability is investigated with SOL 145 of Nastran. Regarding this constraint, the flutter velocity of the candidate design needs to be higher than the dive speed of the aircraft with a factor of safety of 1.2. The dive speed is taken as 221 m/s Equivalent Air Speed (EAS) from reference designs. The flutter analysis parameters are shown in table 4.4.
Since the flutter analysis is calculated at an altitude of $35000ft$, the flutter velocity needs to be compared to the True Air Speed (TAS) at that altitude rather than the EAS which corresponds to sea level. This is done by using the formula $EAS = TAS \times \sqrt{\frac{\rho}{\rho_0}}$, which results in a velocity of 464.4 m/s.

The P-K method is chosen to calculate the flutter velocity with Nastran, and 50 user defined Mach-reduced frequency (MK) pairs are employed to interpolate the values of the frequency dependent aerodynamic loads. The MK pairs represent conditions where the AIC matrices used in the flutter analysis are explicitly computed. They are selected as such to include the

Table 4.4: Flutter Parameters

Condition	Dive Speed(TAS)	Mach Number	Altitude (ft)
Flutter	464.4 m/s	0.8	35 000

entire range of the first 16 eigenfrequencies of the wing and velocities up to 800 m/s, while the Mach number is set at 0.8.

To summarize, the objective function as well as the complete set of constraints is depicted in table 4.5.

Table 4.5: Objective Function and Constraints summary

Objective function: Minimize Wing Mass	
Under the constraints :	
Constraint Type	Limit Value
KS(FI), Composite skins	≤ 1
KS(FI), Aluminum Spars	≤ 1
KS(FI), Aluminum Ribs	≤ 1
KS(FI), Aluminum Spar Caps	≤ 1
KS(FI), Aluminum Stringers	≤ 1
First Buckling Eigenvalue	≥ 1.5
Flutter Speed	$\geq 1.2 \cdot V_{dive}$

4.3 Optimization Parameters

As previously discussed, the optimization algorithm that has been chosen for the current work is the MIDACO solver. The optimization process followed involves several runs with different starting points depending on the last solution found by the solver. In addition, the internal parameter FOCUS is utilized to achieve faster convergence. This parameter is increased during the optimization process to allow the optimizer to 'focus' more near the current best solution and find a refined solution.

The accuracy remains constant throughout the optimization process, set at 0.01. The accu-

racy set determines how the constraint violations are handled. This accuracy is chosen due to the fact that a safety factor of 1.5 is already used, therefore solutions that only violate a constraint by a maximum of 0.01 shouldn't be excluded as useful and viable solutions.
The optimization procedure and the parameters are summarized in table 4.6.

Table 4.6: Optimization procedure and parameters

Run	Iterations	Accuracy	FOCUS	Starting Point
1	400	0.01	0	Random
2	200	0.01	30	Run 1
3	200	0.01	60	Run 2
4	200	0.01	100	Run 3

4.4 Optimization Results

Objective function and constraints result

The results of the optimization procedure described in the previous section for the wing mass optimization are depicted in table 4.7.

Table 4.7: Optimization results

Property	Value
Wing Mass	16920 kg
KS(FI), Upper skin	0.945
KS(FI), Lower skin	0.985
KS(FI), Aluminum Spars	0.959
KS(FI), Aluminum Ribs	0.856
KS(FI), Aluminum Spar Caps	1.007
KS(FI), Aluminum Stringers	1.000
First Buckling Eigenvalue	1.492
Flutter Speed	474 m/s

It can be seen from the above results that the critical constraints for the mass optimization are the buckling constraint and bar failure. In addition, these values slightly violate the constraints due to the chosen accuracy level discussed earlier.
To visualize the results, starting with the 2.5g maneuver analysis, the composite failure index

distribution of the upper (left) and lower (right) skins is presented in figure 4.3.
Similarly, figure 4.2 shows the Von Mises stress failure index distribution of the aluminum components. Note that the real values of the failure indices are shown, without the safety factor.

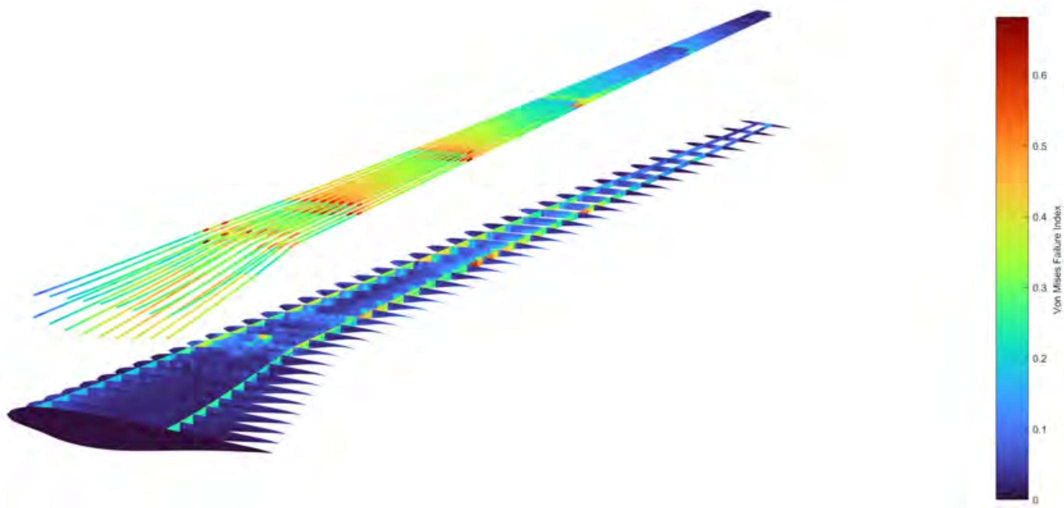


Figure 4.2: Von Mises stress distribution for aluminum components

Figure 4.3: Composite failure index for upper (left) and lower (right) skins

From the above figures, several observations can be made.

- The regional division of the wing is visible through the stress distribution, where high stresses occur in the bounds of adjacent regions.

- The Yehudi break of the wing features high stress concentrations not associated with regional division.

- The most critical region for all components except the spars is region 2, which includes the Yehudi break. However for the spars, the highest stresses appear near the limits of subsequent regions (4 and 5).

- For all constraints it can be seen that individual highly stressed elements are driving the design process, rather than a very smooth stress distribution. This is presumably due to the 6 region division of the wing that only allows for step-wise variations of the thickness between regions.

The resulting deflected shape of the wing in the 2.5g flight condition is depicted in figure 4.4. The maximum deflection is at the wing tip as expected, with a value above 5m, which is natural for a very flexible high aspect ratio wing under extreme loading.

Additionally, the coefficient of pressure distribution along the aerodynamic surface is presented in figure 4.5. The distribution of the coefficient of pressure seems consistent with

Figure 4.4: Deflected shape of the wing

that of a typical wing. The resulting lift distribution is nearly elliptical, aligning well with theoretical expectations.

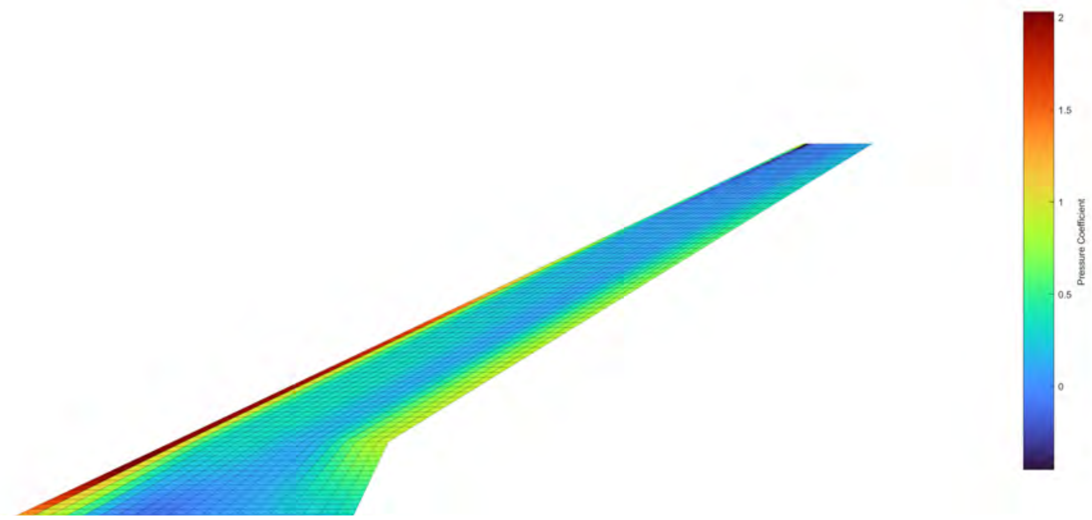


Figure 4.5: Coefficient of pressure distribution

Regarding the buckling constraint, the upper skin is generally the primary component prone to buckling due to the large compressive forces that act there during flight. During the optimization process, the critical buckling mode shifted between regions depending on the skin thickness.
In the optimal design, the most critical buckling mode is observed in the first region of the upper skin. The shape of the mode is extracted from Nastran and depicted in figure 4.6.

Figure 4.6: Critical buckling mode of optimal design

Finally, concerning the flutter constraint, it is useful to first visualize the eigenmodes of the wing. The first 9 eigenmodes and their respective eigenfrequencies are illustrated in figure 4.7.
The first mode is a torsion mode, while the rest are mainly different types of bending modes. Due to the flexibility of the wing along with the increased mass (also fuel and engine mass), the eigenfrequencies are generally low, with the first one being at just 0.629 Hz.

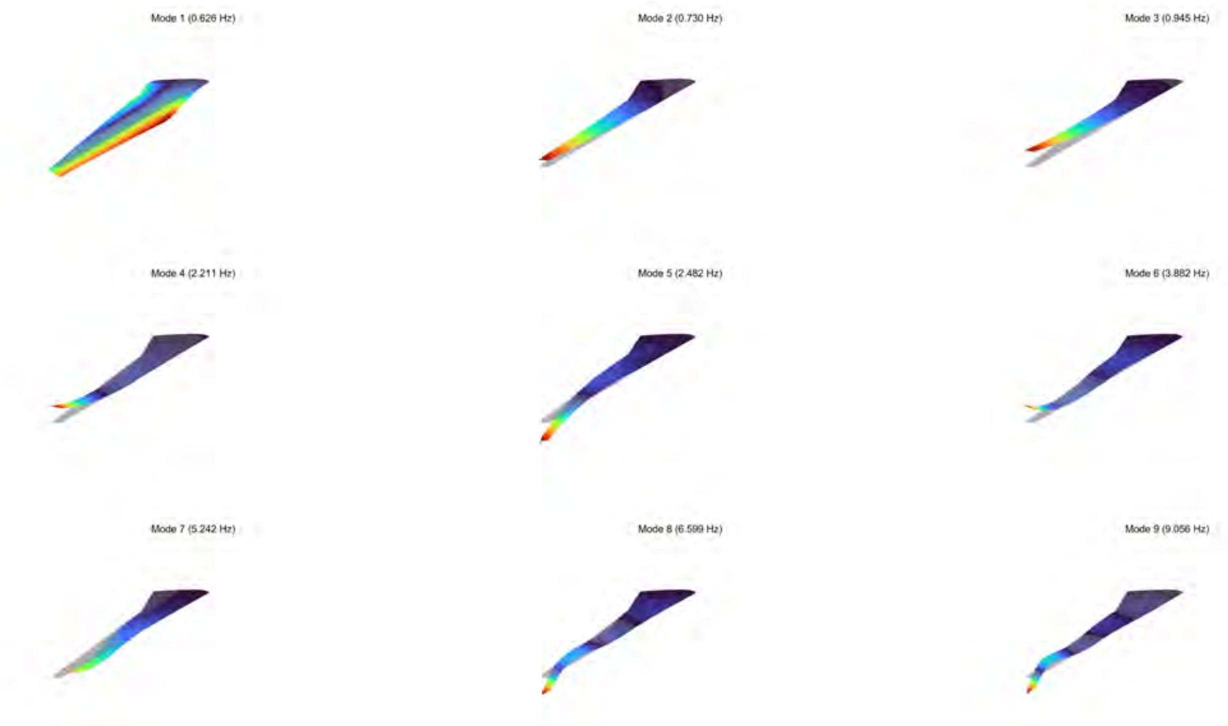
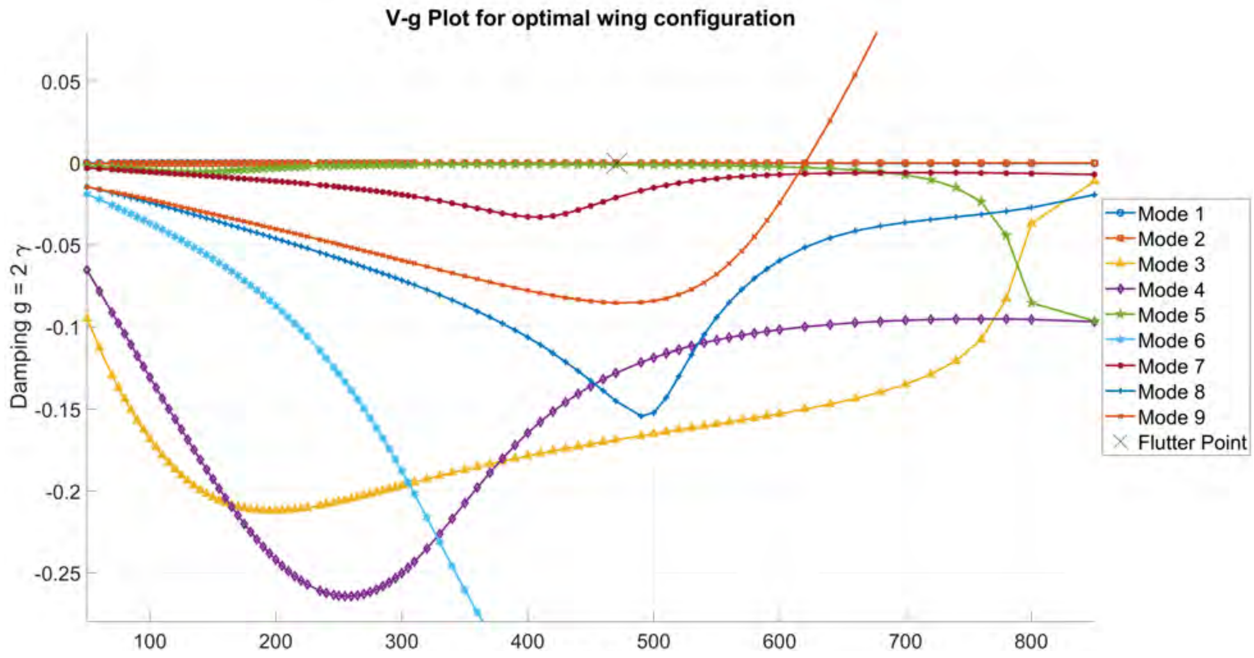


Figure 4.7: First 9 eigenmodes of optimal wing configuration

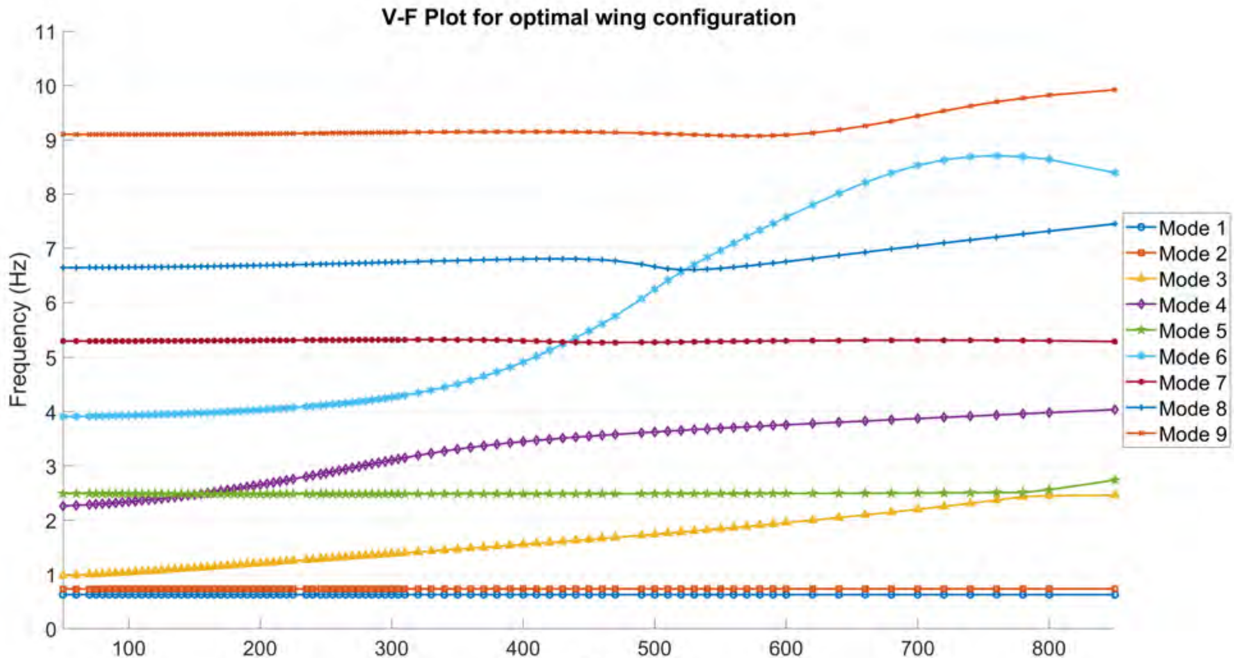
To better investigate the behavior of the wing during flutter, the velocity-damping factor (V-g) and velocity-frequency (V-F) plots are presented for the first 9 modes in figure 4.8.

From the V-g and V-F plots we can deduce the following :

- The mode that becomes unstable first is the second mode (bending mode) when the damping factor becomes positive at a velocity of 474 m/s and a frequency of 0.731 Hz. The mode remains unstable, however only by a small amount.
- The first mode (torsion mode) is stable throughout the whole range of velocities, however the damping factor retains only a small negative value.
- Mode 9 becomes unstable at a higher velocity of 640 m/s much more violently. presumably due to a convergence of the frequencies with the sixth mode.



(a) V-g Plot



(b) V-F Plot

Figure 4.8: V-g and V-F plots for optimal wing configuration

Design variables result

The configuration of the wing with the lowest mass is summarized in tables 4.8 for the aluminum components and table 4.9 for the composite skins.

Table 4.8: Optimal wing configuration - aluminum components

	Region 1	Region 2	Region 3	Region 4	Region 5	Region 6
Spars thickness	6.8 mm	7 mm	4.7 mm	2.5 mm	1.5 mm	1 mm
Ribs thickness	24.8 mm	16.6 mm	5.3 mm	2.8 mm	2 mm	1.5 mm
Spar caps thickness	8.5 mm	10 mm	5.8 mm	4 mm	2 mm	1 mm
Stringers thickness	9.8 mm	14.6 mm	6.5 mm	4 mm	3.8 mm	1 mm

Table 4.9: Optimal wing configuration - composite skins

Upper Skin						
	Region 1	Region 2	Region 3	Region 4	Region 5	Region 6
No. of 0° plies	7	5	3	1	2	1
No. of 90° plies	10	2	3	4	2	1
No. of ±45° plies	4	7	7	5	3	3
Total in stack	100	84	80	60	40	32

Lower Skin						
	Region 1	Region 2	Region 3	Region 4	Region 5	Region 6
No. of 0° plies	4	7	5	3	0	0
No. of 90° plies	2	4	3	1	1	1
No. of ±45° plies	2	5	3	1	3	2
Total in stack	40	84	56	24	28	20

The thickness distribution of the skins based on the above results is better visualized in figure 4.9.

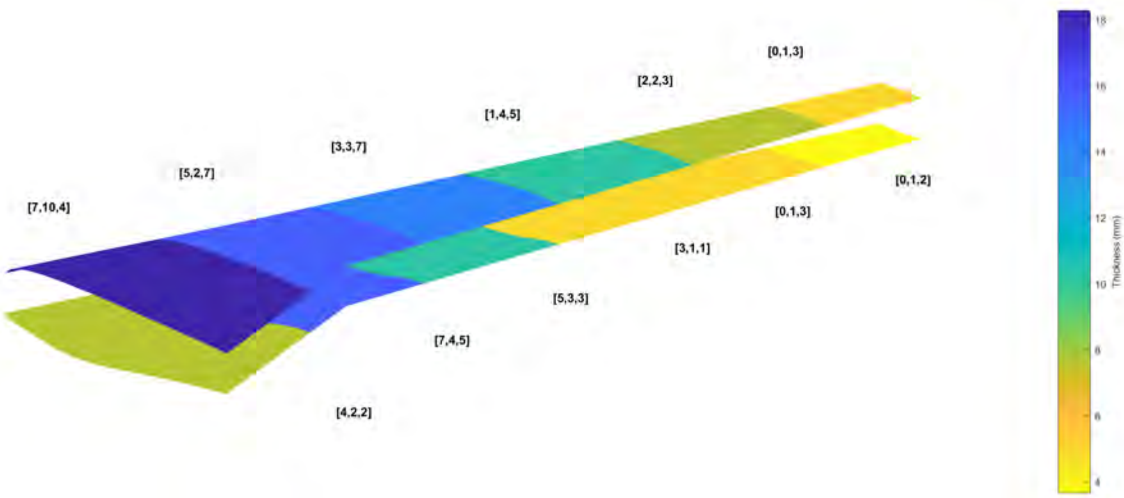


Figure 4.9: Thickness distribution of composite skins for optimal design

Several insights can be drawn from the resulting thickness distribution based on the constraints:

- The thickness distribution features the expected tapper from root to tip, with region 2 near the Yehudi break requiring the most material.
- Stringers and spar caps, especially in region 2, are very prone to failure and are one of the main drivers of the weight optimization.
- Ribs in the first 2 regions require a lot of material to prevent buckling failure. Additionally, it was observed that an excessive reduction in the thickness of the ribs in these regions could potentially trigger flutter of the first mode.
- Regarding the skins, the upper part requires higher thickness, especially in region 1, to meet buckling constraints. As for the lower skins, region 2 is prone to composite ply failure, while the rest regions are mostly limited because of bar failure due to load redistribution.

Chapter 5

Aeroelastic Tailoring of the uCRM-13.5 via Tow-steering

Throughout this study, the composite plies comprising the skins were previously assumed to be made out of unidirectional CFRP material and tailoring was a matter of determining the number and orientation of these plies for each region. However, in this chapter, we will introduce tow-steering into the design problem, allowing the fibers to follow curvilinear paths along the wings.

The impact of tow-steering will be explored across various objectives. Initially, we will incorporate tow-steering into the weight minimization problem addressed in the previous chapter. Subsequently, we will conduct single-objective optimizations of flutter velocity and the KS function of the composite skins, while keeping the wing weight fixed. This will allow us to evaluate the influence of fiber orientation alone on certain design constraints.

5.1 Skins Tow-steering Definition

To implement tow-steering on the composite skins, we need to determine the paths of the fibers. In reality, each ply can be steered individually, however this would make the design space impractically large while also increasing the complexity. In addition, it would be convenient to retain the symmetric and balanced laminate with the same variables as defined previously.

For these reasons, we adopt a simplified approach where the 'core' laminate for each region is defined in the same manner as before, specifying the number of plies oriented at 0°, 90°, and ±45°, and tow-steering is considered to uniformly rotate all plies in the stacking sequence

by a certain angle defined by a single steering path. This means that all 0° plies are aligned parallel to the steering path, while the 90° plies are perpendicular to it. Locally, plies are constrained to orientations of 0°, 90°, and ±45° relative to the steering path, but globally, the orientation can take any value.
Regarding the steering path, we use an one-dimensional piecewise linear variation of the orientation angle between the six regions of the upper and lower skins, as defined by equation 5.1, which was also introduced in Chapter 1.

$$\theta(x) = \frac{(\theta_2 - \theta_1)x}{L} + \theta_1 \tag{5.1}$$

The axis used for the steering coincides with the leading edge of the wing, which is the same as the one used to define the ply orientations in general. In essence, a 0° ply with no local steering will be aligned with the leading edge.
To implement steering within the Finite Element model of the wing, we individually determine the orientation angle of the composite material for each element of the skins. This is achieved by computing the appropriate angle at the centroid of each element based on the steering path, using equation 5.1.
The reference length L is determined for each region as the maximum distance of an element centroid from one boundary to the next.However, it's important to note that the region boundaries are not necessarily perpendicular to the steering axis and therefore the steering angle design variables are satisfied only on the outermost elements of each region.
The resulting steering variables are the angles at the bounds of each region of the upper and lower skins. This makes up a total of 7 design variables for the upper skin path and another 7 for the lower.

5.2 Wing Mass Minimization with Tow-Steering

5.2.1 Optimization Problem Definition

The optimization problem for this section is formulated in a similar manner as presented in Chapter 4, albeit with the inclusion of 14 additional design variables related to tow-steering. These variables expand the existing design space, resulting in a total of 74 design variables under consideration.

The objective function is the wing structural mass and the constraints remain consistent with those outlined in Table 4.5. Table 5.1 provides a summary of the design variables for this study along with their numerical bounds.

Regarding the boundaries set for the steering angles, it's important to recall that the local ply orientations are constrained to values with increments of 45°. Consequently, it is reasonable for the steering angles to fall within the range of $[-45°, 45°]$, which is also beneficial in terms of decreasing the feasible design space.

The steering angle design variables are chosen to be of type integer, essentially discretizing the angle values with a step size of 1° for simplicity purposes. This discrete nature of the design variables does not limit the angle values that each individual element can have, since the latter is calculated using equation 5.1.

Table 5.1: Design variables for wing mass minimization with tow-steering

Design Variable	Lower Bound	Upper bound	Type	Total in Model
Spars thickness	1 mm	25 mm	Continuous	6
Ribs thickness	1 mm	35 mm	Continuous	6
Stringers thickness	1 mm	20 mm	Continuous	6
Spar caps thickness	1 mm	20 mm	Continuous	6
No. of 0° plies	0	10	Integer	12
No. of 90° plies	0	10	Integer	12
No. of ±45° plies	1	10	Integer	12
Upper skin steering angles	$-45°$	45°	Integer	7
Bottom skin steering angles	$-45°$	45°	Integer	7

Finally, in regard to the optimization parameters, these are the same as the ones used in the previous Chapter shown in table 4.6 and won't be repeated here.

5.2.2 Optimization Results

The resulting wing minimized mass and constraint values are summarized for the current optimization in table 5.2.

Table 5.2: Weight optimization with tow-steering results

Property	Value
Wing Mass	14805 kg
KS(FI), Upper skin	0.605
KS(FI), Lower skin	0.775
KS(FI), Aluminum Spars	0.951
KS(FI), Aluminum Ribs	0.856
KS(FI), Aluminum Spar Caps	1.009
KS(FI), Aluminum Stringers	0.999
First Buckling Eigenvalue	1.491
Flutter Speed	492 m/s

The addition of tow-steering has led to a 12.5% improvement in weight, which is a considerable improvement. The optimizer is able to decrease the weight mainly by improving the buckling constraint on the upper skins and redistributing the load in a way to prevent failure of the spar caps and stringers. However, the critical constraints still revolve around the buckling of the upper skin and potential bar failure, as observed in the unsteered case of Chapter 4.

Interestingly, although the number of plies has been reduced (see table 5.5 below), the peak composite failure index of both the upper and lower skins have been improved by a significant margin compared to the unsteered case. In addition, the resulting wing configuration also features a small improvement on the flutter velocity.

The Von Mises failure index distribution on the aluminum parts is illustrated in figure 5.1, while the composite first ply failure failure index distribution in figure 5.2.

Macroscopically, it is evident that there are no significant differences observed in the stress distribution in either case, with the critical region once again being near the Yehudi break (Region 2) and stress concentrations appearing on region boundaries.

Regarding the buckling response, the critical buckling mode remains the same as the one illustrated in figure 4.6 of Chapter 4, with buckling occurring on the first region on the upper skin owing to significant compressive loads.

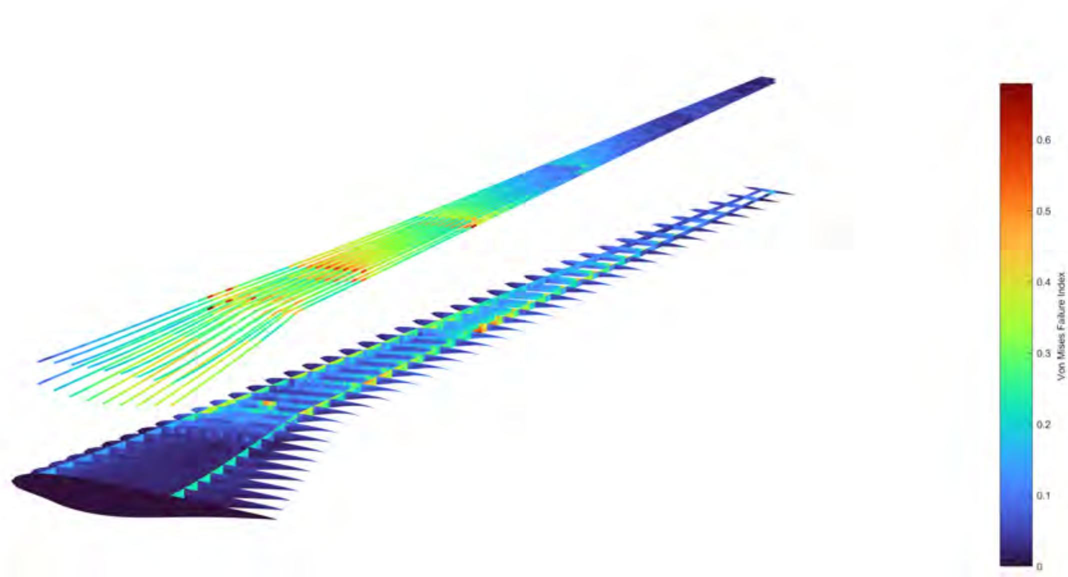


Figure 5.1: Von Mises stress distribution for aluminum components - Mass optimization with steering

Figure 5.2: Composite failure index for upper (left) and lower (right) skins - Mass optimization with steering

It is worth mentioning here that while the critical buckling mode occurs in the first region, significant progress in weight reduction was achieved by the optimizer through the dropping

of plies in subsequent regions of the upper skin. This wouldn't have been possible without tow-steering, due to buckling of those regions becoming more critical.
Moving to the dynamic behavior of the wing, the first nine eigenmodes of the optimal steered wing configuration are visualized in figure 5.3 along with their respective eigenfrequencies.

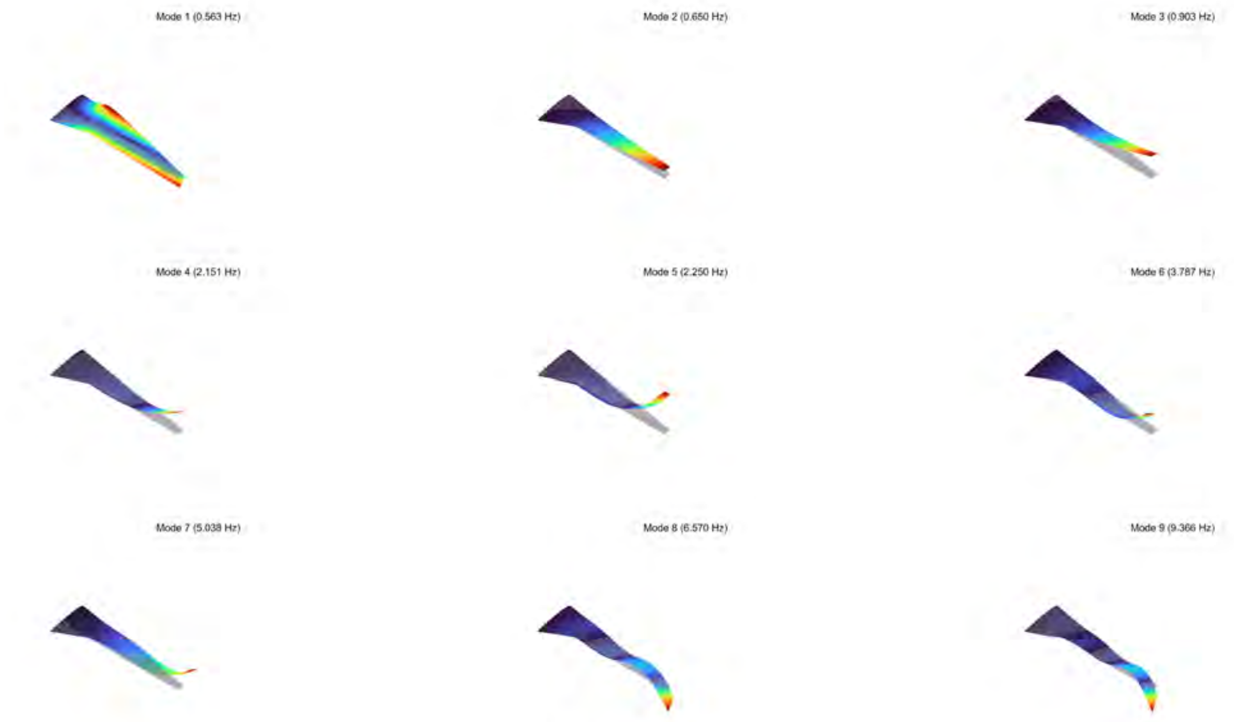
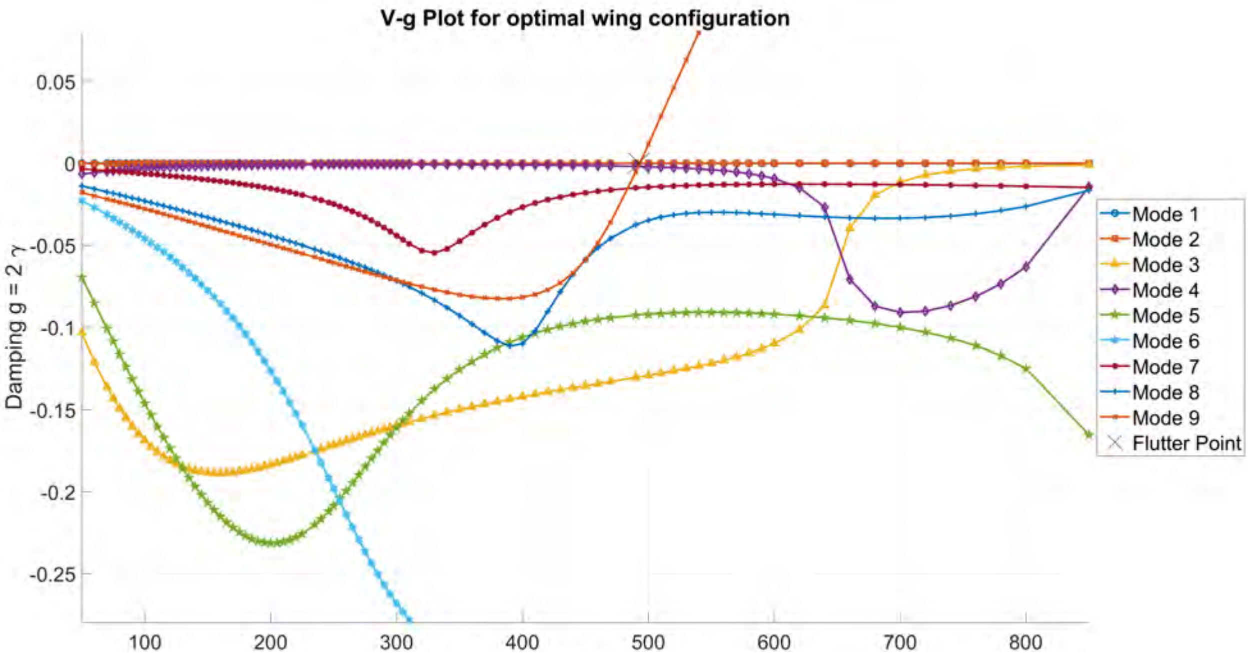


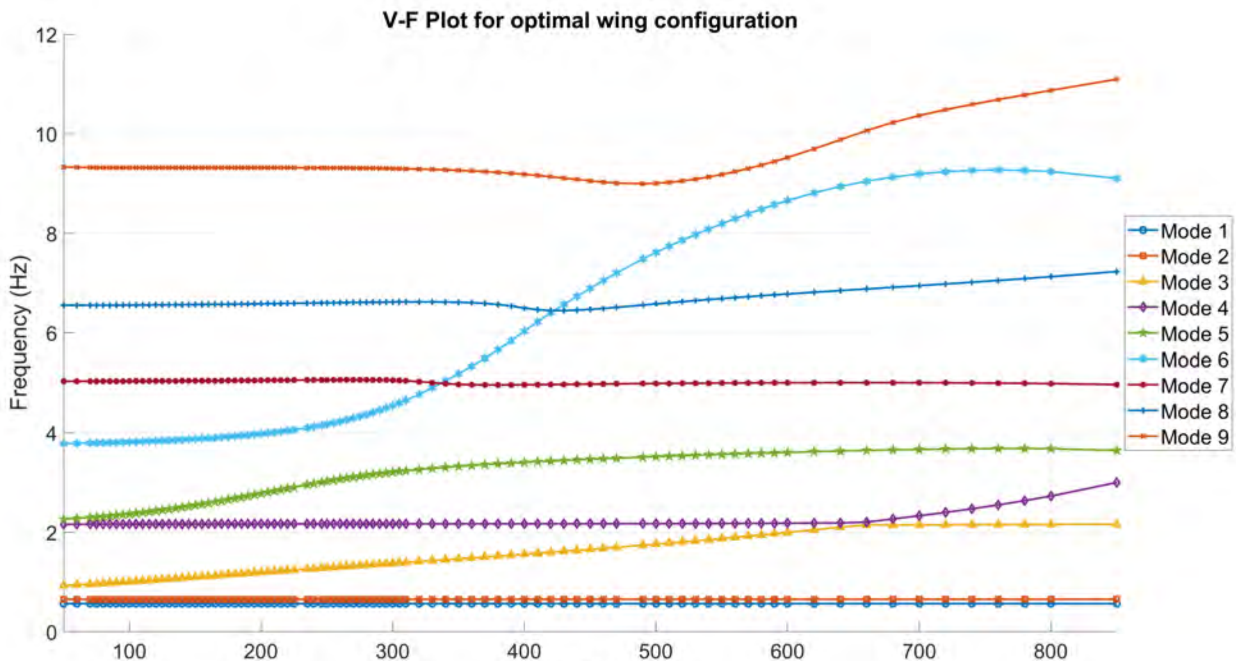
Figure 5.3: First 9 eigenmodes of optimal steered wing configuration

The mode shapes of the wing closely resemble the ones of the previous Chapter, with the most significant difference being that the fourth and fifth mode have been swapped. Concerning the eigenfrequencies, most of them have dropped in value, owning to the reduced stiffness of the structure. The ninth mode features the largest change in frequency with an increase of $0.3Hz$.
Regarding the flutter response of the wing, the V-g and V-F plots of the first 9 modes are shown for this case in figure 5.4.
From the V-g and V-F plots we can deduce the following :

- Flutter occurs once again due to instability of the second mode, which is the first bending mode. The flutter velocity is 492 m/s at a frequency of 0.651 Hz. In higher velocities the damping factor remains positive, however only by a small value.



(a) V-g Plot



(b) V-F Plot

Figure 5.4: V-g and V-F plots for optimal steered wing configuration

- One of the most significant differences in the flutter response compared to the previous results is the instability of the ninth mode shortly after the second mode becomes unstable, at nearly the same velocity. This flutter mechanism is a result of the convergence of frequencies with the sixth mode. Notably, the ninth mode features a much more violent instability compared to the second mode, with the damping factor values becoming increasingly higher.

- The behavior of the fourth and fifth mode have been reversed compared to the optimum unsteered configuration, due to changes in the eigenfrequencies as mentioned above.

Design variable results

Regarding the design variables, firstly we summarize the ones not associated with tow steering in tables 5.3 for the aluminum components and table 5.4 for the composite skins as in Chapter 4.

Table 5.3: Optimal steered wing configuration - aluminum components

	Region 1	Region 2	Region 3	Region 4	Region 5	Region 6
Spars thickness	6 mm	6.5 mm	4 mm	2 mm	1.5 mm	1 mm
Ribs thickness	20.1 mm	15.2 mm	5.3 mm	2 mm	1.8 mm	1.5 mm
Spar caps thickness	8 mm	9.4 mm	5.3 mm	3 mm	2 mm	1 mm
Stringers thickness	9 mm	12.5 mm	6 mm	3.8 mm	3 mm	1 mm

The thickness distribution of the skins based on the above results is better visualized in figure 5.5.

The following conclusions can be drawn about the resulting thickness values compared to the unsteered optimal wing configuration.

- The upper skin thickness has dropped significantly in all regions. This is enabled by the steering patterns that allow for better tailoring against buckling constraints.

- In terms of the bottom skins, thickness has also decreased, particularly in regions 2 and 3, although not to the same extent as observed for the upper skins. Region 2 of the bottom skins is critical for composite ply failure and weight reduction is achieved by adjusting the orientation of the composite plies to improve the local failure index.

Table 5.4: Optimal steered wing configuration - composite skins

Upper Skin						
	Region 1	Region 2	Region 3	Region 4	Region 5	Region 6
No. of 0° plies	7	1	1	1	0	0
No. of 90° plies	9	3	3	2	2	2
No. of ±45° plies	4	7	6	5	2	2
Total in stack	96	72	64	52	24	24

Lower Skin						
	Region 1	Region 2	Region 3	Region 4	Region 5	Region 6
No. of 0° plies	3	4	4	3	1	0
No. of 90° plies	2	4	2	1	1	1
No. of ±45° plies	2	5	3	1	2	2
Total in stack	36	72	48	24	24	20

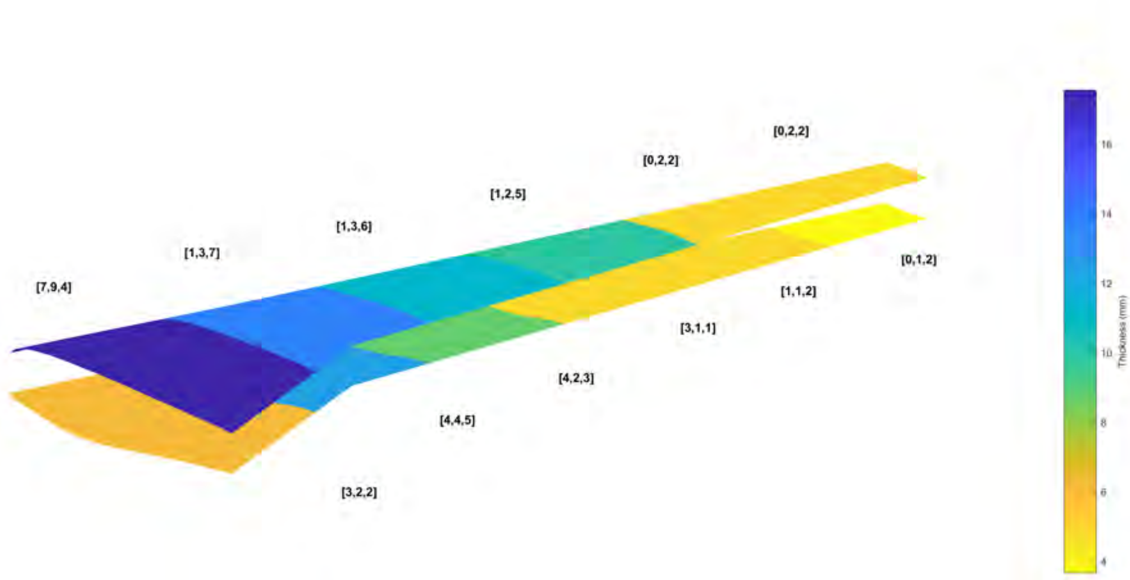


Figure 5.5: Thickness distribution of composite skins for optimal steered design

- In addition to skin thickness improvements, the buckling improvement has allowed for further reduction on rib thickness, especially in region 1. Furthermore, considerable weight savings are achieved by stringer thickness reduction on regions 1 and 2 by means of redistributing the load.

Regarding tow-steering, the results for the angle design variables are depicted in table 5.5 for the upper and lower skins.

Table 5.5: Optimal steering angles for wing mass optimization

			Upper Skin				
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	$39°$	$-29°$	$-14°$	$-24°$	$-39°$	$-33°$	$10°$
			Lower Skin				
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	$16°$	$-45°$	$12°$	$43°$	$-2°$	$-44°$	$-8°$

The main path of the fibers that is created with the above steering angles is better visualized in figure 5.6, where at the left is the top skin and at the right the bottom skin. Recall that the main fiber path is in fact the path that the $0°$ plies follow, while the rest of the plies are rotated by $90°$, $45°$ or $-45°$ from the main path.

It can be observed from the resulting paths that the optimizer heavily utilizes steering to improve the design performance. This is especially true for the bottom skin, where curvature concerns may arise, however this isn't examined in this work. Notably, many angles take extreme values near $-45°$ or $45°$, which is effectively similar to changing the number of plies at each orientation locally.

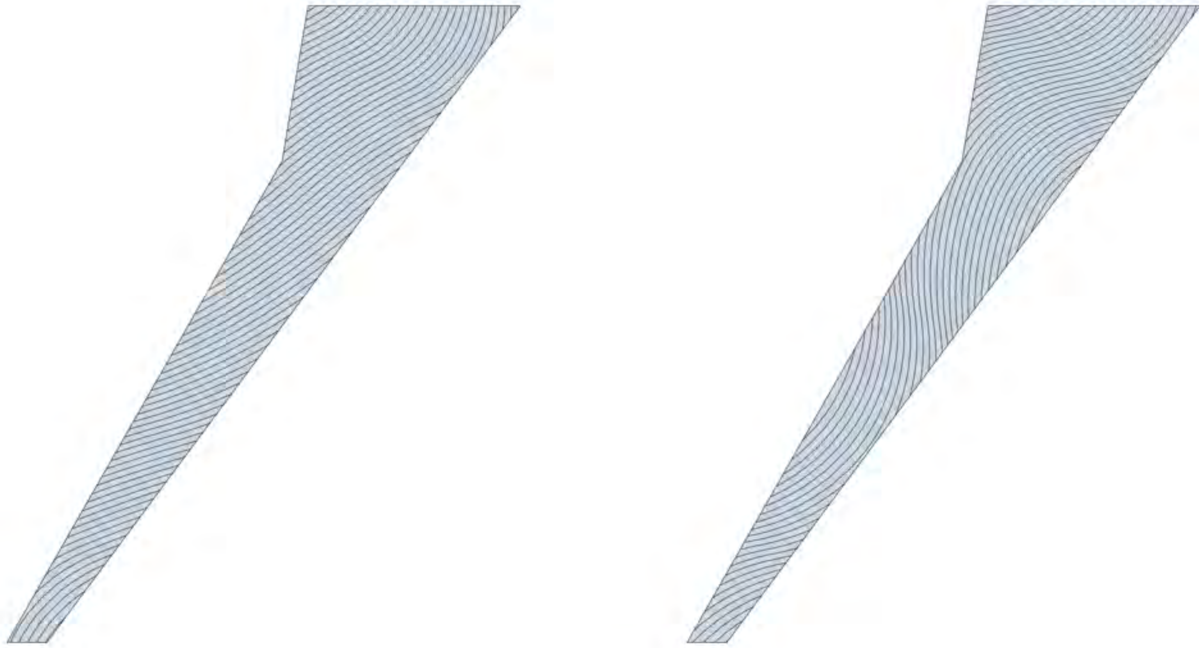


Figure 5.6: Main fiber path for upper (left) and lower (right) skins - Mass optimization with steering

5.3 Composite KS Function Minimization

In this section, we will deal with the optimization of the wing with respect to the KS function of the skins. Even though stress skin failure has not been critical so far, with bar failure being more prominent, there is still value in minimizing it due to the inherent uncertainty involved in calculating failure of laminate structures. In addition, tow-steering directly affects the stresses on the skin, which is a local metric and can be greatly improved by proper fiber orientation.

5.3.1 Optimization Problem Definition

The wing configuration considered is the optimal unsteered one from Chapter 4 (see tables 4.8 and 4.9) and will be kept constant for this optimization problem, meaning that the wing structural mass remains unchanged. This is done in order to isolate the impact of tow-steering on the composite failure index and reduce the complexity, aiming to reach a global optimal solution.
The optimization problem objective and constraints are summarized in table 5.6.

Table 5.6: Objective function and constraints summary for KS function optimization

Objective function: Minimize composite KS function	
Under the constraints :	
Constraint Type	Limit Value
KS(FI), Aluminum Spars	≤ 1
KS(FI), Aluminum Ribs	≤ 1
KS(FI), Aluminum Spar Caps	≤ 1
KS(FI), Aluminum Stringers	≤ 1
First Buckling Eigenvalue	≥ 1.5
Flutter Speed	$\geq 1.2 \cdot V_{dive}$

It's worth noting that the composite skins KS function no longer appears in the constraints, since now it is the objective function. The starting point for the optimization already satisfies this constraint, and thus it becomes unnecessary to include skin failure as a constraint.
The design variables of the problem comprise of the 14 tow-steering angles on the composite skins, depicted in table 5.7.

Table 5.7: Design variables for composite KS function optimization

Design Variable	Lower Bound	Upper bound	Type	Total in Model
Upper skin steering angles	$-45°$	$45°$	Integer	7
Bottom skin steering angles	$-45°$	$45°$	Integer	7

Finally, the optimization parameters for the current study are summarized in table 5.8. As previously mentioned, the starting point for the optimization is the optimal unsteered wing configuration, and therefore all steering angles are initially set to 0°. Moreover, following the approach of previous optimizations, the FOCUS parameter is gradually increased in subsequent runs to improve convergence.

Table 5.8: Optimization parameters for KS function optimization

Run	Iterations	Accuracy	FOCUS	Starting Point
1	300	0.01	0	Un-steered
2	200	0.01	50	Run 1
3	200	0.01	100	Run 2

Optimization Results

The results of the optimization for the objective function and the constraints are outlined in table 5.9. The critical composite KS function appears on the upper skins, thus making it the objective function value. For completeness, the lower skin KS function is also included in the table. It can be observed that the KS function of the upper and lower skins is very close in value.

Table 5.9: Composite KS function optimization results

Property	Value
Objective KS(FI), Composite skins	0.5215
KS(FI), Lower skin	0.5203
KS(FI), Aluminum Spars	0.963
KS(FI), Aluminum Ribs	0.872
KS(FI), Aluminum Spar Caps	1.009
KS(FI), Aluminum Stringers	0.967
First Buckling Eigenvalue	1.494
Flutter Speed	469 m/s

In order to visualize the results, the failure index distribution of the composite wings with optimal steering is presented in figure 5.8. Additionaly, the Von Mises failure index distribution of the aluminum components is also presented in figure 5.7.

The aluminum components have a stress distribution very similar to the previous results. Peak bar failure indices appear on the Yehudi break in region 2, and the same is true for the ribs. Conversely, for the spars, critical regions are observed along the boundaries of subsequent regions and at the rear part.

Regarding the skins, the peak failure index values have decreased significantly, as anticipated since that was the objective of the optimization. The overall distribution remains consistent with past results, with stress concentrations appearing on the same spots. Notably, the

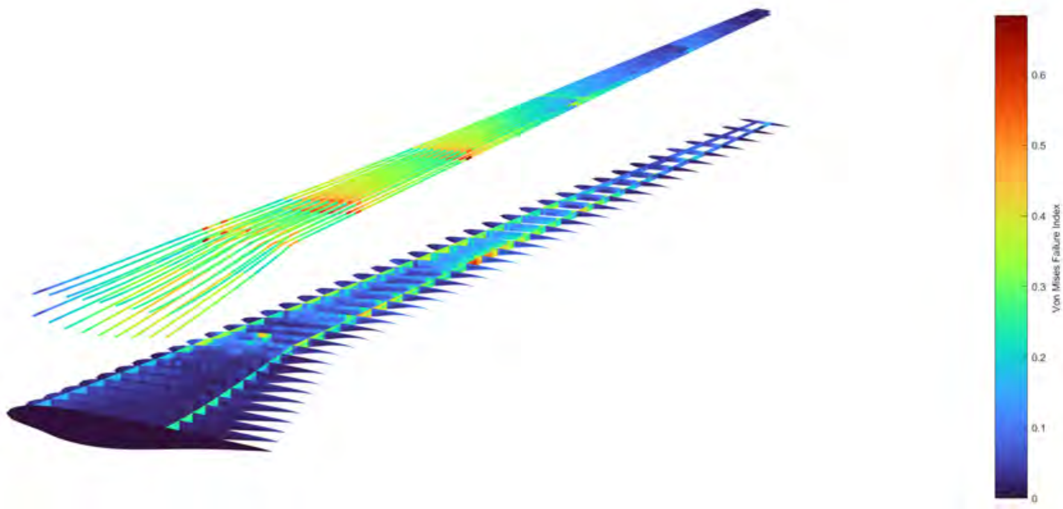


Figure 5.7: Von Mises stress distribution for aluminum components - Composite KS function optimization

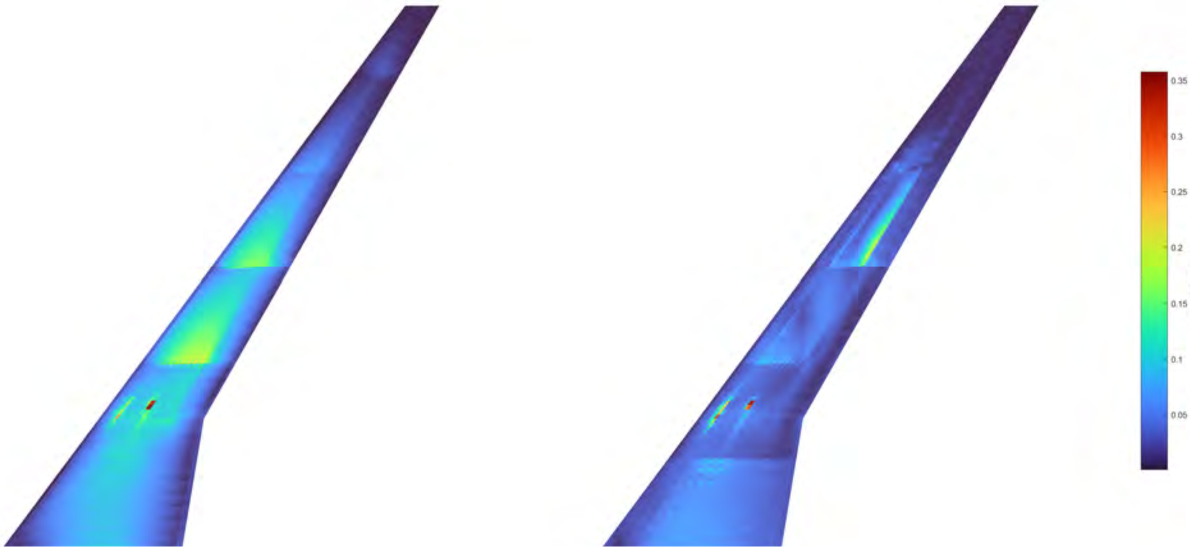


Figure 5.8: Composite failure index for upper (left) and lower (right) skins - Composite KS function optimization

heatmap appears brighter due to the reduction in peak values.

The values of the optimal steering angles found by the optimization are summarized in table 5.10 for the upper and lower skins.

Table 5.10: Optimal steering angles for composite KS function optimization

Upper Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	40°	−14°	−22°	−18°	−26°	−28°	8°

Lower Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	37°	−1°	−37°	14°	15°	−5°	−6°

The fibers steering path which corresponds to the above angles is illustrated in figure 5.9, where at the left is the upper skin and at the right the bottom one.

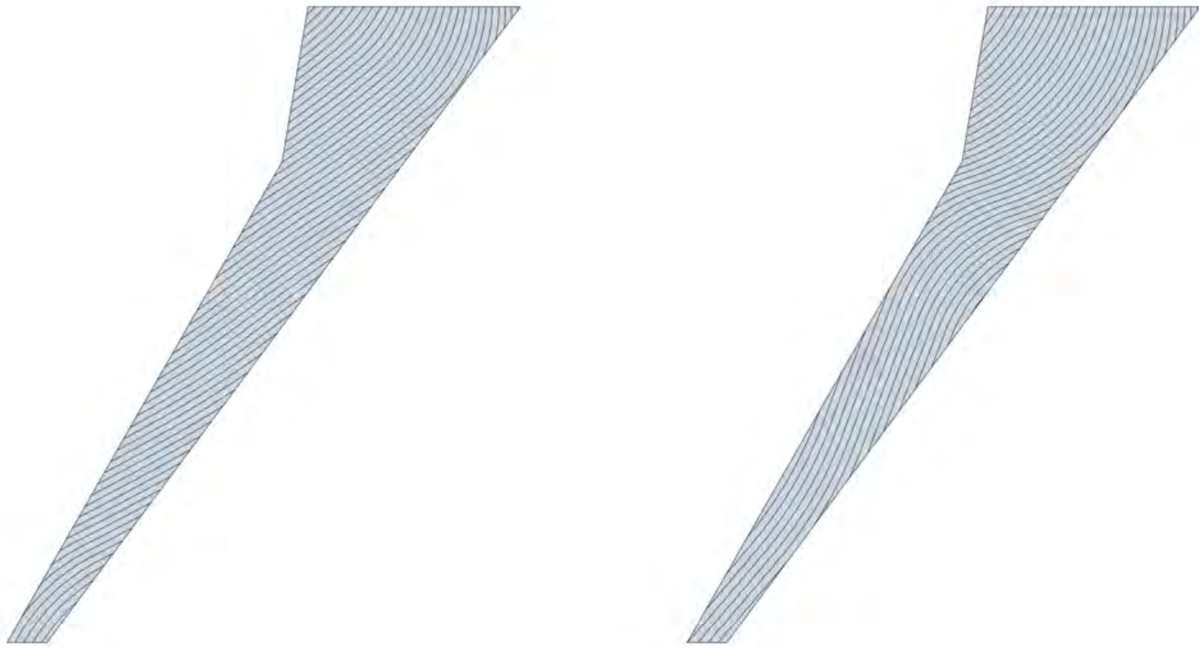


Figure 5.9: Main fiber path for upper (left) and lower (right) skins - Composite KS function optimization

Some observations made from the resulting angles and steering paths are as follows:

- The design variables corresponding to the first regions are the most influential to the optimization. More specifically, steering of the second region affects the objective function the most, while the rest are mainly determined in order to satisfy constraints.

- The critical constraint that impedes the further reduction of the failure index is the flutter instability, greatly impacted by steering angles of the second region. While buckling and bar failure also impose limitations, their influence is comparatively lesser.

- The upper skin paths are similar to the optimized paths found in the mass optimization section, possibly due to buckling constraints, with the main differences appearing on the second region. On the other hand, the bottom skin paths are quite different.

5.4 Flutter Velocity Maximization

This section deals with the topic of optimizing the composite skins of the wing in order to maximize the flutter velocity. Similarly to the previous section, the wing configuration and mass remain constant, and the optimizer relies on determining the optimum steering paths that achieve this objective without adding mass.
Unlike stress, which is a local metric, flutter is a complex phenomenon primarily influenced by the macroscopic dynamic behavior of the structure. The purpose of this study is to examine the effect of tow-steering in the flutter constraint and its ability to improve it.

Optimization Problem Definition

As in the optimization of the previous section, the optimal unsteered wing structure from Chapter 4 will be used for this study. The optimization problem objective and constraints are summarized in table 5.11.
The flutter velocity is not included in the constraints for this study, since it is the objective function, and the constraint will be automatically satisfied. The design variables for the current optimization problem consist of the identical set of 14 tow-steering angles introduced in the previous section, which were listed in Table 5.7.
Lastly, the optimization parameters chosen for this study are the same as those set for the failure index optimization in the previous section in table 5.8.

Table 5.11: Objective function and constraints summary for flutter velocity optimization

Objective function: Maximize flutter velocity	
Under the constraints :	
Constraint Type	Limit Value
KS(FI), Composite skins	≤ 1
KS(FI), Aluminum Spars	≤ 1
KS(FI), Aluminum Ribs	≤ 1
KS(FI), Aluminum Spar Caps	≤ 1
KS(FI), Aluminum Stringers	≤ 1
First Buckling Eigenvalue	≥ 1.5

Optimization Results

The results of the current optimization for the objective function and the constraints are summarized in table 5.12.

Table 5.12: Flutter velocity optimization results

Property	Value
Flutter Velocity	667 m/s
KS(FI), Upper skin	0.942
KS(FI), Lower skin	0.999
KS(FI), Aluminum Spars	0.934
KS(FI), Aluminum Ribs	0.869
KS(FI), Aluminum Spar Caps	1.004
KS(FI), Aluminum Stringers	0.991
First Buckling Eigenvalue	1.503

The advent of tow-steering has raised the flutter velocity to 667 m/s, which is a substantial improvement compared to the un-steered configuration. In order to better examine the dynamic behavior of the optimized for flutter wing, the first 9 normal modes are presented in figure 5.10.

The resulting mode shapes are the the same as the ones of the unsteered wing, exhibiting the only noticeable difference in the torsion mode. Additionally, the eigenfrequencies have been shifted slightly, with the eighth and ninth modes featuring the largest differences.

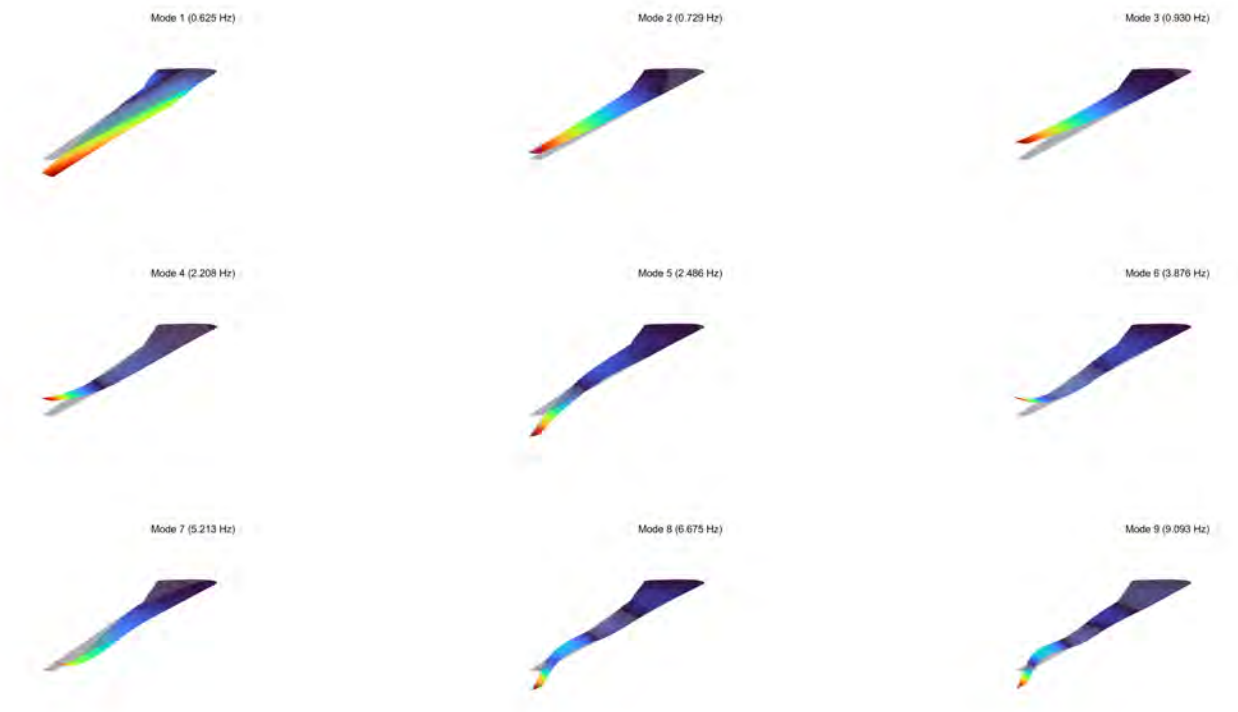


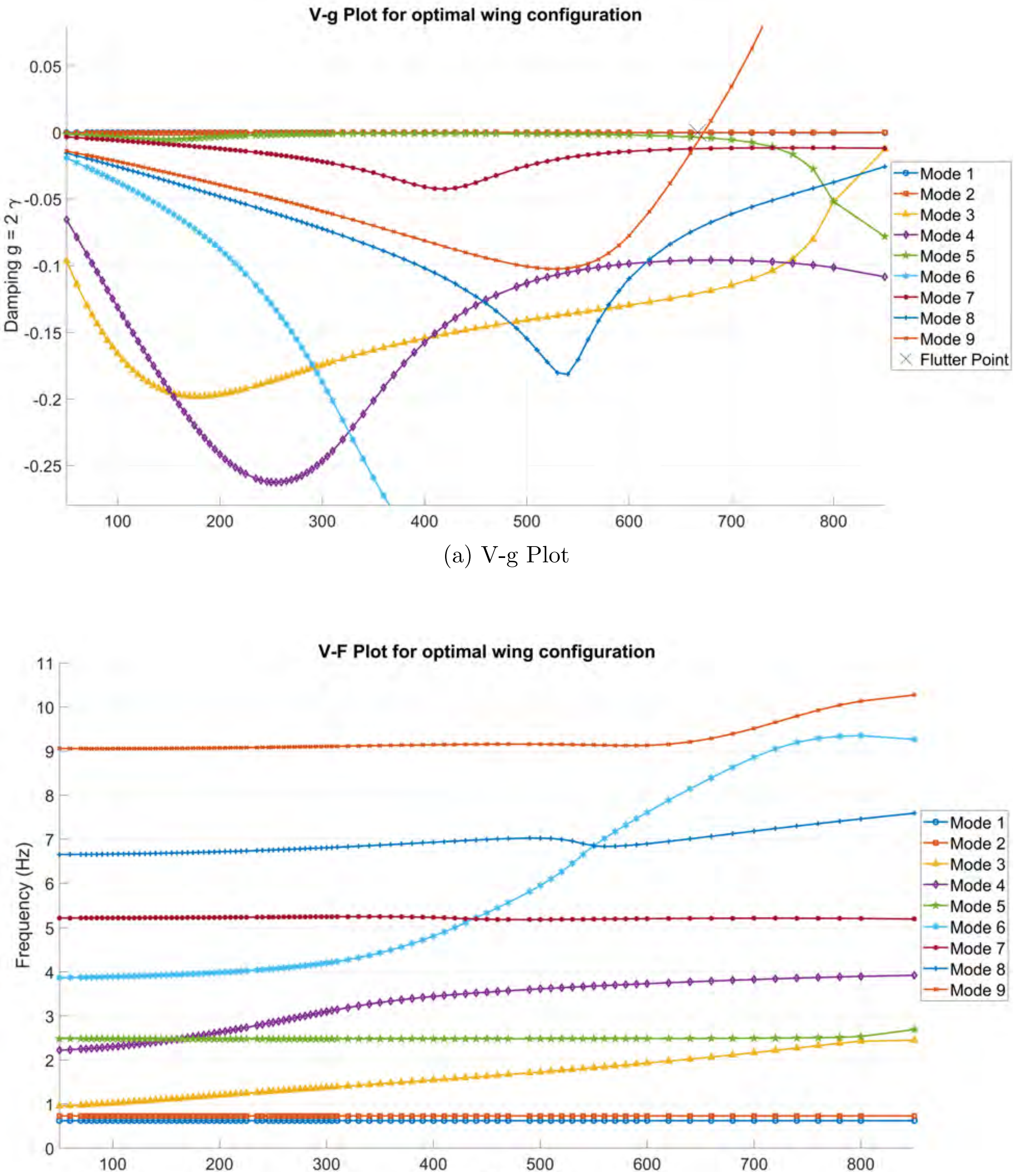
Figure 5.10: First 9 eigenmodes of the optimized for flutter wing configuration

Concerning the flutter response of the wing, Figure 5.11 illustrates the V-g and V-F plots for the first 9 modes of the optimized wing.
The following remarks can be derived from the V-g and V-F plots:

- Consistent with previous results, flutter occurs due to an instability of the second mode. The flutter velocity is 667 m/s at a frequency of 0.7307 Hz, which is effectively the same as in the un-steered case.

- The ninth mode also becomes unstable at only a slightly higher velocity of around 675 m/s. This is similar to the result of the mass optimization with steering, albeit here in a much higher velocity. Tow-steering not only improved the instability of the second mode, but it also improved the flutter velocity of the ninth mode (previously at 640 m/s).

- The flutter mechanism of the ninth mode remains the same, i.e a convergence of frequencies with the sixth mode near the instability velocity. Regarding the second mode, only slight differences are noticed in terms of frequencies, however this is enough to



(a) V-g Plot

(b) V-F Plot

Figure 5.11: V-g and V-F plots of the optimized for flutter wing configuration

alter the flutter velocity significantly in this case, presumably due to the fact that the damping values of this mode are generally very small throughout the range of velocities.

The values of the steering variables for the design with the highest flutter velocity are presented in table 5.13 for the upper and lower skins.

Table 5.13: Optimal steering angles for flutter velocity optimization

Upper Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	$33°$	$5°$	$-2°$	$-25°$	$-24°$	$-7°$	$19°$

Lower Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	$-31°$	$-36°$	$36°$	$17°$	$24°$	$3°$	$-13°$

The core path of the fibers resulting from the aforementioned steering angles is more clearly illustrated in Figure 5.12, with the top skin depicted on the left and the bottom skin on the right.

Regarding the steering angles and resulting paths, the following remarks are made :

- Similarly to the failure index optimization, the design variables of the first regions have the greatest impact on the flutter velocity and constraints. The second region including the Yehudi break once again proves to be pivotal.
- Although buckling and bar failure constraints are active in some runs, the constraint posing the greatest limitation is the skin failure constraint within the second region. Both the flutter velocity and skin failure indices are notably sensitive to changes made on the relevant steering angles.
- Both the upper and lower skin paths differ significantly from previous optimizations. While the upper skin path does feature some similarities, the lower skin path seems almost opposite to the one optimized for skin failure, indicating a trade-off between flutter velocity and skin failure.

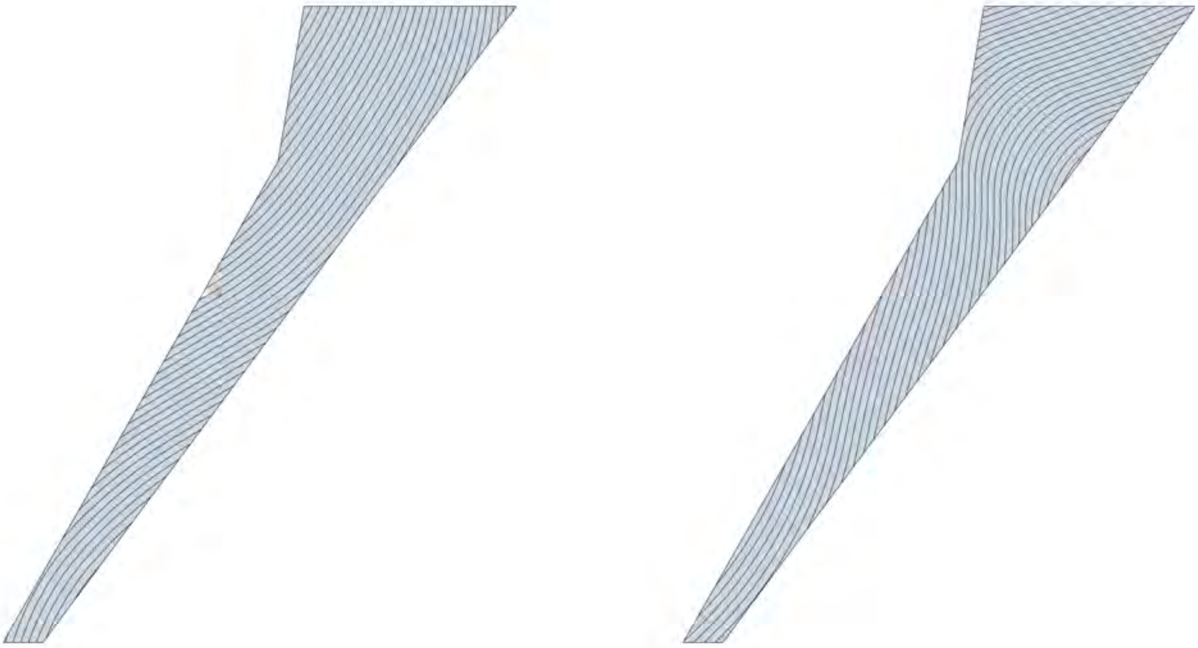


Figure 5.12: Main fiber path for upper (left) and lower (right) skins -Flutter velocity with steering

Chapter 6

Multi-objective Optimization of the uCRM-13.5 wing

6.1 Introduction

In the previous Chapter, we considered single-objective optimization of the wing, using the weight, the skin KS function and finally the flutter velocity as the objective functions to be optimized.
In this Chapter we will conduct multi-objective optimization of the wing, meaning that multiple objectives will be optimized simultaneously. In this case, there isn't a single solution to exist as a global optimum, but rather a set of non-dominated solutions. A solution is considered as a non-dominated solution if there is no other solution that is better than it in every objective and at least as good in one objective. This set of non-dominated solutions form a trade-off curve, also called a Pareto front, between the individual objectives, where no objective can be improved without degrading the other.
As an example, consider a multi-objective optimization toy problem with two objective functions $f_1(x)$ and $f_2(x)$ taken directly from MIDACO site:

$$\text{Minimize} \quad \begin{cases} f_1(x) & = (x_1 - 0)^2 + (x_2 - 0)^2 \\ f_2(x) & = (x_1 - 1)^2 + (x_2 - 1)^2 \end{cases} \quad \text{with} \quad x_1, x_2 \in [0, 1]$$

The Pareto front of this toy problem using the MIDACO solver is the following:

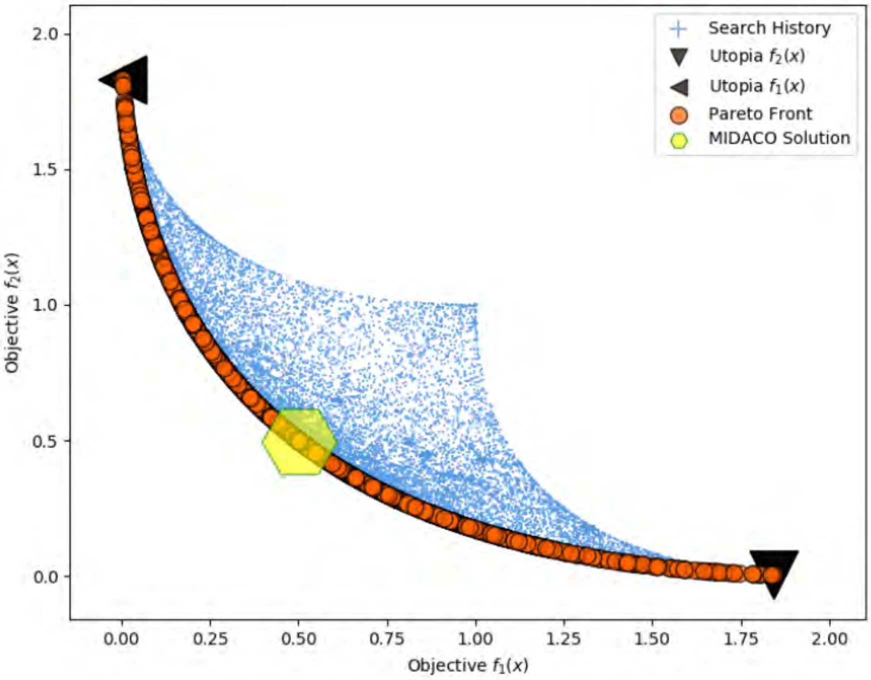


Figure 6.1: Multi-objective example Pareto front

In the graph, the Pareto front is represented by the orange circles. It's evident that no solution found performs better in both objectives simultaneously. This demonstrates a clear trade-off between the two objectives: the best result for $f_1(x)$ corresponds to the lowest value for $f_2(x)$, and vice versa.

6.2 Problem Definition

The multi-objective problem that we will explore considers 3 objectives. These are the skin KS function, the flutter velocity and the first buckling eigenvalue. Although it would certainly be interesting to include the wing weight as an objective, it was chosen not to for several reasons.

First and foremost, this would require the use of the component thicknesses as design variables, since tow-steering doesn't directly affect weight, and thereby considerably increase the complexity.

Moreover, by excluding thickness variables from our problem, we can isolate the effect of tow-steering on wing performance with respect to our other objectives. This approach allows us to get a better understanding of the influence of tow-steering and the underlying

trade-offs.
Throughout the current optimization process, we will utilize the optimal unsteered wing configuration obtained in Chapter 4, and hence the wing mass remains constant.
To summarize, only the 14 steering angles outlined in table 5.7 from the previous chapter are used as design variables for the multi-objective optimization problem.
Furthermore, for better clarity, the formulation of the problem regarding the objective functions is given in table 6.1.

Table 6.1: Multi-objective Problem formulation

Objective Functions	
Minimize:	Skin KS function
Maximize:	Flutter Velocity
Maximize:	First Buckling Eigenvalue

For the optimization problem to be complete, the constraints need to be defined as well. These are the same constraints as the ones considered in the previous Chapters for the 2.5g maneuver and flutter condition. Due to the nature of multi-objective optimization, these constraints need to also be imposed on the objectives, in order to ensure a feasible design. The full set of constraints is presented in table 6.2.

Table 6.2: Multi-objective Optimization Constraints summary

Constraint Type	Limit Value
KS(FI), Composite skins	≤ 1
KS(FI), Aluminum Spars	≤ 1
KS(FI), Aluminum Ribs	≤ 1
KS(FI), Aluminum Spar Caps	≤ 1
KS(FI), Aluminum Stringers	≤ 1
First Buckling Eigenvalue	≥ 1.5
Flutter Speed	$\geq 1.2 \cdot V_{dive}$

6.3 Optimization Parameters

Regarding the optimization parameters, a different approach is followed compared to the previous single-objective optimizations. Rather than conducting multiple runs where the best result from one run initiates the next, we opt for a single run with a larger number of evaluations.

This decision stems from the fact that, unlike in single-objective optimization, in multi-objective optimization transferring the entire Pareto front from one run to the next isn't feasible, resulting in information loss.

Consistent with previous optimizations, the accuracy is once again set to 0.01. Additionally, the FOCUS parameter is turned-off (set to 0) because the initial point is random, and it is desirable to explore the entire design space.

Finally, for multi-objective optimization there is an additional parameter that can be used to determine the relative importance of each objective. Typically, MIDACO provides the entire Pareto front and highlights a certain solution as the best equally balanced among the objectives. By tuning the BALANCE parameter the user can instead focus on a different part of the Pareto front. In this study, the BALANCE parameter will be set to 0, indicating equal weight for all objectives.

Therefore, the optimization parameters used for the optimization are depicted in table 6.3.

Table 6.3: Optimization parameters for multi-objective optimization

Run	Iterations	Accuracy	FOCUS	BALANCE	Starting Point
1	800	0.01	0	0	Unsteered

6.4 Optimization Results

6.4.1 Pareto Fronts

The Pareto front is the trade-off curve between the objectives. Although we're dealing with three objectives, visualizing the Pareto front in three dimensions can be challenging. Hence, for better clarity, we'll focus on the two-dimensional Pareto fronts formed between each pair of objectives. This approach allows us to gain insights into the trade-offs between specific pairs of objectives.

A total of number of 47 Pareto-optimal solutions is found by MIDACO. First, we start with the Pareto front between the skin KS function and the flutter velocity in figure 6.2.
The objective function values are normalized with respect to the unsteered design. The red dot on the graph represents the most balanced Pareto point found among the three objectives.

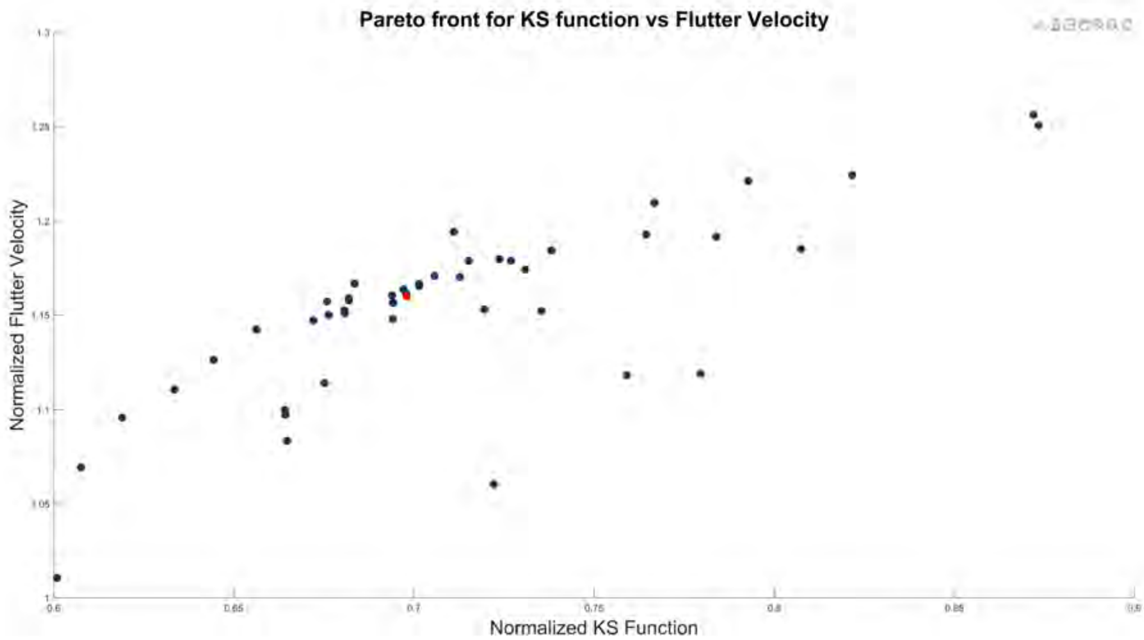


Figure 6.2: Pareto front for skin KS function vs Flutter Velocity

The resulting graph reveals a trade-off between flutter velocity and skin KS function. The design with the highest flutter velocity has the worst KS function and vice versa.
This is consistent with the results of the single-objective optimizations of the previous Chapter, where it was seen that flutter velocity and skin KS function required different steering patterns and were the limiting constraints for the improvement of each objective.
Next, the Pareto front between the skin KS function and the first buckling eigenvalue is presented in figure 6.3.
The relationship between buckling eigenvalue and the KS function is not as straightforward. The design with the lowest KS function has a relatively high buckling eigenvalue and in return the design with the highest buckling eigenvalue has a decently low KS function.
Additionally, there are solutions with both high KS function and low buckling eigenvalue. These solutions are Pareto-optimal owing to their high flutter velocity.

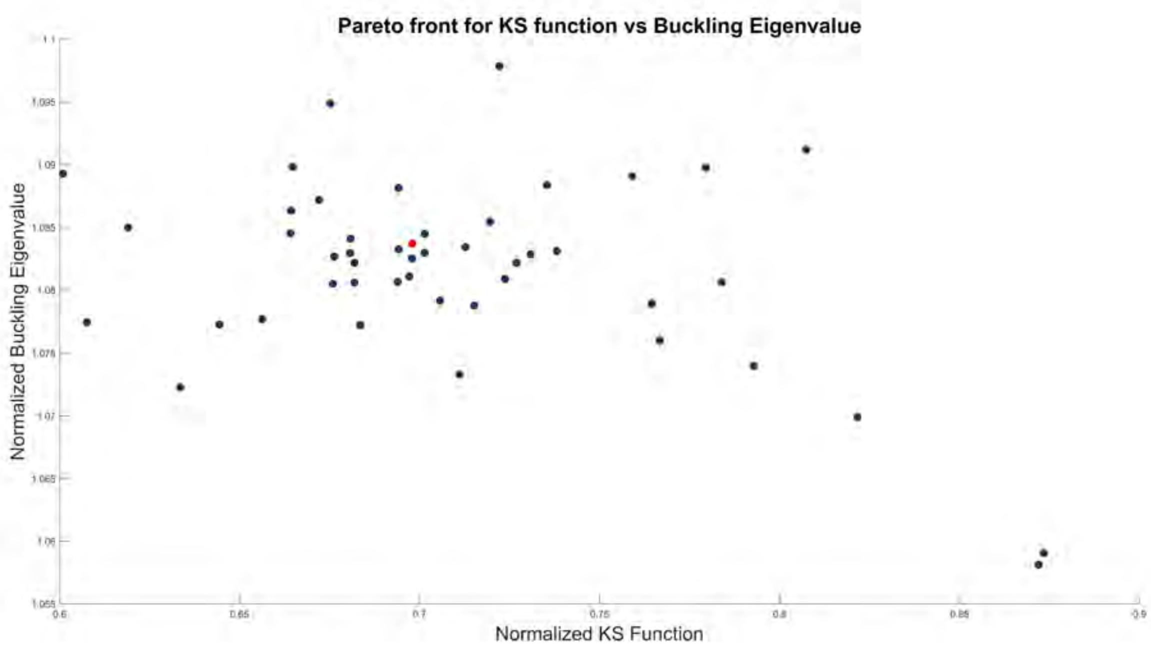


Figure 6.3: Pareto front for skin KS function vs Buckling Eigenvalue

Finally, the Pareto front between buckling eigenvalue and flutter velocity is shown in figure 6.4.
From the resulting graph, it appears that there is a potential trade-off between flutter velocity and buckling eigenvalue, as designs with higher flutter velocities tend to exhibit lower buckling eigenvalues, and vice versa.
It's worth noting however that this trade-off is less clear compared to that between flutter velocity and skin KS function.

6.4.2 Steering Patterns for Pareto-optimal Solutions

Presenting the steering patterns for every single one of the 47 Pareto points is not practical. Instead, the designs with the best individual objective performance will be demonstrated, along with the best equally balanced design.

Pareto point with best skin KS function

We start with the point of the Pareto front that features the lowest skin KS function. It's worth noting that a superior design was identified for the KS function (and flutter velocity) in

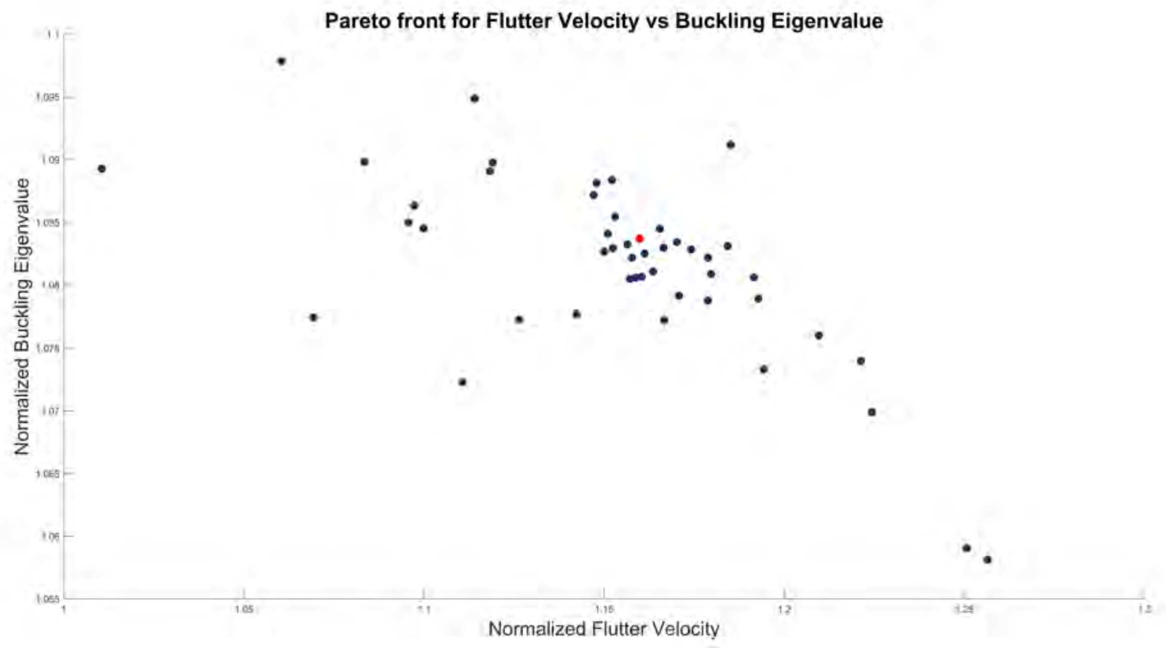


Figure 6.4: Pareto front for skin KS function vs Buckling Eigenvalue

the previous chapter through single-objective optimization, however is is not Pareto optimal with respect to the other objectives.
The values of the objective functions for this Pareto point are shown in table 6.4.

Table 6.4: Objective functions values for Pareto point with highest KS function

Objective	Value
KS(FI), Composite skins	0.592
First Buckling Eigenvalue	1.624
Flutter Speed	480 m/s

The low flutter speed in this design is attributed to the underlying trade-off observed in the Pareto front. On the other hand, the design exhibits a high buckling eigenvalue, which is close to the highest value found in the Pareto front.
The values for the steering angles of this design are depicted in table 6.5.

Table 6.5: Design variables values for Pareto point with highest KS function

Upper Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	23°	−28°	−6°	−13°	−35°	−39°	19°

Lower Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	6°	−44°	11°	13°	13°	−8°	−17°

The steering patterns for the upper and lower skin for this design are illustrated in figure 6.5.

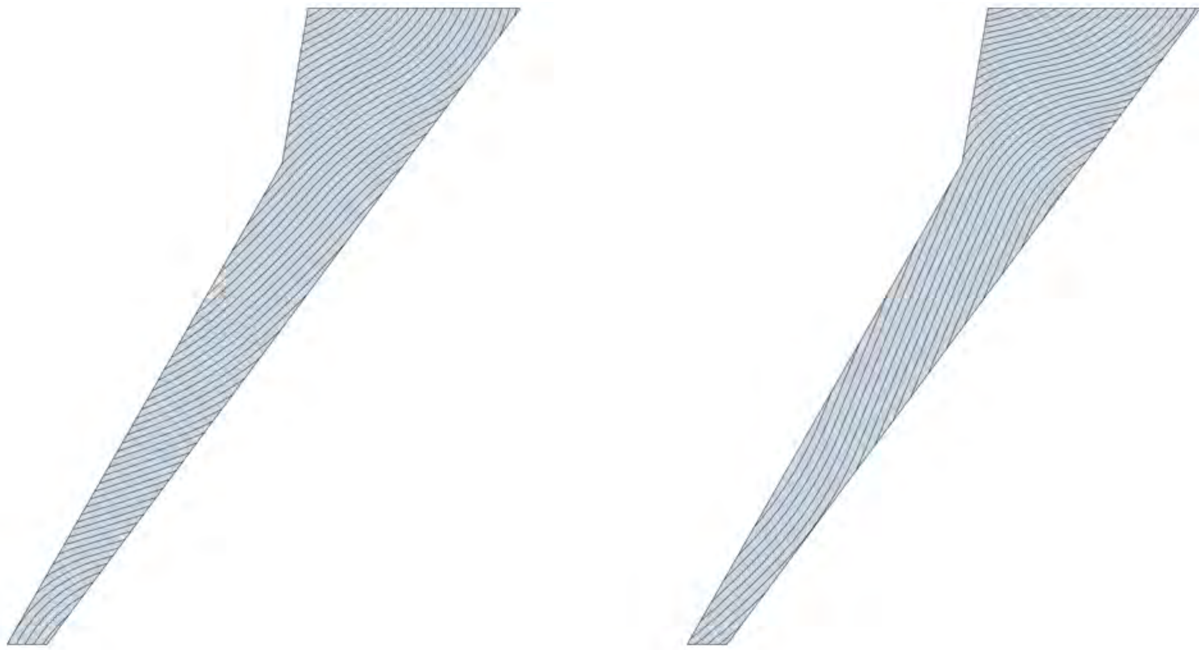


Figure 6.5: Main fiber path for upper (left) and lower (right) skins - Pareto point with best KS function

The resulting steering patterns are quite different than the ones found in single-objective optimization of KS function, despite marginal differences in the KS function value itself. However this can be explained if someone considers the orientation of the fibers in the critical

spots of the upper and lower skin between the 2 designs, which lies around the middle of Region 2. Remarkably, the fiber orientation remains roughly the same in that spot for both designs, leading to similar failure index values due to stress being a local phenomenon, irrespective of the macroscopic fiber patterns.
Additionally, the current design offers a higher flutter velocity and a significantly higher buckling eigenvalue than the one optimized only for skin KS function in the previous Chapter.

Pareto Point with best flutter velocity

Next, we show the Pareto point with the highest flutter velocity. Table 6.6 contains the values of the objectives for this design.

Table 6.6: Objective functions values for Pareto point with highest flutter velocity

Objective	Value
KS(FI), Composite skins	0.859
First Buckling Eigenvalue	1.578
Flutter Speed	596 m/s

The trade-off between flutter velocity and skin KS function is once again evident in this design, with the KS function taking its highest value. Additionally, we can observe the trade-off between flutter and buckling eigenvalue, as this design exhibits the worst buckling performance.
For this design, the values for the design variables are shown in table 6.7.

Table 6.7: Design variables values for Pareto point with highest flutter velocity

Upper Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	30°	−3°	−8°	−17°	−25°	−41°	14°

Lower Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	2°	−29°	11°	16°	14°	6°	−10°

The steering paths of the upper and lower skin of this design are shown in figure 6.6.

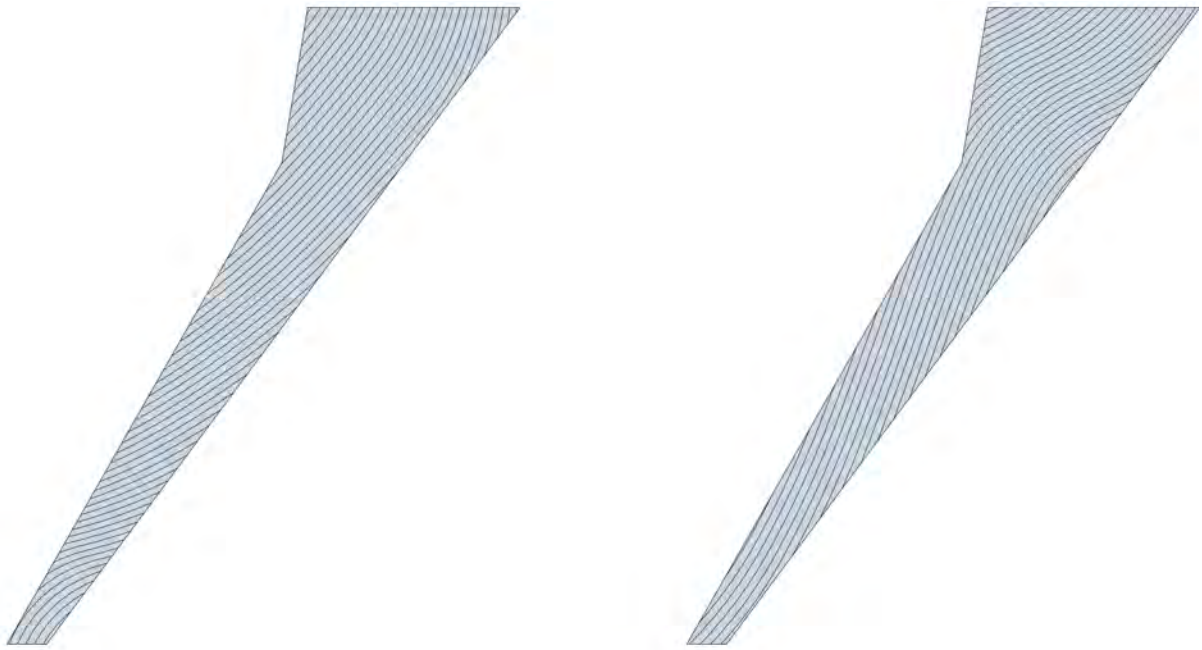


Figure 6.6: Main fiber path for upper (left) and lower (right) skins - Pareto point with best flutter velocity

As it was observed from single-objective flutter velocity optimization, the design variables that have the most impact are the ones pertaining to the first 3 regions, and particularly the second one. The steering patterns of this design don't differ by a lot from the rest of the Pareto points. The main difference lies on the steering angles of the second region, where for flutter different fiber orientations seem to work better, albeit at the expense of a higher skin failure index, thus illustrating the observed trade-off. Presumably, the same is true for the trade-off between flutter speed and buckling eigenvalue on the upper skin patterns.
In comparison to the design optimized solely for flutter in the preceding chapter, similarities are evident in the patterns of the upper skin, while substantial distinctions emerge in those of the lower skin. It's worth noting, however, that the highest flutter velocity found during the multi-objective optimization is significantly lower than the one found in Chapter 5. On the other hand, as expected, this designs offers better performance on the other 2 objectives.

Pareto Point with best buckling eigenvalue

Following this, we move on to the Pareto point with the highest buckling eigenvalue. The values for the objectives for this case are shown in table 6.8.

Table 6.8: Objective functions values for Pareto point with highest buckling eigenvalue

Objective	Value
KS(FI), Composite skins	0.711
First Buckling Eigenvalue	1.637
Flutter Speed	503 m/s

The buckling eigenvalue doesn't vary significantly among the Pareto optimal designs, and similarly the highest buckling eigenvalue found is not considerably higher than that of the rest Pareto points. Regarding the other objectives, the flutter velocity is low, as expected due to the observed trade-off, while the skin KS function takes an intermediate value.
The values for the design variables for this case are depicted in table 6.9.

Table 6.9: Design variables values for Pareto point with highest buckling eigenvalue

Upper Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	33°	−22°	−16°	−17°	−18°	−41°	26°

Lower Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	−2°	−40°	15°	17°	14°	−1°	−34°

Then the steering paths for this case are shown for the upper and lower skin in figure 6.7.
The buckling eigenvalue remains relatively consistent across all designs within the Pareto front, ranging from 1.578 to 1.637. Moreover, since buckling occurs on the upper skin, adjustments to the steering of the lower skin have minimal impact on buckling performance. Consequently, the steering patterns of the lower skin primarily aim to improve the other two objectives, whereas steering adjustments in the upper skin also contribute to buckling performance.

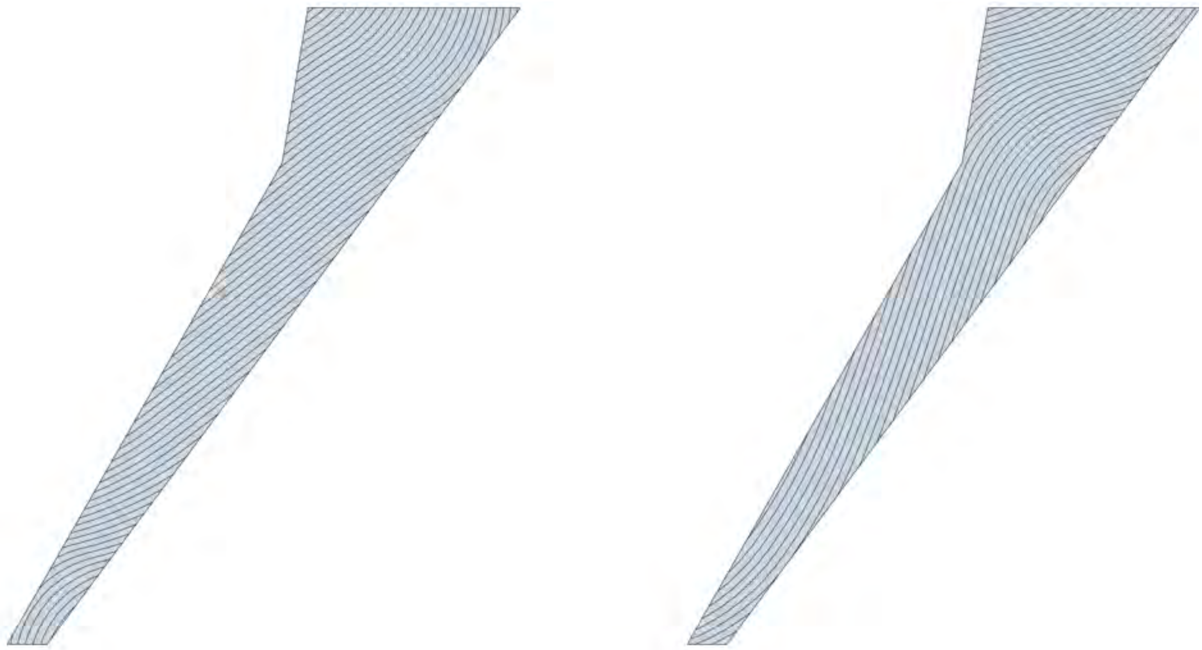


Figure 6.7: Main fiber path for upper (left) and lower (right) skins - Pareto point with best buckling eigenvalue

Similarly to the other objectives, the most critical regions are the first 3, with buckling usually occurring on the first region. The steering angles of the current design exhibit more similarities to the design with the best KS function rather than the one with the highest flutter velocity, resulting in poor flutter performance but relatively low peak failure indices.

Best equally balanced Pareto Point

Last, but certainly not least, is the Pareto point that is found by MIDACO as the best equally balanced solution. The values of the objectives for this solution are shown in table 6.10.

Table 6.10: Objective functions values for the best equally balanced Pareto point

Objective	Value
KS(FI), Composite skins	0.687
First Buckling Eigenvalue	1.612
Flutter Speed	550 m/s

As expected, this designs features intermediate values of the Pareto front for all 3 of the objectives.
The values for the design variables for this case are depicted in table 6.11.

Table 6.11: Design variables values for the best equally balanced Pareto point

Upper Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	38°	−17°	−7°	−21°	−19°	−39°	19°

Lower Skin							
	Bound 1	Bound 2	Bound 3	Bound 4	Bound 5	Bound 6	Bound 7
Steering Angle	0°	−45°	17°	18°	13°	7°	−22°

The steering patterns of the best balanced design are shown for the upper and lower skin in figure 6.8.

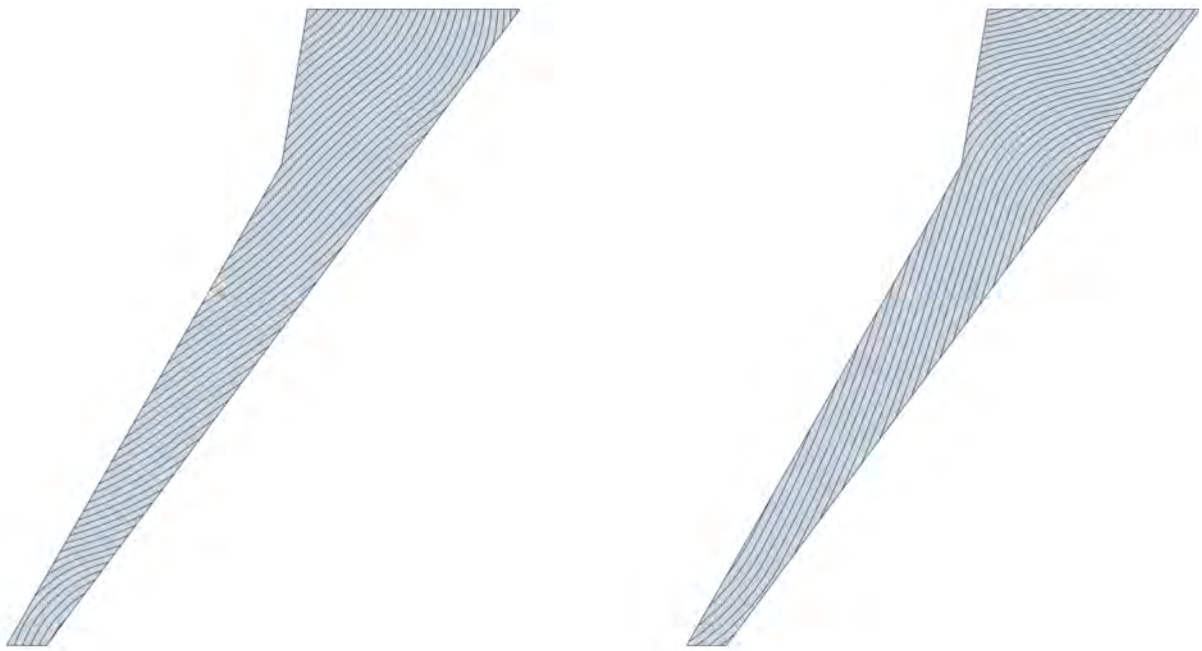


Figure 6.8: Main fiber path for upper (left) and lower (right) skins - Best equally balanced Pareto point

The steering patterns of the equally balanced design generally fall between the values observed in the previous designs. This outcome aligns with expectations, given that the equally balanced design aims to achieve intermediate values for all objectives and thus the steering variables also tend to take intermediate values between the extremes represented by the designs with the best performance in each individual objective.

Chapter 7

Concluding Remarks & Future Work

7.1 Concluding Remarks

This thesis deals with the topic of aeroelastic optimization of the uCRM wing. Chapter 3 provides a detailed development of the aeroelastic model for the wing. Regarding the geometry, the wing configuration includes 41 evenly distributed ribs, a front and a rear spar including spar caps and 7 pairs of upper and lower stringers along the skin. The Finite Element Method is employed for the structural representation of the wing, with a total discretization into 16,355 elements, resulting in 84,222 structural DoF. Representative masses for fuel and the engine are also incorporated to the model to make it more realistic. On the other hand, a DLM model is developed to model the wing aerodynamics, comprising a mesh of 1600 rectangular aerodynamic boxes. Finally, an infinite plate spline is employed to couple the two models.

In Chapter 4, a mass optimization problem is formulated to determine the optimal wing configuration in terms of component thicknesses. The wing's interior is constructed from aluminum, while the skins are composed of composite CFRP material, and the wing is divided into 6 regions along the span of the wing. One design variable is used for each aluminum component thickness, while for the laminate structure of the skins a balanced and symmetric set-up is used and the design variables are the number of $0°$, $\pm 45°$ and $90°$ plies of each laminate. The total number of design variables is 60. The core of the framework involves static and dynamic aeroelastic analyses of the wing using Nastran. More specifically, the constraints imposed during the optimization include stress failure and buckling under an equivalent load of a 2.5g maneuver along with flutter velocity.

The optimizer successfully converged to a structural mass of 16,920 kg while meeting all imposed constraints. The most critical constraints of this optimization are buckling of the upper skin and bar failure near the Yehudi break. Regarding the thickness distribution, the thickness values decrease gradually along the wing, except for the second region near the Yehudi break, where higher values are needed due to stress concentrations. Finally, for the dynamic behavior of the wing, flutter occurred due to an instability of the second mode, which is the first bending mode, at a velocity of 474 m/s. It's worth noting that the damping factor of this mode remains near 0 throughout the range of velocities, and flutter velocity can be greatly improved with changes in the stiffness of the first 2 regions.

Chapter 5 dives into the concept of aeroelastic tailoring via the use of tow-steering of the composite skins, i.e the fibers are allowed to follow curvilinear paths along the span of the wing. Tow-steering is incorporated to the model with 14 new design variables that represent the angles at the limits of each region of the upper and lower skin, and a linear variation is used for the ply orientation between them. For simplicity, the same balanced and symmetric laminate is used for the stacking sequence and tow-steering is considered to uniformly rotate all plies by certain angles defined from the 'core' steering path.

Initially, a mass optimization problem is set-up in the same way as in the previous Chapter but with the addition of the tow-steering variables, increasing the total number of design variables to 74. The optimal wing configuration found by MIDACO has a structural mass of 14805 kg, which is a 12.5% improvement compared to the unsteered case. Although the critical constraints remain the same as before, tow-steering has allowed for further reduction in thickness without constraint violations. The most significant weight savings are achieved through thickness reduction on the first regions, and specifically the second one. The upper skin ply count and rib thickness have dropped mainly by means of improving the buckling constraint, whereas for the bottom skin steering achieves better performance in terms of the failure indices KS function. Additionally, a great reduction is observed in the thickness of the stringers of the first 2 regions, due to load redistribution. Regarding the flutter constraint, although the critical mode is once again the second mode, in this case the ninth mode appears to flutter at nearly the same velocity.

Subsequently, the effect of tow-steering in the composite skin KS function is investigated. For this optimization problem, the objective function is the critical skin KS function, while the mass of the wing remains fixed at the unsteered optimum. The only design variables used are the ones associated with tow-steering of the skins, i.e 14 design variables. Additionally, the constraints of the optimization remain consistent with previous runs.

A significant reduction in peak composite failure indices was achieved in the optimized tow-steered wing, with the critical skin KS function reaching a value of 0.5215 for the upper skin, marking an improvement of 47% compared to the unsteered wing of the same mass. This result was expected, because stress is a local metric and can be greatly improved in a laminate structure by carefully adjusting the orientation of the plies. The most critical failure indices appear on the laminates near the Yehudi break, and therefore the steering variables of the first regions are the most important. Further reduction of the KS function is impeded due to constraint violation, with the most prominent being flutter instability, also greatly impacted by steering of the second region.

The last single-objective optimization problem considered the flutter velocity as the objective function to be maximized. Similarly to the KS function optimization, the mass of the wing once again remains fixed and the only design variables active are the 14 steering angles of the skins. The optimizer converged to a flutter velocity of 667 m/s, which is a 29% improvement compared to the unsteered wing. The unstable mode is once again the second mode, however the ninth mode also becomes unstable at nearly the same velocity.

Unlike stress, flutter is a much more complex phenomenon influenced by the general dynamic behavior of the wing. Despite that, in this study tow-steering was able to significantly improve the flutter velocity. This is presumably the case here owing to the low values of the damping factor g of the second mode throughout the entire velocity range, resulting in considerable shifts in the flutter velocity from relatively small differences in the stiffness of the wing. The steering angles that have the greatest impact on flutter are the ones pertaining to the first 2 regions. Although buckling and bar failure constraints impose some limitations, the most critical one is skin failure. Interestingly, the steering patterns on the lower skin for optimum KS function and optimum flutter velocity are almost the opposite, indicating an underlying trade-off between flutter and peak stresses.

Finally, Chapter 6 dives into the topic of multi-objective optimization of the wing. The wing weight is not considered as an objective, because that would require the addition of thickness variables and increase the design space considerably. Instead, only the 14 design variables associated with steering are used, and therefore the mass of the wing once again remains fixed. Three objective functions are considered to be optimized simultaneously. These are the skin KS function, the flutter velocity, and finally the buckling eigenvalue, all of which can be directly influenced from skin fiber steering. The objective functions are also constrained as in previous optimizations, along with the rest of the constraints.

A single optimization run of 800 iterations was conducted and the three-dimensional Pareto front between the three objectives was found by MIDACO. The two-dimensional projections

of the Pareto front were illustrated in Chapter 6 for each pair of objectives. A clear trade-off between flutter velocity and skin KS function is observed, as it was expected from the results of the respective single-objective optimizations. The design with the best KS function has the lowest flutter velocity and vice versa. A potential trade-off is also observed between flutter velocity and buckling eigenvalue, however it is not as clear, with some solutions having decent values on both objectives. Finally, the relationship between KS function and buckling doesn't offer a specific trend or trade-off.
Regarding tow-steering, most of the Pareto optimal solutions have roughly the same fiber patterns, with the most significant differences being on the second region, which was found throughout this study to be the most influential. The skin KS function value is mainly determined by the critical failure indices near the Yehudi break, where specific orientations are required locally to reduce peak stresses. Interestingly, although the steering patterns between the design of the Pareto point with the lowest KS function and the single-objective optimum have significant differences overall, the local steering angle in the critical area is roughly the same. Regarding the Pareto point with the highest flutter velocity, the steering patterns of the upper skin bear some similarities with the design solely optimized for flutter, whereas in the lower skin more differences are observed. It's worth noting that the best flutter velocity found in the Pareto front is significantly lower than the one found in single-objective optimization. Finally, the buckling objective is primarily impacted by steering of the upper skin, and more so from the first regions that are more prone to buckling. Thereby, the lower skin patterns are determined more by the performance on the other two objectives. The steering angles of the Pareto point with the best buckling eigenvalue are closer to the ones found in the design with the best KS function rather than those of the design with the best flutter velocity, resulting in poor flutter performance and relatively low peak stresses.

7.2 Future Work and Considerations

Although this thesis provided valuable insights into the topic of aeroelastic tailoring and the use of tow-steered composites to enhance wing performance, there are several areas that merit further exploration and improvement. The following bullet points outline potential topics for future research and development:

- *Inclusion of high-fidelity aerodynamics such as RANS or Euler CFD.*
  The weakness of the medium fidelity DLM model used throughout this thesis is the inability to properly account for viscosity and compressibility. On the other hand,

high-fidelity CFD solvers are more accurate because they can capture more complex phenomena such as flow separation and shock waves. Alternatively, CFD corrections may be incorporated to the model to improve accuracy without compromising the computational time, as shown in *Dillinger [12]*.

- *Inclusion of geometric shape design variables and increased thickness resolution.*
  The optimization framework developed in this thesis employed only structural sizing design variables of the 6 regions that the wing was divided. Including geometric shape design variables would allow for a more holistic aerostructural optimization framework that can also take advantage of topology to improve wing performance. Additionally, a more detailed thickness variation can be employed to further enhance the aeroelastic tailoring capabilities.

- *Investigation of different parameterizations for the laminates.*
  In this thesis, due to computational limitations, a balanced and symmetric laminate $[(45_{n_{45}}/0_{n_0}/-45_{n_{-45}}/90_{n_{90}})_S]_S$ has been used, which only requires the definition of 3 design variables per laminate. A more detailed approach may be used to take advantage of the additional tailorability offered by unbalanced and assymetric lay-ups. Interestingly, such lay-ups have been identified to perform better due to bend-torsion coupling in several works, such as *Dillinger [12]*, *Guo et al. [15]* and *Stanford et al. [38]*.

- *Consideration of additional load cases for the aeroelastic analysis of the wing.*
  This thesis used only two cases for the structural sizing of the wing: a static case with a critical load equivalent to a 2.5g maneuver and a flutter analysis during normal flight conditions. Other important load cases for structural sizing include a static -1g maneuever and the dynamic response of the wing to gusts or turbulence. Additionally, other load cases within the flight envelope can be considered for tailoring to increase aerodynamic performance. Finally, control effectiveness can also be explored with the incorporation of a control surface to the model.

- *Considerations for geometrical non-linearities due to large deformations.*
  The structural analysis of this work assumed only small linear deformations. However as the aspect ratio of the wing increases, so do the deformations, and geometric non-linearities become more important. Although linear analysis may still be used in these

cases to explore some basic trends, a geometrically non-linear analysis will produce more accurate results.

- *Utilization of gradient-based optimization techniques.*
  Gradient-based algorithms can handle a larger design space and allow for faster convergence, however they can get stuck in local optima. Initially, a gradient-free optimizer can search for global optimums in the design space, and then a gradient-based algorithm can be incorporated to explore solutions around these optimums.

- *Inclusion of different angle variations for tow-steering.*
  A linear one-dimensional variation of the steering angles is used throughout this work. A relatively straightforward improvement to this approach could involve increasing the order of variation between control points, for instance, a parabolic variation of steering angles. A potentially more rewarding approach might involve a two-dimensional variation of the steering angles to incorporate chordwise steering.

- *Individual steering for each ply or a set of plies.*
  In this work, although globally the plies can take any angle value, locally they are limited to angles of $0°$, $90°$ and $\pm 45°$ between them, and thus remaining specially orthotropic. Assigning a design variable to each ply or a set of plies would increase the tailorability, albeit at the expense of a much larger design space that can be explored only with gradient-based algorithms.

- *Considerations for manufacturing constraints for the AFP process to ensure physically realizable designs.*
  The most common constraint imposed is the limitation of the path curvature in order to avoid fiber wrinkling and twisting. Another possible constraint is concerned with the presence of gaps or overlaps that may appear due to non-parallel steering paths of adjacent tows during the AFP process and influence the response of the laminate. Finally, thickness gradient constraints may be imposed to avoid large thickness differences between adjacent regions.

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